

SEVEN-DIGIT TABLES OF LANE-EMDEN FUNCTIONS

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Abstract. In Table I we present seven digit numerical solutions of the Lane-Emden equation for the plane-parallel, cylindrical, and spherical case for polytropic indices of $n = -10, -5, -4, -3, -2, -1.5, -1.01, -0.9, -0.5, 0, 0.5, 1, 1.5, 2, 3, 4, 5, 6, 10, 20, \pm \infty$, supplemented by $n = 2.5, 3.5, 4.5$, and 4.99 for the spherical case.

In Table II some finite boundary values of polytropic slabs, cylinders, and spheres are summarized. For polytropic spheres ($N = 3$) we have also quoted boundary values near the minimum of the dimensionless mass $-\xi_1^2 \theta_1$, occurring at $n \simeq 4.823$ (Seidov and Kuzakhmedov, 1978).

Introduction

The well-known Lane-Emden equation

$$\begin{aligned} d^2\theta/d\xi^2 + (N-1)(d\theta/d\xi)/\xi &= \mp \theta^n, \\ (N = 1, 2, 3, \dots; -1 < n < \infty; -\infty < n < -1), \end{aligned} \quad (1)$$

admits of solutions of physical interest when $N = 1$ (polytropic slab), $N = 2$ (polytropic cylinder), and $N = 3$ (polytropic sphere) (e.g., Emden, 1907; Chandrasekhar, 1939; Ostriker, 1964; Harrison and Lake, 1972; Viala and Horedt, 1974a). The polytropic indices associated with physical solutions are $-1 < n < \infty$ with the upper minus sign on the right-hand side of Equation (1), and $-\infty < n < -1$ with the lower plus sign.

If $n = \pm \infty$ the Lane-Emden equation has the form

$$d^2\theta/d\xi^2 + (N-1)(d\theta/d\xi)/\xi = \exp(-\theta), \quad (N = 1, 2, 3, \dots; n = \pm \infty). \quad (2)$$

Tabulations are presented for solutions of Equations (1) and (2) obeying the initial conditions $\theta = 1$ and $d\theta/d\xi = 0$ if $\xi = 0$ for Equation (1), and $\theta = d\theta/d\xi = 0$ if $\xi = 0$ for Equation (2). It is relatively simple, and takes at most several seconds on an IBM-360, to solve Equations (1) and (2) by using a Runge-Kutta routine with self-adapting step. To avoid any numerical errors all integrations have been carried out with a precision $\varepsilon = 10^{-12}$, and even with $\varepsilon = 10^{-15}$ in the spherical case $N = 3$ for the polytropic indices

$1.5 < n < \infty$. A precision of ε has been assured by requiring the iterations obtained for θ and $d\theta/d\xi$ with the full step length, and with the halved step lengths, to differ at most by $\varepsilon = 10^{-12}$ or 10^{-15} . A slight difficulty occurs with the first integration step, where we have to take into account that $d\theta/d\xi = \mp \xi/N$ ($\xi \ll 1$), for Equation (1) and $d\theta/d\xi = \xi/N$ ($\xi \ll 1$), for Equation (2). Thus we have to start the Runge–Kutta routine at the origin with the equation

$$d^2\theta/d\xi^2 = \mp 1/N, \quad (\xi = 0 \text{ and } -1 < n < \infty; -\infty < n < -1), \quad (3)$$

instead of Equation (1), and with the equation

$$d^2\theta/d\xi^2 = 1/N, \quad (\xi = 0 \text{ and } n = \pm \infty), \quad (4)$$

instead of Equation (2).

The zeros of Lane–Emden functions at the finite boundary occurring if $-1 < n < \infty$ for $N = 1, 2$ and $-1 < n < 5$ for $N = 3$ have been obtained by halving the length of the ξ -interval where θ changes its sign, and restricting to positive values of θ . The boundary values of ξ and $d\theta/d\xi$ are attained if concomitantly $|\theta_{k+1}| < \varepsilon$, $|\xi_{k+1} - \xi_k| < \varepsilon$ and $|(d\theta/d\xi)_{k+1} - (d\theta/d\xi)_k| < \varepsilon$, where symbols indexed with k and $k + 1$ denote values for the k th and $(k + 1)$ th iteration, respectively. This elementary procedure may require up to several dozen of additional iteration steps, but has the great advantage that the whole integration can be effected with the same Runge–Kutta routine. Since most large computers set numbers smaller than about 10^{-80} equal to zero, our simple procedure for calculating finite boundary values is inexact when ξ and/or $d\theta/d\xi$ change drastically as $\theta < 10^{-80}$. This is the case for values of the polytropic index $n \simeq -1$, ($n > -1$), as can be shown explicitly for the plane-symmetric case ($N = 1$): a first integration of Equation (1) with the canonical initial conditions yields (Harrison and Lake, 1972; Viala and Horedt, 1974b)

$$d\theta/d\xi = \theta' = \mp \{2(1 - \theta^{n+1})/[(\pm(n+1))]\}^{1/2}, \quad N = 1; \quad -1 < n < \infty \\ \text{and } -\infty < n < -1). \quad (5)$$

At the finite or infinite boundary we have $\theta^{n+1} = 0$ and, therefore,

$$\theta' = \mp \{2/[(\pm(n+1))]\}^{1/2}, \quad (N = 1; -1 < n < \infty \text{ and } -\infty < n < -1); \quad (6)$$

providing an additional analytical check for the correctness of our iteration when $-1 < n < \infty$ and $N = 1$.

Comparing Equations (5) and (6) when $\theta = 10^{-80}$ and $n = -0.99$ we obtain $\theta' = -12.9731313$ and -14.1421356 , respectively; this shows the inability of purely numerical iterations to reproduce even the second digit for the boundary value of θ' when $n \simeq -1$, ($n > -1$). If the polytropic index departs from $n = -1$ to $n = -0.9$, the values of θ' from Equations (5) and (6) for $\theta = 10^{-80}$ become -4.47213593 and -4.47213595 , showing coincidence within the first eight digits. Our tables show that the behaviour of solutions is similar for planar, cylindrical, and spherical symmetry if $-1 < n < 0$. Thus, we have excluded from our seven-digit tabulations the polytropic

indices $-1 < n < -0.9$, since in this case the boundary value of the derivative θ' may be incorrect, as shown previously for the case $N = 1$, $-1 < n < -0.9$. A combination of series expansions with numerical methods (e.g., Sadler and Miller, 1932; Linnell, 1975), eventually provides more exact values, but seems not worthwhile to be pursued in the present context.

When $-\infty < n < -1$, $n = \pm \infty$ ($N = 1, 2, 3$), and $5 \leq n < \infty$ ($N = 3$), the values for the points at infinity, have been inserted by hand as they result from analytical solutions ($n = \pm \infty$, $N = 1, 2$; $n = 5$, $N = 3$), asymptotic expansions, and from the topology of Lane-Emden functions (e.g., Chandrasekhar, 1939; Ostriker, 1964; Harrison and Lake, 1972; Viala and Horedt, 1974a, b; Kimura, 1981).

The correctness of our integration has been checked with Equations (5) and (6) and with the analytical solutions for $n = 0, 1$ ($N = 1, 2, 3$), $n = 5$ ($N = 3$), as well as with the seven digit numerical tables of Sadler and Miller (1932), Chandrasekhar and Wares (1949), strengthening the admiration for the huge amount of work done by these authors without the aid of electronic computers. Our tables for the spherical case ($n \geq 1$, $N = 3$), are practically a computer reprint of those of Sadler and Miller (1932) and Chandrasekhar and Wares (1949).

I made efforts to follow closely the tables already published, dropping merely occasionally some intermediate ξ -values spaced apparently too narrowly. There have been omitted the values $n = 2.5, 3.5, 25$ if $N = 1$ from Tascione's (1972) tables, $n = 7, 8, 12, 16$ if $N = 2$ from Ostriker's (1965) tables, and $n = -0.2$ if $N = 1, 2, 3$ from the tables of Viala and Horedt (1974b), since they seem to be too close to other tabulated polytropic indices. In order to be much closer to the special values $n = 5$ ($N = 3$) and $n = 1$ ($N = 1, 2, 3$), I have replaced the value $n = 4.9$ from Emden's (1907) tabulation by $n = 4.99$, and the values $n = -0.8, -1.2$ from the tables of Viala and Horedt (1974b) by $n = -0.9, -1.01$.

Last, but not least, I wish to justify this new publication of the subsequent numerical tables. (i) A first obvious reason stems from the fact that the tables of Lane-Emden functions are spread out over a variety of books, journals, and journal supplements that are not readily accessible to most people: Emden (1907) if $0 \leq n \leq 6$ ($N = 3$), Sadler and Miller (1932) if $1 \leq n \leq 5$ ($N = 3$), Chandrasekhar and Wares (1949) if $n = \pm \infty$ ($N = 3$), Ostriker (1965) if $1 \leq n \leq 20$ ($N = 2$), Tascione (1972) if $0.5 \leq n \leq 25$ ($N = 1$), Viala and Horedt (1974b) if $-10 \leq n < 0$ ($N = 1, 2, 3$). (ii) A second reason is the fact that in the tables of Ostriker (1965) and Tascione (1972), the derivative $d\theta/d\xi$ is not explicitly quoted. (iii) A drawback of most tables, except those of Sadler and Miller (1932) or of Chandrasekhar and Wares (1949), consists in the fact that they are generally only five-digit tables, showing often only three or even fewer places, with the last digit being sometimes inexact (e.g., Ostriker, 1965; Tascione, 1972; Viala and Horedt, 1974b).

Finally it should be stressed that the subsequent tables are computer printouts, and should contain merely errors inherent to the Runge-Kutta method.

TABLE 1. VALUES OF LANE-EMDEN FUNCTIONS.

ξ	θ	θ'	θ^{n+1}
Slab of polytropic index $n = -10$			
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00
1.000E-01	1.004959E+00	9.836838E-02	9.564565E-01
2.000E-01	1.019369E+00	1.877189E-01	8.414273E-01
3.000E-01	1.042004E+00	2.622462E-01	6.905213E-01
4.000E-01	1.071268E+00	3.203581E-01	5.381680E-01
5.000E-01	1.105577E+00	3.635537E-01	4.052290E-01
6.000E-01	1.143577E+00	3.946944E-01	2.989594E-01
7.000E-01	1.184215E+00	4.167769E-01	2.183365E-01
8.000E-01	1.226715E+00	4.323149E-01	1.589672E-01
9.000E-01	1.270524E+00	4.432410E-01	1.159185E-01
1.000E+00	1.315256E+00	4.509497E-01	8.489933E-02
1.200E+00	1.406489E+00	4.603314E-01	4.642740E-02
1.400E+00	1.499093E+00	4.651991E-01	2.615428E-02
1.600E+00	1.592420E+00	4.678111E-01	1.518731E-02
1.800E+00	1.686141E+00	4.692601E-01	9.077262E-03
2.000E+00	1.780083E+00	4.700893E-01	5.572329E-03
2.200E+00	1.874154E+00	4.705775E-01	3.505516E-03
2.400E+00	1.968302E+00	4.708727E-01	2.255147E-03
2.600E+00	2.062496E+00	4.710554E-01	1.480683E-03
2.800E+00	2.156719E+00	4.711710E-01	9.905060E-04
3.000E+00	2.250962E+00	4.712456E-01	6.740423E-04
3.200E+00	2.345216E+00	4.712947E-01	4.659661E-04
3.400E+00	2.439478E+00	4.713275E-01	3.268339E-04
3.600E+00	2.533746E+00	4.713498E-01	2.323442E-04
3.800E+00	2.628018E+00	4.713651E-01	1.672419E-04
4.000E+00	2.722292E+00	4.713758E-01	1.217883E-04
4.200E+00	2.816563E+00	4.713834E-01	8.964330E-05
4.400E+00	2.910845E+00	4.713888E-01	6.665426E-05
4.600E+00	3.005123E+00	4.713927E-01	5.003100E-05
4.800E+00	3.099402E+00	4.713956E-01	3.788772E-05
5.000E+00	3.193682E+00	4.713977E-01	2.893180E-05
5.500E+00	3.429381E+00	4.714009E-01	1.524256E-05
ξ	θ	θ'	θ^{n+1}
6.000E+00	3.665082E+00	4.714025E-01	8.380047E-06
6.500E+00	3.900784E+00	4.714034E-01	4.782259E-06
7.000E+00	4.136486E+00	4.714039E-01	2.820437E-06
7.500E+00	4.372188E+00	4.714041E-01	1.712817E-06
8.000E+00	4.607890E+00	4.714043E-01	1.067779E-06
8.500E+00	4.843592E+00	4.714044E-01	6.815412E-07
9.000E+00	5.079294E+00	4.714044E-01	4.443956E-07
9.500E+00	5.314996E+00	4.714045E-01	2.954427E-07
1.000E+01	5.550699E+00	4.714045E-01	1.999269E-07
1.100E+01	6.022103E+00	4.714045E-01	9.599893E-08
1.200E+01	6.493508E+00	4.714045E-01	4.871760E-08
1.300E+01	6.964912E+00	4.714045E-01	2.592742E-08
1.400E+01	7.436317E+00	4.714045E-01	1.438067E-08
1.500E+01	7.907721E+00	4.714045E-01	8.270617E-09
1.600E+01	8.379126E+00	4.714045E-01	4.911472E-09
1.700E+01	8.850530E+00	4.714045E-01	3.001072E-09
1.800E+01	9.321935E+00	4.714045E-01	1.881246E-09
1.900E+01	9.793339E+00	4.714045E-01	1.206765E-09
2.000E+01	1.026474E+01	4.714045E-01	7.904366E-10
2.500E+01	1.262177E+01	4.714045E-01	1.230039E-10
3.000E+01	1.497879E+01	4.714045E-01	2.634570E-11
3.500E+01	1.733581E+01	4.714045E-01	7.071351E-12
4.000E+01	1.969283E+01	4.714045E-01	2.245050E-12
4.500E+01	2.204986E+01	4.714045E-01	8.116113E-13
5.000E+01	2.440688E+01	4.714045E-01	3.253792E-13
6.000E+01	2.912092E+01	4.714045E-01	6.639778E-14
7.000E+01	3.383497E+01	4.714045E-01	1.720711E-14
8.000E+01	3.854901E+01	4.714045E-01	5.319618E-15
9.000E+01	4.326306E+01	4.714045E-01	1.883410E-15
1.000E+02	4.797711E+01	4.714045E-01	7.424952E-16
1.200E+02	9.511756E+01	4.714045E-01	1.569111E-18
1.422580E+02	1.422580E+02	4.714045E-01	4.190921E-20
1.893985E+02	1.893985E+02	4.714045E-01	3.188688E-21
2.365389E+02	2.365389E+02	4.714045E-01	4.313963E-22
4.722412E+02	4.722412E+02	4.714045E-01	8.561017E-25
∞	∞	4.714045E-01	0.000000E+00

ξ	θ	θ'	θ^{n+1}	ξ	θ	θ'	θ^{n+1}
Slab of polytropic index $n = -5$							
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	7.500E+00	5.903181E+00	7.068156E-01	8.234846E-04
1.000E-01	1.004979E+00	9.917613E-02	9.803282E-01	8.000E+00	6.256605E+00	7.068760E-01	6.525970E-04
2.000E-01	1.019677E+00	1.936256E-01	9.250183E-01	8.500E+00	6.610055E+00	7.069216E-01	5.238172E-04
3.000E-01	1.043420E+00	2.795962E-01	8.436520E-01	9.000E+00	6.963525E+00	7.069564E-01	4.252884E-04
4.000E-01	1.075237E+00	3.548664E-01	7.481397E-01	9.500E+00	7.317010E+00	7.069834E-01	3.488714E-04
5.000E-01	1.114013E+00	4.187533E-01	6.492914E-01	1.000E+01	7.670507E+00	7.070046E-01	2.888708E-04
6.000E-01	1.158624E+00	4.717417E-01	5.549195E-01	1.100E+01	8.377528E+00	7.070350E-01	2.030189E-04
7.000E-01	1.208036E+00	5.150009E-01	4.695481E-01	1.200E+01	9.084574E+00	7.070549E-01	1.468188E-04
8.000E-01	1.261348E+00	5.499738E-01	3.950576E-01	1.300E+01	9.791636E+00	7.070683E-01	1.087875E-04
9.000E-01	1.317803E+00	5.781059E-01	3.315872E-01	1.400E+01	1.049871E+01	7.070777E-01	8.231072E-05
1.000E+00	1.376785E+00	6.007019E-01	2.783144E-01	1.500E+01	1.120579E+01	7.070844E-01	6.342056E-05
1.200E+00	1.500442E+00	6.335226E-01	1.972982E-01	1.600E+01	1.191288E+01	7.070892E-01	4.965161E-05
1.400E+00	1.629445E+00	6.550368E-01	1.418537E-01	1.700E+01	1.261997E+01	7.070928E-01	3.942457E-05
1.600E+00	1.761985E+00	6.694211E-01	1.037508E-01	1.800E+01	1.332706E+01	7.070956E-01	3.170022E-05
1.800E+00	1.896912E+00	6.792516E-01	7.723441E-02	1.900E+01	1.403416E+01	7.070977E-01	2.577831E-05
2.000E+00	2.033489E+00	6.861183E-01	5.848339E-02	2.000E+01	1.474126E+01	7.070993E-01	2.117687E-05
2.200E+00	2.171229E+00	6.910151E-01	4.499635E-02	2.500E+01	1.827677E+01	7.071036E-01	8.961950E-06
2.400E+00	2.309807E+00	6.945748E-01	3.513157E-02	3.000E+01	2.181229E+01	7.071052E-01	4.417687E-06
2.600E+00	2.448998E+00	6.972087E-01	2.780011E-02	3.500E+01	2.534782E+01	7.071059E-01	2.422354E-06
2.800E+00	2.588647E+00	6.991890E-01	2.226942E-02	4.000E+01	2.888335E+01	7.071063E-01	1.436845E-06
3.000E+00	2.728642E+00	7.007000E-01	1.803905E-02	4.500E+01	3.241888E+01	7.071065E-01	9.053322E-07
3.200E+00	2.868904E+00	7.018683E-01	1.476169E-02	5.000E+01	3.595441E+01	7.071066E-01	5.983995E-07
3.400E+00	3.009373E+00	7.027828E-01	1.219259E-02	6.000E+01	4.302548E+01	7.071067E-01	2.918080E-07
3.600E+00	3.150004E+00	7.035067E-01	1.015676E-02	7.000E+01	5.009655E+01	7.071067E-01	1.587702E-07
3.800E+00	3.290766E+00	7.040855E-01	8.527312E-03	8.000E+01	5.716761E+01	7.071067E-01	9.362671E-08
4.000E+00	3.431631E+00	7.045527E-01	7.211035E-03	9.000E+01	6.423868E+01	7.071068E-01	5.872372E-08
4.200E+00	3.572581E+00	7.049331E-01	6.138633E-03	1.000E+02	7.130975E+01	7.071068E-01	3.867269E-08
4.400E+00	3.713600E+00	7.052454E-01	5.257988E-03	2.000E+02	1.420204E+02	7.071068E-01	2.458048E-09
4.600E+00	3.854676E+00	7.055035E-01	4.529484E-03	3.000E+02	2.127311E+02	7.071068E-01	1.549380E-10
4.800E+00	3.995798E+00	7.057185E-01	3.922706E-03	4.000E+02	2.834418E+02	7.071068E-01	1.549332E-10
5.000E+00	4.136961E+00	7.058987E-01	3.414086E-03	5.000E+02	3.541525E+02	7.071068E-01	6.358806E-11
5.500E+00	4.490000E+00	7.062363E-01	2.460450E-03	1.000E+03	7.077059E+02	7.071068E-01	3.986473E-12
6.000E+00	4.843179E+00	7.064639E-01	1.817515E-03	∞	∞	7.071068E-01	0.000000E+00
6.500E+00	5.196452E+00	7.066217E-01	1.371425E-03	Slab of polytropic index $n = -4$			
7.000E+00	5.549793E+00	7.067340E-01	1.054127E-03	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00
				1.000E-01	1.004983E+00	9.933960E-02	9.851975E-01

ξ	θ	θ'	θ^{n+1}	ξ	θ	θ'	θ^{n+1}
2.000E-01	1.019740E+00	1.948613E-01	9.430436E-01	1.000E+01	8.599533E+00	8.158544E-01	1.572445E-03
3.000E-01	1.043722E+00	2.834083E-01	8.795196E-01	1.100E+01	9.415468E+00	8.160073E-01	1.198043E-03
4.000E-01	1.076118E+00	3.629019E-01	8.024533E-01	1.200E+01	1.023153E+01	8.161153E-01	9.338367E-04
5.000E-01	1.115969E+00	4.324181E-01	7.195219E-01	1.300E+01	1.104769E+01	8.161938E-01	7.416272E-04
6.000E-01	1.162270E+00	4.919951E-01	6.369112E-01	1.400E+01	1.186391E+01	8.162521E-01	5.988473E-04
7.000E-01	1.214059E+00	5.423214E-01	5.588312E-01	1.500E+01	1.268019E+01	8.162963E-01	4.904817E-04
8.000E-01	1.270460E+00	5.844316E-01	4.876595E-01	1.600E+01	1.349650E+01	8.163305E-01	4.067581E-04
9.000E-01	1.330710E+00	6.194755E-01	4.243751E-01	1.700E+01	1.431285E+01	8.163573E-01	3.410530E-04
1.000E+00	1.394157E+00	6.485715E-01	3.690325E-01	1.800E+01	1.512922E+01	8.163787E-01	2.887691E-04
1.200E+00	1.528568E+00	6.928241E-01	2.799921E-01	1.900E+01	1.594560E+01	8.163959E-01	2.466477E-04
1.400E+00	1.670395E+00	7.236219E-01	2.145571E-01	2.000E+01	1.676201E+01	8.164099E-01	2.123352E-04
1.600E+00	1.817418E+00	7.453926E-01	1.665848E-01	2.500E+01	2.084417E+01	8.164515E-01	1.104195E-04
1.800E+00	1.968147E+00	7.610660E-01	1.311678E-01	3.000E+01	2.492648E+01	8.164702E-01	6.456793E-05
2.000E+00	2.121568E+00	7.725629E-01	1.047199E-01	3.500E+01	2.900886E+01	8.164799E-01	4.096452E-05
2.200E+00	2.276979E+00	7.811497E-01	8.470767E-02	4.000E+01	3.309128E+01	8.164853E-01	2.759683E-05
2.400E+00	2.433891E+00	7.876725E-01	6.935805E-02	4.500E+01	3.717371E+01	8.164886E-01	1.946669E-05
2.600E+00	2.591949E+00	7.927053E-01	5.742757E-02	5.000E+01	4.125616E+01	8.164908E-01	1.424077E-05
2.800E+00	2.750900E+00	7.966442E-01	4.803697E-02	5.500E+01	4.542108E+01	8.164932E-01	8.284442E-06
3.000E+00	2.910553E+00	7.997676E-01	4.055772E-02	6.000E+01	4.942108E+01	8.164944E-01	5.236593E-06
3.200E+00	3.070766E+00	8.022738E-01	3.453503E-02	6.500E+01	5.357509E+01	8.164951E-01	3.517981E-06
3.400E+00	3.231431E+00	8.043069E-01	2.963571E-02	7.000E+01	5.758602E+01	8.164956E-01	2.476202E-06
3.600E+00	3.392464E+00	8.059724E-01	2.561263E-02	7.500E+01	6.157509E+01	8.164958E-01	1.808315E-06
3.800E+00	3.553801E+00	8.073495E-01	2.228029E-02	8.000E+01	6.575097E+01	8.164965E-01	1.278301E-07
4.000E+00	3.715389E+00	8.084974E-01	1.949787E-02	8.500E+01	6.991592E+01	8.164966E-01	6.768332E-08
4.200E+00	3.877187E+00	8.094618E-01	1.715733E-02	9.000E+01	7.408088E+01	8.164966E-01	2.859157E-08
4.400E+00	4.039164E+00	8.102778E-01	1.517489E-02	9.500E+01	7.824580E+01	8.164966E-01	1.465047E-08
4.600E+00	4.201291E+00	8.109727E-01	1.348503E-02	1.000E+02	8.240808E+01	8.164966E-01	1.834210E-09
4.800E+00	4.363546E+00	8.115680E-01	1.203598E-02	1.000E+03	8.657278E+01	8.164966E-01	0.000000E+00
5.000E+00	4.525912E+00	8.120811E-01	1.078653E-02				
5.500E+00	4.932219E+00	8.130870E-01	8.334375E-03				
6.000E+00	5.338952E+00	8.138096E-01	6.571006E-03				
6.500E+00	5.745997E+00	8.143418E-01	5.271132E-03				
7.000E+00	6.153272E+00	8.147424E-01	4.292216E-03				
7.500E+00	6.560724E+00	8.150496E-01	3.541154E-03				
8.000E+00	6.968311E+00	8.152891E-01	2.955408E-03				
8.500E+00	7.376004E+00	8.154786E-01	2.491934E-03				
9.000E+00	7.783783E+00	8.156305E-01	2.120449E-03				
9.500E+00	8.191630E+00	8.157535E-01	1.819236E-03				

Slab of polytropic index $n = -3$

ξ	θ	θ'	θ^{n+1}	ξ	θ	θ'	θ^{n+1}
7.000E-01	1.220656E+00	5.734623E-01	6.711409E-01	1.500E+01	1.503330E+01	9.977852E-01	4.424779E-03
8.000E-01	1.280625E+00	6.246950E-01	6.097561E-01	1.600E+01	1.603122E+01	9.980526E-01	3.891051E-03
9.000E-01	1.345362E+00	6.689647E-01	5.524862E-01	1.700E+01	1.702939E+01	9.982744E-01	3.448276E-03
1.000E+00	1.414214E+00	7.071088E-01	5.000000E-01	1.800E+01	1.802776E+01	9.984604E-01	3.076923E-03
1.200E+00	1.562050E+00	7.682213E-01	4.098361E-01	1.900E+01	1.902630E+01	9.986178E-01	2.762431E-03
1.400E+00	1.720465E+00	8.137335E-01	3.378378E-01	2.000E+01	2.002498E+01	9.987523E-01	2.493766E-03
1.600E+00	1.886786E+00	8.479983E-01	2.808989E-01	2.500E+01	2.501999E+01	9.992010E-01	1.597444E-03
1.800E+00	2.059126E+00	8.741573E-01	2.358491E-01	3.000E+01	3.001668E+01	9.994449E-01	1.109878E-03
2.000E+00	2.236068E+00	8.944272E-01	2.000000E-01	3.500E+01	3.501428E+01	9.995921E-01	8.156607E-04
2.200E+00	2.416609E+00	9.103665E-01	1.712329E-01	4.000E+01	4.001250E+01	9.996876E-01	6.246096E-04
2.400E+00	2.600000E+00	9.230769E-01	1.479290E-01	4.500E+01	4.501111E+01	9.997532E-01	4.935834E-04
2.600E+00	2.785678E+00	9.333456E-01	1.288660E-01	5.000E+01	5.001000E+01	9.998001E-01	3.998401E-04
2.800E+00	2.973214E+00	9.417419E-01	1.131222E-01	6.000E+01	6.000833E+01	9.998611E-01	2.777006E-04
3.000E+00	3.162278E+00	9.486833E-01	1.000000E-01	7.000E+01	7.000714E+01	9.998980E-01	2.040400E-04
3.200E+00	3.352611E+00	9.544800E-01	8.896797E-02	8.000E+01	8.000625E+01	9.999219E-01	1.562256E-04
3.400E+00	3.544009E+00	9.593655E-01	7.961763E-02	9.000E+01	9.000556E+01	9.999383E-01	1.234416E-04
3.600E+00	3.736308E+00	9.635179E-01	7.163324E-02	1.000E+02	1.000050E+02	9.999500E-01	9.999000E-05
3.800E+00	3.929377E+00	9.670745E-01	6.476684E-02	2.000E+02	2.00025E+02	9.999875E-01	2.499938E-05
4.000E+00	4.123106E+00	9.701425E-01	5.882353E-02	3.000E+02	3.000017E+02	9.999944E-01	1.111099E-05
4.200E+00	4.317407E+00	9.728062E-01	5.364807E-02	4.000E+02	4.000012E+02	9.999969E-01	6.249961E-06
4.400E+00	4.512206E+00	9.751329E-01	4.911591E-02	5.000E+02	5.000010E+02	9.999980E-01	3.999984E-06
4.600E+00	4.707441E+00	9.771764E-01	4.512635E-02	1.000E+03	1.000000E+03	9.999995E-01	9.999990E-07
4.800E+00	4.903060E+00	9.789804E-01	4.159734E-02	∞	∞	1.000000E+00	0.000000E+00
5.000E+00	5.099020E+00	9.805807E-01	3.846154E-02	Slab of polytropic index $n = -2$			
5.500E+00	5.590170E+00	9.838699E-01	3.200000E-02				
6.000E+00	6.082763E+00	9.863939E-01	2.702703E-02	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00
6.500E+00	6.576473E+00	9.883717E-01	2.312139E-02	1.000E-01	1.004992E+00	9.968849E-02	9.950331E-01
7.000E+00	7.071068E+00	9.899495E-01	2.000000E-02	2.000E-01	1.019869E+00	1.973906E-01	9.805185E-01
8.000E+00	8.062258E+00	9.922779E-01	1.538462E-02	3.000E-01	1.043446E+00	2.914216E-01	9.575367E-01
8.500E+00	8.558621E+00	9.931506E-01	1.365188E-02	4.000E-01	1.077983E+00	3.803726E-01	9.276583E-01
9.000E+00	9.055385E+00	9.938837E-01	1.218512E-02	5.000E-01	1.120219E+00	4.632873E-01	8.926824E-01
9.500E+00	9.552487E+00	9.945055E-01	1.095890E-02	6.000E-01	1.170421E+00	5.396423E-01	8.543931E-01
1.000E-01	1.004988E+01	9.950372E-01	9.900990E-03	7.000E-01	1.227924E+00	6.092901E-01	8.143828E-01
1.100E+01	1.104536E+01	9.958932E-01	8.196721E-03	8.000E-01	1.292060E+00	6.723725E-01	7.739576E-01
1.200E+01	1.204159E+01	9.965458E-01	6.896552E-03	9.000E-01	1.362191E+00	7.292302E-01	7.341116E-01
1.300E+01	1.303840E+01	9.970545E-01	5.882353E-03	1.000E+00	1.437714E+00	7.803223E-01	6.955485E-01
1.400E+01	1.403567E+01	9.974587E-01	5.076142E-03	1.200E+00	1.602789E+00	8.672805E-01	6.239123E-01

ξ	θ	θ'	θ^{n+1}	ξ	θ	θ'	θ^{n+1}
1.400E+00	1.783501E+00	9.373421E-01	5.606949E-01	2.000E+01	2.646756E+01	1.387240E+00	3.778210E-02
1.600E+00	1.976843E+00	9.941256E-01	5.058571E-01	2.500E+01	3.341899E+01	1.392894E+00	2.992310E-02
1.800E+00	2.180463E+00	1.040559E+00	4.586182E-01	3.000E+01	4.039331E+01	1.396598E+00	2.475658E-02
2.000E+00	2.392529E+00	1.078918E+00	4.179677E-01	3.500E+01	4.738318E+01	1.399211E+00	2.110454E-02
2.200E+00	2.611609E+00	1.110940E+00	3.829058E-01	4.000E+01	5.438431E+01	1.401151E+00	1.838766E-02
2.400E+00	2.836572E+00	1.137947E+00	3.525382E-01	4.500E+01	6.139396E+01	1.402649E+00	1.628255E-02
2.600E+00	3.066521E+00	1.160946E+00	3.261025E-01	5.000E+01	6.841028E+01	1.403839E+00	1.461769E-02
2.800E+00	3.300735E+00	1.180709E+00	3.029628E-01	6.000E+01	8.245809E+01	1.405612E+00	1.212737E-02
3.000E+00	3.538629E+00	1.197835E+00	2.825953E-01	7.000E+01	9.652082E+01	1.406869E+00	1.036046E-02
3.200E+00	3.779724E+00	1.212791E+00	2.645696E-01	8.000E+01	1.105944E+02	1.407803E+00	9.042049E-03
3.400E+00	4.023625E+00	1.225943E+00	2.485321E-01	9.000E+01	1.246762E+02	1.408531E+00	8.020775E-03
3.600E+00	4.270001E+00	1.237585E+00	2.341920E-01	1.000E+02	1.387645E+02	1.409109E+00	7.208453E-03
3.800E+00	4.518574E+00	1.247951E+00	2.213088E-01	2.000E+02	2.798338E+02	1.411684E+00	3.573550E-03
4.000E+00	4.769109E+00	1.257233E+00	2.096828E-01	3.000E+02	4.210504E+02	1.412533E+00	2.375012E-03
4.200E+00	5.021405E+00	1.265585E+00	1.991474E-01	4.000E+02	5.623289E+02	1.412956E+00	1.778325E-03
4.400E+00	5.275290E+00	1.273135E+00	1.895630E-01	5.000E+02	7.036360E+02	1.413208E+00	1.421189E-03
4.600E+00	5.530613E+00	1.279991E+00	1.808118E-01	1.000E+03	1.410393E+03	1.413712E+00	7.090213E-04
4.800E+00	5.787246E+00	1.286240E+00	1.727938E-01	∞	∞	1.414214E+00	0.000000E+00
5.000E+00	6.045074E+00	1.291957E+00	1.654240E-01	Slab of polytropic index $n = -1.5$			
5.500E+00	6.694246E+00	1.304314E+00	1.493820E-01	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00
6.000E+00	7.349024E+00	1.314479E+00	1.360725E-01	1.000E-01	1.004994E+00	9.975112E-02	9.975124E-01
6.500E+00	8.008448E+00	1.322975E+00	1.248681E-01	2.000E-01	1.019901E+00	1.980353E-01	9.901955E-01
7.000E+00	8.671784E+00	1.330176E+00	1.153165E-01	3.000E-01	1.044507E+00	2.935109E-01	9.784628E-01
7.500E+00	9.338453E+00	1.336350E+00	1.070841E-01	4.000E-01	1.078472E+00	3.850622E-01	9.629318E-01
8.000E+00	1.000800E+01	1.341700E+00	9.992010E-02	5.000E-01	1.121360E+00	4.718568E-01	9.443378E-01
8.500E+00	1.068904E+01	1.346378E+00	9.363259E-02	6.000E-01	1.172667E+00	5.533589E-01	9.234485E-01
9.000E+00	1.135428E+01	1.350502E+00	8.807249E-02	7.000E-01	1.231847E+00	6.293061E-01	9.009935E-01
9.500E+00	1.203047E+01	1.354162E+00	8.312230E-02	8.000E-01	1.298342E+00	6.996628E-01	8.776180E-01
1.000E+01	1.270838E+01	1.357433E+00	7.868823E-02	9.000E-01	1.371597E+00	7.645653E-01	8.538600E-01
1.100E+01	1.406870E+01	1.363026E+00	7.107975E-02	1.000E+00	1.451081E+00	8.242685E-01	8.301454E-01
1.200E+01	1.543410E+01	1.367632E+00	6.479158E-02	1.200E+00	1.626751E+00	9.294251E-01	7.840422E-01
1.300E+01	1.680372E+01	1.371488E+00	5.951084E-02	1.400E+00	1.821749E+00	1.018050E+00	7.408933E-01
1.400E+01	1.817689E+01	1.374762E+00	5.501492E-02	1.600E+00	2.033061E+00	1.093007E+00	7.013339E-01
1.500E+01	1.955309E+01	1.377576E+00	5.114281E-02	1.800E+00	2.258209E+00	1.156799E+00	6.654538E-01
1.600E+01	2.093191E+01	1.380019E+00	4.777394E-02	2.000E+00	2.495175E+00	1.211500E+00	6.330667E-01
1.700E+01	2.231303E+01	1.382160E+00	4.481687E-02	2.200E+00	2.742315E+00	1.258782E+00	6.038670E-01
1.800E+01	2.369615E+01	1.384051E+00	4.220095E-02				
1.900E+01	2.508106E+01	1.385734E+00	3.987072E-02				

ξ	θ	θ'	θ^{n+1}	ξ	θ	θ'	θ^{n+1}
2.400E+00	2.99233E+00	1.29976E+00	5.775156E-01	4.500E+01	7.936044E+01	1.884407E+00	1.122530E-01
2.600E+00	3.261971E+00	1.336142E+00	5.536814E-01	5.000E+01	8.879917E+01	1.890905E+00	1.061196E-01
2.800E+00	3.532461E+00	1.368122E+00	5.320609E-01	6.000E+01	1.077623E+02	1.901230E+00	9.63109E-02
3.000E+00	3.808986E+00	1.396591E+00	5.123837E-01	7.000E+01	1.268158E+02	1.909136E+00	8.880008E-02
3.200E+00	4.090899E+00	1.422092E+00	4.944138E-01	8.000E+01	1.459397E+02	1.915434E+00	8.277767E-02
3.400E+00	4.377654E+00	1.445065E+00	4.779465E-01	9.000E+01	1.651207E+02	1.920603E+00	7.782143E-02
3.600E+00	4.668781E+00	1.465871E+00	4.628052E-01	1.000E+02	1.843490E+02	1.924940E+00	7.365116E-02
3.800E+00	4.963878E+00	1.484806E+00	4.488378E-01	2.000E+02	3.781951E+02	1.947900E+00	5.142118E-02
4.000E+00	5.262595E+00	1.502114E+00	4.359132E-01	3.000E+02	5.735314E+02	1.957799E+00	4.175624E-02
4.200E+00	5.564629E+00	1.518001E+00	4.239180E-01	4.000E+02	7.696239E+02	1.963623E+00	3.604630E-02
4.400E+00	5.869713E+00	1.532639E+00	4.127543E-01	5.000E+02	9.661945E+02	1.967566E+00	3.217123E-02
4.600E+00	6.177611E+00	1.546174E+00	4.023368E-01	1.000E+03	1.952817E+03	1.977241E+00	2.262920E-02
4.800E+00	6.488117E+00	1.558729E+00	3.925913E-01	*	*	2.000000E+00	0.000000E+00
5.000E+00	6.801044E+00	1.570410E+00	3.834530E-01	Slab of polytropic index $n = -1.01$			
5.500E+00	7.592916E+00	1.596362E+00	3.629073E-01				
6.000E+00	8.396775E+00	1.618519E+00	3.450990E-01	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00
6.500E+00	9.210938E+00	1.637688E+00	3.294944E-01	1.000E-01	1.004996E+00	9.983226E-02	9.99502E-01
7.000E+00	1.003406E+01	1.654460E+00	3.156905E-01	2.000E-01	1.019933E+00	1.986720E-01	9.998026E-01
7.500E+00	1.086507E+01	1.669278E+00	3.033777E-01	3.000E-01	1.044666E+00	2.95936E-01	9.995631E-01
8.000E+00	1.170307E+01	1.682481E+00	2.923143E-01	4.000E-01	1.078961E+00	3.897943E-01	9.992403E-01
8.500E+00	1.254733E+01	1.694333E+00	2.823088E-01	5.000E-01	1.123513E+00	4.806305E-01	9.988450E-01
9.000E+00	1.339722E+01	1.705042E+00	2.732076E-01	6.000E-01	1.174959E+00	5.676329E-01	9.983890E-01
1.000E+01	1.425221E+01	1.714776E+00	2.648859E-01	7.000E-01	1.235901E+00	6.505019E-01	9.978842E-01
1.100E+01	1.511185E+01	1.723669E+00	2.572416E-01	8.000E-01	1.304917E+00	7.290890E-01	9.973421E-01
1.200E+01	1.684358E+01	1.739357E+00	2.436592E-01	9.000E-01	1.381575E+00	8.03712E-01	9.967730E-01
1.300E+01	1.858982E+01	1.752789E+00	2.319329E-01	1.000E+00	1.465450E+00	8.734220E-01	9.961857E-01
1.400E+01	2.034857E+01	1.764445E+00	2.216834E-01	1.200E+00	1.653197E+00	1.001448E+00	9.948855E-01
1.500E+01	2.211823E+01	1.774677E+00	2.126301E-01	1.400E+00	1.865048E+00	1.114763E+00	9.937865E-01
1.600E+01	2.389753E+01	1.783747E+00	2.045613E-01	1.600E+00	2.092444E+00	1.215205E+00	9.926164E-01
1.700E+01	2.568541E+01	1.791855E+00	1.973135E-01	1.800E+00	2.350391E+00	1.304562E+00	9.914906E-01
1.800E+01	2.748098E+01	1.799157E+00	1.907585E-01	2.000E+00	2.619437E+00	1.384441E+00	9.904166E-01
1.900E+01	2.928349E+01	1.805775E+00	1.847943E-01	2.200E+00	2.903629E+00	1.456228E+00	9.893970E-01
2.000E+01	3.109233E+01	1.811807E+00	1.793384E-01	2.400E+00	3.201468E+00	1.521095E+00	9.884313E-01
2.500E+01	3.290694E+01	1.817335E+00	1.743236E-01	2.600E+00	3.511671E+00	1.580023E+00	9.875176E-01
3.000E+01	4.205199E+01	1.839339E+00	1.542079E-01	2.800E+00	3.833136E+00	1.633827E+00	9.865530E-01
3.500E+01	5.129010E+01	1.855121E+00	1.396314E-01	3.000E+00	4.164906E+00	1.683190E+00	9.858344E-01
4.000E+01	6.059697E+01	1.867124E+00	1.284620E-01	3.200E+00	4.506154E+00	1.728681E+00	9.850583E-01
	6.995722E+01	1.876636E+00	1.195594E-01				

ξ	θ	θ'	θ^{n+1}	ξ	θ	θ'	θ^{n+1}
3.400E+00	4.856152E+00	1.770777E+00	9.843217E-01	9.000E+01	2.615037E+02	3.290701E+00	9.458564E-01
3.600E+00	5.214265E+00	1.809863E+00	9.836216E-01	1.000E+02	2.945844E+02	3.324743E+00	9.447304E-01
3.800E+00	5.579929E+00	1.846338E+00	9.829552E-01	2.000E+02	6.390526E+02	3.537160E+00	9.374425E-01
4.000E+00	5.952643E+00	1.880436E+00	9.823198E-01	3.000E+02	9.990162E+02	3.653396E+00	9.332635E-01
4.200E+00	6.331963E+00	1.912424E+00	9.817132E-01	4.000E+02	1.368536E+03	3.732804E+00	9.303309E-01
4.400E+00	6.717487E+00	1.942519E+00	9.811331E-01	5.000E+02	1.744939E+03	3.792806E+00	9.280731E-01
4.600E+00	7.108856E+00	1.970904E+00	9.805777E-01	6.000E+02	3.691542E+03	3.971276E+00	9.211448E-01
4.800E+00	7.505746E+00	1.997744E+00	9.800451E-01	7.000E+02	∞	1.414214E-01	0.000000E+00
5.000E+00	7.907860E+00	2.023178E+00	9.795338E-01	Slab of polytropic index $n = -0.9$			
5.500E+00	8.943430E+00	2.081391E+00	9.783391E-01	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00
6.000E+00	9.983174E+00	2.133130E+00	9.772488E-01	1.000E-01	9.949962E-01	-1.001505E-01	9.994985E-01
6.500E+00	1.106655E+01	2.179570E+00	9.762474E-01	2.000E-01	9.799395E-01	-2.012161E-01	9.979756E-01
7.000E+00	1.216701E+01	2.221607E+00	9.753223E-01	3.000E-01	9.546901E-01	-3.041749E-01	9.953739E-01
7.500E+00	1.328754E+01	2.259936E+00	9.744635E-01	4.000E-01	9.190044E-01	-4.101430E-01	9.915891E-01
8.000E+00	1.442642E+01	2.295106E+00	9.736624E-01	5.000E-01	8.725164E-01	-5.204771E-01	9.864552E-01
8.500E+00	1.558219E+01	2.327559E+00	9.729123E-01	6.000E-01	8.147056E-01	-6.369347E-01	9.797157E-01
9.000E+00	1.675359E+01	2.357651E+00	9.722074E-01	7.000E-01	7.448455E-01	-7.619474E-01	9.709718E-01
9.500E+00	1.793950E+01	2.385678E+00	9.715427E-01	8.000E-01	6.619126E-01	-8.991383E-01	9.595775E-01
1.000E+01	1.913896E+01	2.411882E+00	9.709141E-01	9.000E-01	5.641196E-01	-1.054427E+00	9.444092E-01
1.100E+01	2.05646E+01	2.459614E+00	9.697515E-01	1.000E+00	4.500706E-01	-1.238801E+00	9.232686E-01
1.200E+01	2.240564E+01	2.502143E+00	9.686964E-01	1.100E+00	3.149007E-01	-1.477320E+00	8.908762E-01
1.300E+01	2.465780E+01	2.540423E+00	9.677313E-01	1.200E+00	1.501963E-01	-1.858464E+00	8.273055E-01
1.400E+01	2.731361E+01	2.575174E+00	9.668424E-01	1.220E+00	1.118305E-01	-1.983634E+00	8.032598E-01
1.500E+01	3.037242E+01	2.606954E+00	9.660190E-01	1.240E+00	7.053085E-02	-2.158355E+00	7.670751E-01
1.600E+01	3.434920E+01	2.636199E+00	9.652523E-01	1.260E+00	2.457023E-02	-2.468755E+00	6.903048E-01
1.700E+01	3.999909E+01	2.663261E+00	9.645352E-01	1.265E+00	1.170153E-02	-2.679738E+00	6.409502E-01
1.800E+01	3.967508E+01	2.688423E+00	9.638619E-01	1.267E+00	6.212595E-03	-2.822669E+00	6.016270E-01
1.900E+01	4.237538E+01	2.711919E+00	9.63275E-01	1.268E+00	3.332822E-03	-2.948549E+00	5.650299E-01
2.000E+01	4.509843E+01	2.733943E+00	9.626278E-01	1.269E+00	2.508040E-04	-3.357234E+00	4.364489E-01
2.500E+01	5.901107E+01	2.826908E+00	9.600430E-01	1.26907230E+00	0.000000E+00	-4.472136E+00	0.000000E+00
3.000E+01	7.333421E+01	2.895689E+00	9.579590E-01	Slab of polytropic index $n = -0.5$			
3.500E+01	8.798599E+01	2.959200E+00	9.562157E-01	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00
4.000E+01	1.029107E+02	3.009365E+00	9.547186E-01	1.000E-01	9.949979E-01	-1.000835E-01	9.974958E-01
4.500E+01	1.180682E+02	3.052614E+00	9.534077E-01	2.000E-01	9.799664E-01	-2.006734E-01	9.899325E-01
5.000E+01	1.334280E+02	3.090552E+00	9.522424E-01	3.000E+00	0.000000E+00	0.000000E+00	1.000000E+00
6.000E+01	1.646653E+02	3.154632E+00	9.502415E-01	1.000E-01	9.949979E-01	-1.000835E-01	9.974958E-01
7.000E+01	1.964831E+02	3.207359E+00	9.485642E-01	2.000E-01	9.799664E-01	-2.006734E-01	9.899325E-01
8.000E+01	2.287857E+02	3.252029E+00	9.471215E-01				

ξ	θ	θ'	θ^{n+1}	ξ	θ	θ'	θ^{n+1}
3.000E-01	9.548287E-01	-3.023022E-01	9.771533E-01	1.41421356E+00	0.000000E+00	-1.414214E+00	0.000000E+00
4.000E-01	9.194518E-01	-4.055388E-01	9.588806E-01	Slab of polytropic index $n = 0.5$			
5.000E-01	8.736400E-01	-5.111277E-01	9.346871E-01	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00
6.000E-01	8.171216E-01	-6.198459E-01	9.039478E-01	1.000E-01	9.950021E-01	-9.981662E-02	9.925125E-01
7.000E-01	7.495303E-01	-7.327914E-01	8.657542E-01	2.000E-01	9.800334E-01	-1.993320E-01	9.702001E-01
8.000E-01	6.703740E-01	-8.514371E-01	8.187637E-01	3.000E-01	9.551693E-01	-2.977397E-01	9.335133E-01
9.000E-01	5.798924E-01	-9.779388E-01	7.609089E-01	4.000E-01	9.205362E-01	-3.946228E-01	8.33046E-01
1.000E+00	4.744121E-01	-1.115749E+00	6.887758E-01	5.000E-01	8.763133E-01	-4.894474E-01	8.203309E-01
1.100E+00	3.552528E-01	-1.271171E+00	5.960300E-01	6.000E-01	8.227340E-01	-5.816548E-01	7.462582E-01
1.200E+00	2.191387E-01	-1.458599E+00	4.681226E-01	7.000E-01	7.600893E-01	-6.706520E-01	6.626694E-01
1.300E+00	6.083209E-02	-1.735924E+00	2.466416E-01	8.000E-01	6.887319E-01	-7.557974E-01	5.715777E-01
1.310E+00	4.325920E-02	-1.779901E+00	2.079883E-01	9.000E-01	6.090817E-01	-8.363812E-01	4.753499E-01
1.320E+00	2.519988E-02	-1.834399E+00	1.587447E-01	1.000E+00	5.216345E-01	-9.115944E-01	3.787467E-01
1.330E+00	6.484284E-03	-1.917785E+00	8.052505E-02	1.100E+00	4.269736E-01	-9.804772E-01	2.789983E-01
1.331E+00	4.559943E-03	-1.931292E+00	6.752735E-02	1.200E+00	3.257898E-01	-1.041823E+00	1.859543E-01
1.332E+00	2.620615E-03	-1.948136E+00	5.119194E-02	1.300E+00	2.189148E-01	-1.093967E+00	1.02466E-01
1.333E+00	6.609196E-04	-1.974124E+00	2.570836E-02	1.400E+00	1.636569E-01	-1.115822E+00	6.20661E-02
1.33333333E+00	0.000000E+00	-2.000000E+00	0.000000E+00	1.450E+00	1.073903E-01	-1.134200E+00	3.519232E-02
Slab of polytropic index $n = 0$				1.460E+00	5.030978E-02	-1.148167E+00	1.128440E-02
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	1.470E+00	3.881733E-02	-1.150277E+00	7.647835E-03
1.000E-01	9.950000E-01	-1.000000E-01	9.950000E-01	1.480E+00	2.730522E-02	-1.152093E+00	4.511995E-03
2.000E-01	9.800000E-01	-2.000000E-01	9.800000E-01	1.490E+00	1.577665E-02	-1.153556E+00	1.981628E-03
3.000E-01	9.550000E-01	-3.000000E-01	9.550000E-01	1.490E+00	4.235670E-03	-1.154541E+00	2.756660E-04
4.000E-01	9.200000E-01	-4.000000E-01	9.200000E-01	1.493686840E+00	0.000000E+00	-1.154701E+00	0.000000E+00
5.000E-01	8.750000E-01	-5.000000E-01	8.750000E-01	Slab of polytropic index $n = 1$			
6.000E-01	8.200000E-01	-6.000000E-01	8.200000E-01	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00
7.000E-01	7.550000E-01	-7.000000E-01	7.550000E-01	1.000E-01	9.950042E-01	-9.983342E-02	9.900333E-01
8.000E-01	6.800000E-01	-8.000000E-01	6.800000E-01	2.000E-01	9.800666E-01	-1.986693E-01	9.605305E-01
9.000E-01	5.950000E-01	-9.000000E-01	5.950000E-01	3.000E-01	9.553365E-01	-2.955202E-01	9.126678E-01
1.000E+00	5.000000E-01	-1.000000E+00	5.000000E-01	4.000E-01	9.210610E-01	-3.894183E-01	8.483534E-01
1.100E+00	3.950000E-01	-1.100000E+00	3.950000E-01	5.000E-01	8.775826E-01	-4.794255E-01	7.701512E-01
1.200E+00	2.800000E-01	-1.200000E+00	2.800000E-01	6.000E-01	8.253356E-01	-5.646425E-01	6.811789E-01
1.300E+00	1.550000E-01	-1.300000E+00	1.550000E-01	7.000E-01	7.648422E-01	-6.442177E-01	5.849836E-01
1.350E+00	8.875000E-02	-1.350000E+00	8.875000E-02				
1.400E+00	2.000000E-02	-1.400000E+00	2.000000E-02				
1.410E+00	5.950000E-03	-1.410000E+00	5.950000E-03				

ξ	θ	θ'	θ^{n+1}
1.640E+00	4.77698E-03	-8.94426E-01	1.577203E-06
1.64534085E+00	0.000000E+00	-8.944272E-01	0.000000E+00
Slab of polytropic index $n = 2$			
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00
1.000E+00	9.950083E-01	-9.966750E-02	9.850996E-01
2.000E-01	9.801324E-01	-1.973598E-01	9.415737E-01
3.000E-01	9.556650E-01	-2.911991E-01	8.728046E-01
4.000E-01	9.220777E-01	-3.794947E-01	7.839757E-01
5.000E-01	8.799988E-01	-4.608187E-01	6.814692E-01
6.000E-01	8.301837E-01	-5.340619E-01	5.721668E-01
7.000E-01	7.734814E-01	-5.984684E-01	4.627534E-01
8.000E-01	7.107981E-01	-6.536466E-01	3.591193E-01
9.000E-01	6.430613E-01	-6.995600E-01	2.659237E-01
1.000E+00	5.711853E-01	-7.365003E-01	1.863509E-01
1.100E+00	4.960413E-01	-7.650471E-01	1.220544E-01
1.200E+00	4.184288E-01	-7.860197E-01	7.325960E-02
1.300E+00	3.390564E-01	-8.004259E-01	3.897765E-02
1.400E+00	2.585244E-01	-8.094120E-01	1.727843E-02
1.500E+00	1.773134E-01	-8.142175E-01	5.574739E-03
1.600E+00	9.577705E-02	-8.161378E-01	8.785863E-04
1.660E+00	4.679718E-02	-8.164547E-01	1.024847E-04
1.680E+00	3.046774E-02	-8.164850E-01	2.82870E-05
1.700E+00	1.413791E-02	-8.164954E-01	2.825894E-06
1.710E+00	5.972952E-03	-8.164965E-01	2.130919E-07
1.71731534E+00	0.000000E+00	-8.164966E-01	0.000000E+00
Slab of polytropic index $n = 3$			
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00
1.000E-01	9.950125E-01	-9.950224E-02	9.801986E-01
2.000E-01	9.801970E-01	-1.960709E-01	9.231124E-01
3.000E-01	9.559859E-01	-2.870282E-01	8.352296E-01
4.000E-01	9.230532E-01	-3.701685E-01	7.259505E-01
5.000E-01	8.822664E-01	-4.439047E-01	6.058972E-01
6.000E-01	8.346167E-01	-5.073309E-01	4.852308E-01
7.000E-01	7.811526E-01	-5.602038E-01	3.723434E-01
Slab of polytropic index $n = 1.5$			
8.000E-01	6.967067E-01	-7.173561E-01	4.854002E-01
9.000E-01	6.216100E-01	-7.83269E-01	3.863990E-01
1.000E+00	5.403023E-01	-8.414710E-01	2.919266E-01
1.100E+00	4.535961E-01	-8.912074E-01	2.057494E-01
1.200E+00	3.623578E-01	-9.320391E-01	1.313031E-01
1.300E+00	2.674988E-01	-9.635582E-01	7.15562E-02
1.400E+00	1.699671E-01	-9.854497E-01	2.88883E-02
1.500E+00	7.073720E-02	-9.974950E-01	5.003752E-03
1.520E+00	5.077448E-02	-9.987101E-01	2.578048E-03
1.540E+00	3.079146E-02	-9.995258E-01	9.481140E-04
1.560E+00	1.079612E-02	-9.999417E-01	1.165561E-04
1.570E+00	7.983267E-04	-9.99997E-01	6.341362E-07
1.57079633E+00	0.000000E+00	-1.000000E+00	0.000000E+00
Slab of polytropic index $n = 1.5$			
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00
1.000E-01	9.950082E-01	-9.975037E-02	9.875623E-01
2.000E-01	9.800996E-01	-1.980120E-01	9.509891E-01
3.000E-01	9.555017E-01	-2.933404E-01	8.924393E-01
4.000E-01	9.215747E-01	-3.843782E-01	8.153167E-01
5.000E-01	8.788103E-01	-4.698946E-01	7.239988E-01
6.000E-01	8.278157E-01	-5.488193E-01	6.234967E-01
7.000E-01	7.692958E-01	-6.202716E-01	5.190789E-01
8.000E-01	7.040330E-01	-6.835824E-01	4.158939E-01
9.000E-01	6.328657E-01	-7.363093E-01	3.186243E-01
1.000E+00	5.56644E-01	-7.842460E-01	2.311978E-01
1.100E+00	4.763086E-01	-8.214258E-01	1.565745E-01
1.200E+00	3.926623E-01	-8.501220E-01	9.661578E-02
1.300E+00	3.065502E-01	-8.708479E-01	5.203002E-02
1.400E+00	2.187334E-01	-8.843636E-01	2.237627E-02
1.500E+00	1.298837E-01	-8.917041E-01	6.079762E-03
1.550E+00	8.524958E-02	-8.93477E-01	2.12128E-03
1.600E+00	4.055217E-02	-8.942791E-01	3.311560E-04
1.610E+00	3.160901E-02	-8.943477E-01	1.776345E-04
1.620E+00	2.266529E-02	-8.943926E-01	7.733985E-05
1.630E+00	1.372123E-02	-8.944173E-01	2.205375E-05

ξ	θ	θ'	θ^{n+1}
8.000E-01	7.229170E-01	-6.023859E-01	2.731202E-01
9.000E-01	6.608932E-01	-6.360900E-01	1.907790E-01
1.000E+00	5.959766E-01	-6.609988E-01	1.261586E-01
1.100E+00	5.289300E-01	-6.788706E-01	7.826952E-02
1.200E+00	4.603925E-01	-6.910400E-01	4.492756E-02
1.300E+00	3.908686E-01	-6.988057E-01	2.334121E-02
1.400E+00	3.207383E-01	-7.033552E-01	1.058287E-02
1.500E+00	2.502703E-01	-7.057184E-01	3.923169E-03
1.600E+00	1.796392E-01	-7.067385E-01	1.041369E-03
1.700E+00	1.089457E-01	-7.070570E-01	1.408772E-04
1.800E+00	3.823656E-02	-7.071060E-01	2.137546E-06
1.820E+00	2.409443E-02	-7.071067E-01	3.370288E-07
1.840E+00	9.952300E-03	-7.071068E-01	9.810560E-09
1.850E+00	2.881232E-03	-7.071068E-01	6.891486E-11
1.85407468E+00	0.000000E+00	-7.071068E-01	0.000000E+00
Slab of polytropic index $n = 4$			
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00
1.000E-01	9.950166E-01	-9.933764E-02	9.753301E-01
2.000E-01	9.802621E-01	-1.948021E-01	9.051303E-01
3.000E-01	9.562994E-01	-2.829998E-01	7.997778E-01
4.000E-01	9.239904E-01	-3.613865E-01	6.734994E-01
5.000E-01	8.844004E-01	-4.284581E-01	5.410590E-01
6.000E-01	8.369624E-01	-4.837491E-01	4.149671E-01
7.000E-01	7.860300E-01	-5.276786E-01	3.038882E-01
8.000E-01	7.335004E-01	-5.613109E-01	2.132533E-01
9.000E-01	6.766630E-01	-5.860946E-01	1.412320E-01
1.000E+00	6.165230E-01	-6.036313E-01	8.907306E-02
1.100E+00	5.552556E-01	-6.154972E-01	5.290786E-02
1.200E+00	4.935646E-01	-6.231244E-01	2.929004E-02
1.300E+00	4.310010E-01	-6.277347E-01	1.487275E-02
1.400E+00	3.680850E-01	-6.303152E-01	6.756797E-03
1.500E+00	3.049802E-01	-6.316206E-01	2.638507E-03
1.600E+00	2.417851E-01	-6.321942E-01	8.263192E-04
1.700E+00	1.785535E-01	-6.323981E-01	1.814856E-04
1.800E+00	1.153105E-01	-6.324491E-01	2.038655E-05
1.900E+00	5.206511E-02	-6.324554E-01	3.825904E-07
Slab of polytropic index $n = 5$			
1.940E+00	2.67689E-02	-6.324555E-01	1.374010E-08
1.960E+00	1.411778E-02	-6.324555E-01	5.608315E-10
1.970E+00	7.793227E-03	-6.324555E-01	2.874660E-11
1.980E+00	1.468671E-03	-6.324555E-01	6.833185E-15
1.98232217E+00	0.000000E+00	-6.324555E-01	0.000000E+00
Slab of polytropic index $n = 5$			
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00
1.000E-01	9.950207E-01	-9.917369E-02	9.704937E-01
2.000E-01	9.803260E-01	-1.935530E-01	8.876117E-01
3.000E-01	9.566058E-01	-2.791065E-01	7.662986E-01
4.000E-01	9.248916E-01	-3.531016E-01	6.259578E-01
5.000E-01	8.864138E-01	-4.142920E-01	4.85084E-01
6.000E-01	8.424578E-01	-4.627776E-01	3.575107E-01
7.000E-01	7.942459E-01	-4.996557E-01	2.510324E-01
8.000E-01	7.428577E-01	-5.266092E-01	1.680482E-01
9.000E-01	6.891914E-01	-5.455390E-01	1.071616E-01
1.000E+00	6.339554E-01	-5.582962E-01	6.491610E-02
1.100E+00	5.778830E-01	-5.665200E-01	3.716527E-02
1.200E+00	5.207571E-01	-5.715640E-01	1.994395E-02
1.300E+00	4.634407E-01	-5.744831E-01	9.907520E-03
1.400E+00	4.059050E-01	-5.760577E-01	4.472462E-03
1.500E+00	3.482555E-01	-5.768351E-01	1.783972E-03
1.600E+00	2.905524E-01	-5.771766E-01	6.016540E-04
1.700E+00	2.328272E-01	-5.773043E-01	1.592951E-04
1.800E+00	1.750945E-01	-5.773420E-01	2.881607E-05
1.900E+00	1.173598E-01	-5.773495E-01	2.612859E-06
2.000E+00	5.962479E-02	-5.773503E-01	4.493254E-08
2.060E+00	2.498377E-02	-5.773503E-01	2.431912E-10
2.080E+00	1.343676E-02	-5.773503E-01	5.885294E-12
2.090E+00	7.663261E-03	-5.773503E-01	2.025265E-13
2.100E+00	1.889759E-03	-5.773503E-01	4.554472E-17
2.10327316E+00	0.000000E+00	-5.773503E-01	0.000000E+00
Slab of polytropic index $n = 6$			
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00

ξ	θ	θ'	θ^{n+1}	ξ	θ	θ'	θ^{n+1}
1.000E-01	9.950248E-01	-9.901040E-02	9.656893E-01	1.000E+00	6.938432E-01	-4.225588E-01	1.794215E-02
2.000E-01	9.803891E-01	-1.923231E-01	8.705413E-01	1.100E+00	6.514810E-01	-4.244842E-01	8.972621E-03
3.000E-01	9.569054E-01	-2.753416E-01	7.346545E-01	1.200E+00	6.089767E-01	-4.254897E-01	4.271767E-03
4.000E-01	9.257593E-01	-3.452719E-01	5.827557E-01	1.300E+00	5.663997E-01	-4.259909E-01	1.924656E-03
5.000E-01	8.883179E-01	-4.012500E-01	4.364947E-01	1.400E+00	5.237872E-01	-4.262278E-01	8.141490E-04
6.000E-01	8.459515E-01	-4.439947E-01	3.100404E-01	1.500E+00	4.811584E-01	-4.263332E-01	3.200141E-04
7.000E-01	7.999030E-01	-4.752331E-01	2.095372E-01	1.600E+00	4.385226E-01	-4.263768E-01	1.153219E-04
8.000E-01	7.512162E-01	-4.971328E-01	1.350065E-01	1.700E+00	3.958840E-01	-4.263935E-01	3.743233E-05
9.000E-01	7.007147E-01	-5.118748E-01	8.294471E-02	1.800E+00	3.532443E-01	-4.263992E-01	1.068628E-05
1.000E+00	6.490148E-01	-5.213980E-01	4.850451E-02	1.900E+00	3.106042E-01	-4.264009E-01	2.595860E-06
1.100E+00	5.965558E-01	-5.272874E-01	2.688795E-02	2.000E+00	2.679641E-01	-4.264013E-01	5.114973E-07
1.200E+00	5.436375E-01	-5.307586E-01	1.403350E-02	2.100E+00	2.253240E-01	-4.264014E-01	7.601195E-08
1.300E+00	4.904550E-01	-5.326949E-01	6.826436E-03	2.200E+00	1.826839E-01	-4.264014E-01	7.563133E-09
1.400E+00	4.371291E-01	-5.337068E-01	3.049778E-03	2.300E+00	1.400437E-01	-4.264014E-01	4.063494E-10
1.500E+00	3.837309E-01	-5.341949E-01	1.225145E-03	2.400E+00	9.740357E-02	-4.264014E-01	7.487268E-12
1.600E+00	3.302992E-01	-5.344078E-01	4.288967E-04	2.500E+00	5.476342E-02	-4.264014E-01	1.328606E-14
1.700E+00	2.768537E-01	-5.344892E-01	1.246671E-04	2.550E+00	3.344335E-02	-4.264014E-01	5.853392E-17
1.800E+00	2.234032E-01	-5.345151E-01	2.777316E-05	2.600E+00	1.212328E-02	-4.264014E-01	8.314211E-22
1.900E+00	1.699513E-01	-5.345214E-01	4.095160E-06	2.610E+00	7.859265E-03	-4.264014E-01	7.066445E-24
2.000E+00	1.164991E-01	-5.345224E-01	2.912440E-07	2.620E+00	3.595250E-03	-4.264014E-01	1.297240E-27
2.100E+00	6.304681E-02	-5.345225E-01	3.959515E-09	2.62843161E+00	0.000000E+00	-4.264014E-01	0.000000E+00
2.150E+00	3.632069E-02	-5.345225E-01	8.338322E-11				
2.200E+00	9.594566E-03	-5.345225E-01	7.484752E-15				
2.210E+00	4.249341E-03	-5.345225E-01	2.501793E-17				
2.21794979E+00	0.000000E+00	-5.345225E-01	0.000000E+00				
Slab of polytropic index $n = 10$							
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00
1.000E-01	9.950412E-01	-9.836362E-02	9.467853E-01	1.000E-01	9.850813E-01	-9.679027E-02	9.016323E-01
2.000E-01	9.806355E-01	-1.875869E-01	8.084613E-01	2.000E-01	9.812105E-01	-1.768945E-01	6.714374E-01
3.000E-01	9.580408E-01	-2.614447E-01	6.240567E-01	3.000E-01	9.650575E-01	-2.331854E-01	4.290580E-01
4.000E-01	9.289332E-01	-3.176174E-01	4.444567E-01	4.000E-01	9.352906E-01	-2.680793E-01	2.454017E-01
5.000E-01	8.950201E-01	-3.579665E-01	2.952300E-01	5.000E-01	9.073945E-01	-2.878603E-01	1.299329E-01
6.000E-01	8.577792E-01	-3.849481E-01	1.849825E-01	6.000E-01	8.780247E-01	-2.983911E-01	6.510901E-02
7.000E-01	8.183539E-01	-4.022113E-01	1.102494E-01	7.000E-01	8.478871E-01	-3.037436E-01	3.126786E-02
8.000E-01	7.775589E-01	-4.127920E-01	6.281540E-02	8.000E-01	8.173658E-01	-3.063644E-01	1.447924E-02
9.000E-01	7.359396E-01	-4.190254E-01	3.429762E-02	9.000E-01	7.866595E-01	-3.076053E-01	6.479216E-03
				1.000E+00	7.558668E-01	-3.081742E-01	2.801240E-03
				1.100E+00	7.250350E-01	-3.084264E-01	1.168257E-03
				1.200E+00	6.941862E-01	-3.085344E-01	4.688086E-04
				1.300E+00	6.633302E-01	-3.085789E-01	1.804369E-04

Slab of polytropic index $n = 20$

ξ	θ	θ'	ρ^{n+1}	ξ	θ	θ'	$\exp(-\theta)$
1.400E+00	6.324713E-01	-3.085965E-01	6.635252E-05	7.000E-01	2.356044E-01	6.479303E-01	7.900932E-01
1.500E+00	6.016113E-01	-3.086031E-01	2.320787E-05	8.000E-01	3.042615E-01	7.243370E-01	7.376860E-01
1.600E+00	5.707508E-01	-3.086055E-01	7.680287E-06	9.000E-01	3.802939E-01	7.954113E-01	6.836605E-01
1.700E+00	5.398902E-01	-3.086063E-01	2.390092E-06	1.000E+00	4.631626E-01	8.610572E-01	6.292903E-01
1.800E+00	5.090296E-01	-3.086066E-01	6.943898E-07	1.200E+00	6.472445E-01	9.762313E-01	5.234863E-01
1.900E+00	4.781689E-01	-3.086067E-01	1.867274E-07	1.400E+00	8.522955E-01	1.071042E+00	4.264349E-01
2.000E+00	4.473082E-01	-3.086067E-01	4.599863E-08	1.600E+00	1.074445E+00	1.147617E+00	3.414874E-01
2.100E+00	4.164476E-01	-3.086067E-01	1.025102E-08	1.800E+00	1.310297E+00	1.208520E+00	2.697398E-01
2.200E+00	3.855869E-01	-3.086067E-01	2.035016E-09	2.000E+00	1.556983E+00	1.256367E+00	2.107711E-01
2.300E+00	3.547262E-01	-3.086067E-01	3.529876E-10	2.200E+00	1.812137E+00	1.293596E+00	1.633048E-01
2.400E+00	3.238656E-01	-3.086067E-01	5.219876E-11	2.400E+00	2.073856E+00	1.322346E+00	1.257001E-01
2.500E+00	2.930049E-01	-3.086067E-01	6.373416E-12	2.600E+00	2.340631E+00	1.344420E+00	9.626686E-02
2.600E+00	2.621442E-01	-3.086067E-01	6.156673E-13	2.800E+00	2.611279E+00	1.361293E+00	7.344059E-02
2.700E+00	2.312836E-01	-3.086067E-01	4.436492E-14	3.000E+00	2.884881E+00	1.374146E+00	5.586144E-02
2.800E+00	2.004229E-01	-3.086067E-01	2.192267E-15	3.200E+00	3.160732E+00	1.383911E+00	4.239472E-02
2.900E+00	1.695622E-01	-3.086067E-01	6.545013E-17	3.400E+00	3.438289E+00	1.391316E+00	3.211961E-02
3.000E+00	1.387015E-01	-3.086067E-01	9.631791E-19	3.600E+00	3.717138E+00	1.396923E+00	2.430341E-02
3.100E+00	1.078409E-01	-3.086067E-01	4.880352E-21	3.800E+00	3.996967E+00	1.401163E+00	1.837128E-02
3.200E+00	7.698020E-02	-3.086067E-01	4.111118E-24	4.000E+00	4.277535E+00	1.404367E+00	1.387683E-02
3.300E+00	4.611953E-02	-3.086067E-01	8.741434E-29	4.200E+00	4.558661E+00	1.406786E+00	1.047607E-02
3.400E+00	1.525886E-02	-3.086067E-01	7.144386E-39	4.400E+00	4.840210E+00	1.408613E+00	7.905395E-03
3.420E+00	9.086728E-03	-3.086067E-01	1.338314E-43	4.600E+00	5.122077E+00	1.409990E+00	5.963626E-03
3.440E+00	2.914594E-03	-3.086067E-01	5.703465E-54	4.800E+00	5.404183E+00	1.411030E+00	4.497726E-03
3.4494436E+00	0.000000E+00	-3.086067E-01	0.000000E+00	5.000E+00	5.686471E+00	1.411813E+00	3.391539E-03
				5.500E+00	6.392718E+00	1.413030E+00	1.673702E-03
				6.000E+00	7.099400E+00	1.413630E+00	8.256002E-04
				6.500E+00	7.806297E+00	1.413926E+00	4.071628E-04
				7.000E+00	8.513301E+00	1.414072E+00	2.007800E-04
				7.500E+00	9.220357E+00	1.414144E+00	9.900335E-05
				8.000E+00	9.927439E+00	1.414179E+00	4.881668E-05
				8.500E+00	1.063453E+01	1.414197E+00	2.407027E-05
				9.000E+00	1.134163E+01	1.414205E+00	1.186837E-05
				9.500E+00	1.204874E+01	1.414209E+00	5.851940E-06
				1.000E+01	1.275584E+01	1.414212E+00	2.885412E-06
				1.100E+01	1.417006E+01	1.414213E+00	7.014928E-07
				1.200E+01	1.558427E+01	1.414213E+00	1.705447E-07
				1.300E+01	1.699848E+01	1.414214E+00	4.146227E-08
				1.400E+01	1.841270E+01	1.414214E+00	1.008017E-08

ξ	θ	θ'	$\exp(-\theta)$
0.000E+00	0.000000E+00	0.000000E+00	0.000000E+00
1.000E-01	4.995839E-03	9.983367E-02	9.950166E-01
2.000E-01	1.993369E-02	1.986772E-01	9.802637E-01
3.000E-01	4.466505E-02	2.955796E-01	9.563164E-01
4.000E-01	7.895555E-02	3.896640E-01	9.240810E-01
5.000E-01	1.224795E-01	4.801582E-01	8.847241E-01
6.000E-01	1.748458E-01	5.664160E-01	8.395865E-01

Slab of polytropic index $n = \infty$

ξ	θ	θ'	$\exp(-\theta)$	ξ	θ	θ'	$\xi\theta'$	θ^{n+1}	$\xi\theta'$
1.500E+01	1.982691E-01	1.414214E+00	2.450658E-09	8.000E-01	1.118290E+00	2.261145E-01	3.656020E-01	3.656020E-01	1.808916E-01
1.600E+01	2.124112E-01	1.414214E+00	5.957961E-10	9.000E-01	1.141067E+00	2.288947E-01	2.288947E-01	2.060052E-01	2.060052E-01
1.700E+01	2.265534E-01	1.414214E+00	1.448480E-10	1.000E-01	1.163980E+00	2.289803E-01	2.549712E-01	2.389803E-01	2.389803E-01
1.800E+01	2.406955E-01	1.414214E+00	3.521497E-11	1.200E-01	1.209361E+00	2.239316E-01	1.807157E-01	1.807157E-01	2.687179E-01
1.900E+01	2.548376E-01	1.414214E+00	8.561349E-12	1.400E-01	1.253306E+00	2.151214E-01	1.310650E-01	1.310650E-01	3.011700E-01
2.000E+01	2.689798E-01	1.414214E+00	2.081407E-12	1.600E-01	1.295314E+00	2.048362E-01	9.741488E-02	3.277379E-01	3.277379E-01
2.500E+01	3.396904E-01	1.414214E+00	1.767793E-15	1.800E-01	1.352223E+00	1.942671E-01	7.413394E-02	3.496808E-01	3.496808E-01
3.000E+01	4.104011E-01	1.414214E+00	1.501432E-18	2.000E-01	1.373042E+00	1.840043E-01	5.765627E-02	3.880087E-01	3.880087E-01
3.500E+01	4.811118E-01	1.414214E+00	1.275205E-21	2.200E-01	1.408663E+00	1.743176E-01	4.572787E-02	3.834988E-01	3.834988E-01
4.000E+01	5.518225E-01	1.414214E+00	1.083064E-24	2.400E-01	1.442814E+00	1.653089E-01	3.690682E-02	3.967415E-01	3.967415E-01
4.500E+01	6.25332E-01	1.414214E+00	9.198741E-28	2.600E-01	1.475033E+00	1.569943E-01	3.025411E-02	4.081851E-01	4.081851E-01
5.000E+01	6.932438E-01	1.414214E+00	7.812727E-31	2.800E-01	1.505656E+00	1.493473E-01	2.514590E-02	4.181723E-01	4.181723E-01
6.000E+01	8.346652E-01	1.414214E+00	5.635743E-37	3.000E-01	1.534813E+00	1.423225E-01	2.115923E-02	4.269674E-01	4.269674E-01
7.000E+01	9.760866E-01	1.414214E+00	4.065367E-43	3.200E-01	1.562623E+00	1.358676E-01	1.800161E-02	4.347763E-01	4.347763E-01
8.000E+01	1.117508E-02	1.414214E+00	2.932569E-49	3.400E-01	1.589195E+00	1.299297E-01	1.546698E-02	4.417611E-01	4.417611E-01
9.000E+01	1.258929E-02	1.414214E+00	2.115421E-55	3.600E-01	1.614626E+00	1.244586E-01	1.340762E-02	4.480310E-01	4.480310E-01
1.000E+02	1.400351E-02	1.414214E+00	1.525968E-61	3.800E-01	1.639006E+00	1.194078E-01	1.171586E-02	4.537497E-01	4.537497E-01
2.000E+02	2.814564E-02	1.414214E+00	0.000000E+00	4.000E-01	1.662415E+00	1.147354E-01	1.031207E-02	4.589414E-01	4.589414E-01
3.000E+02	4.228778E-02	1.414214E+00	0.000000E+00	4.200E-01	1.684923E+00	1.104037E-01	9.136483E-03	4.636954E-01	4.636954E-01
4.000E+02	5.642991E-02	1.414214E+00	0.000000E+00	4.400E-01	1.706597E+00	1.063792E-01	8.143706E-03	4.686885E-01	4.686885E-01
5.000E+02	7.057205E-02	1.414214E+00	0.000000E+00	4.600E-01	1.727493E+00	1.026323E-01	7.298818E-03	4.721084E-01	4.721084E-01
1.000E+03	1.412827E-03	1.414214E+00	0.000000E+00	4.800E-01	1.747666E+00	9.913643E-02	6.574863E-03	4.758549E-01	4.758549E-01
∞	∞	1.414214E+00	0.000000E+00	5.000E-01	1.767163E+00	9.586834E-02	5.949908E-03	4.793417E-01	4.793417E-01
				5.500E-01	1.813223E+00	8.856382E-02	4.719965E-03	4.871010E-01	4.871010E-01
				6.000E-01	1.855899E+00	8.229240E-02	3.828349E-03	4.937544E-01	4.937544E-01
				6.500E-01	1.895654E+00	7.685322E-02	3.163504E-03	4.995460E-01	4.995460E-01
				7.000E-01	1.932865E+00	7.209308E-02	2.655745E-03	5.046516E-01	5.046516E-01
				7.500E-01	1.967840E+00	6.789350E-02	2.259911E-03	5.092012E-01	5.092012E-01
				8.000E-01	2.000836E+00	6.416165E-02	1.945790E-03	5.132932E-01	5.132932E-01
				8.500E-01	2.032068E+00	6.082392E-02	1.692809E-03	5.170033E-01	5.170033E-01
				9.000E-01	2.061716E+00	5.782120E-02	1.485732E-03	5.203908E-01	5.203908E-01
				9.500E-01	2.089337E+00	5.510558E-02	1.314625E-03	5.235030E-01	5.235030E-01
				1.000E-01	2.116863E+00	5.263780E-02	1.171563E-03	5.263780E-01	5.263780E-01
				1.100E-01	2.157280E+00	4.832138E-02	9.479064E-04	5.315352E-01	5.315352E-01
				1.200E-01	2.213727E+00	4.467104E-02	7.832194E-04	5.360525E-01	5.360525E-01
				1.300E-01	2.256796E+00	4.154338E-02	6.585217E-04	5.400640E-01	5.400640E-01
				1.400E-01	2.296953E+00	3.883335E-02	5.618644E-04	5.436669E-01	5.436669E-01
				1.500E-01	2.334575E+00	3.646227E-02	4.854332E-04	5.469341E-01	5.469341E-01

ξ	θ	θ'	$\xi\theta'$	θ^{n+1}
0.000E+00	1.000000E-00	0.000000E+00	0.000000E+00	1.000000E+00
1.000E-01	1.002485E-00	4.938322E-02	9.779147E-01	4.938322E-03
2.000E-01	1.009759E-00	9.525284E-02	9.163095E-01	1.905057E-02
3.000E-01	1.021327E-00	1.349283E-01	8.270204E-01	4.047848E-02
4.000E-01	1.036489E-00	1.670039E-01	7.242958E-01	6.680156E-02
5.000E-01	1.054468E-00	1.912979E-01	6.204118E-01	9.564896E-02
6.000E-01	1.074512E+00	2.084884E-01	5.237202E-01	1.250930E-01
7.000E-01	1.095965E+00	2.196881E-01	4.383573E-01	1.537817E-01

Cylinder of polytropic index $n = -10$

ξ	θ	θ'	θ^{n+1}	$\xi\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi\theta'$
1.600E+01	2.36970E+00	3.437007E-02	4.239496E-04	5.499211E-01	1.600E+00	1.393086E+00	3.314237E-01	2.655146E-01	5.302779E-01
1.700E+01	2.403392E+00	3.251007E-02	3.737471E-04	5.526712E-01	1.800E+00	1.458937E+00	3.266352E-01	2.207261E-01	5.879433E-01
1.800E+01	2.435055E+00	3.084548E-02	3.322161E-04	5.552186E-01	2.000E+00	1.523600E+00	3.197349E-01	1.855734E-01	6.394696E-01
1.900E+01	2.465139E+00	2.934689E-02	2.974595E-04	5.575909E-01	2.200E+00	1.586753E+00	3.116544E-01	1.577476E-01	6.856396E-01
2.000E+01	2.493796E+00	2.799053E-02	2.680718E-04	5.598107E-01	2.400E+00	1.648224E+00	3.029946E-01	1.354987E-01	7.271870E-01
2.500E+01	2.619767E+00	2.276693E-02	1.720425E-04	5.691732E-01	2.600E+00	1.707938E+00	2.941388E-01	1.175199E-01	7.647609E-01
3.000E+01	2.724215E+00	1.921842E-02	1.210116E-04	5.765525E-01	2.800E+00	1.765883E+00	2.853278E-01	1.028378E-01	7.989180E-01
3.500E+01	2.813564E+00	1.664764E-02	9.050844E-05	5.826675E-01	3.000E+00	1.822082E+00	2.767096E-01	9.072523E-02	8.301288E-01
4.000E+01	2.891719E+00	1.469773E-02	7.072871E-05	5.879091E-01	3.200E+00	1.876385E+00	2.683716E-01	8.063561E-02	8.587890E-01
4.500E+01	2.961235E+00	1.316694E-02	5.711412E-05	5.925125E-01	3.400E+00	1.929453E+00	2.603620E-01	7.215449E-02	8.852307E-01
5.000E+01	3.023880E+00	1.193259E-02	4.730634E-05	5.966293E-01	3.600E+00	1.980753E+00	2.537037E-01	6.496483E-02	9.097332E-01
6.000E+01	3.133310E+00	1.006313E-02	3.435343E-05	6.037878E-01	3.800E+00	2.030588E+00	2.454031E-01	5.882179E-02	9.325319E-01
7.000E+01	3.226855E+00	8.713034E-03	2.636240E-05	6.099124E-01	4.000E+00	2.078938E+00	2.384565E-01	5.353448E-02	9.538261E-01
8.000E+01	3.308656E+00	7.691262E-03	2.104434E-05	6.153010E-01	4.200E+00	2.125964E+00	2.318537E-01	4.895270E-02	9.737854E-01
9.000E+01	3.381412E+00	6.890420E-03	1.730282E-05	6.201378E-01	4.400E+00	2.171702E+00	2.255806E-01	4.495720E-02	9.925546E-01
1.000E+02	3.446982E+00	6.245449E-03	1.455624E-05	6.245449E-01	4.600E+00	2.216217E+00	2.196214E-01	4.145252E-02	1.010258E+00
2.000E+02	3.890570E+00	3.281578E-03	4.896444E-06	6.363156E-01	4.800E+00	2.259570E+00	2.139591E-01	3.836160E-02	1.027004E+00
3.000E+02	4.161011E+00	2.260570E-03	2.674299E-06	6.781709E-01	5.000E+00	2.301819E+00	2.085767E-01	3.562175E-02	1.042884E+00
4.000E+02	4.358592E+00	1.739400E-03	1.761506E-06	6.957600E-01	5.500E+00	2.402959E+00	1.962375E-01	2.999261E-02	1.079306E+00
5.000E+02	4.515506E+00	1.421744E-03	1.281274E-06	7.108722E-01	6.000E+00	2.498291E+00	1.853032E-01	2.567013E-02	1.111819E+00
1.000E+03	5.027160E+00	7.684533E-04	4.876358E-07	7.684533E-01	6.500E+00	2.588462E+00	1.755647E-01	2.227578E-02	1.141170E+00
∞	∞	∞	0.000000E+00	∞	7.000E+00	2.674025E+00	1.668456E-01	1.955862E-02	1.167919E+00
Cylinder of polytropic index $n = -5$									
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	0.000E+00	2.978703E+00	1.395789E-01	1.270256E-02	1.256210E+00
1.000E-01	1.002492E+00	4.968970E-02	9.900929E-02	4.968970E-03	0.500E+00	3.047126E+00	1.341931E-01	1.159947E-02	1.274834E+00
2.000E-01	1.009877E+00	9.756879E-02	9.614747E-01	1.951376E-01	1.000E-01	3.112968E+00	1.292417E-01	1.064882E-02	1.292417E+00
3.000E-01	1.021893E+00	1.420667E-01	9.170196E-01	4.262001E-02	1.00E+01	3.776999E+00	1.204747E-01	9.100266E-03	1.324919E+00
4.000E-01	1.038139E+00	1.820262E-01	8.609507E-01	7.281048E-02	1.200E-01	3.354268E+00	1.128719E-01	7.899670E-03	1.354463E+00
5.000E-01	1.058124E+00	2.167741E-01	7.977266E-01	1.083871E-01	1.300E-01	3.463769E+00	1.062767E-01	6.947113E-03	1.381597E+00
6.000E-01	1.081312E+00	2.460940E-01	7.314682E-01	1.476564E-01	1.400E-01	3.567087E+00	1.004806E-01	6.176537E-03	1.406728E+00
7.000E-01	1.107166E+00	2.701343E-01	6.655007E-01	1.890940E-01	1.500E-01	3.664950E+00	9.534783E-02	5.542783E-03	1.430172E+00
8.000E-01	1.135176E+00	2.892919E-01	6.022084E-01	2.314335E-01	1.600E-01	3.757961E+00	9.076077E-02	5.014079E-03	1.452172E+00
9.000E-01	1.164880E+00	3.041031E-01	5.430947E-01	2.736928E-01	1.700E-01	3.846627E+00	8.664272E-02	4.567514E-03	1.472926E+00
1.000E+00	1.195872E+00	3.151563E-01	4.889470E-01	3.151563E-01	1.800E-01	3.931378E+00	8.292177E-02	4.186207E-03	1.492592E+00
1.200E+00	1.260390E+00	3.282706E-01	3.962396E-01	3.939247E-01	1.900E-01	4.025584E+00	7.954205E-02	3.857479E-03	1.511299E+00
1.400E+00	1.326602E+00	3.326773E-01	3.228775E-01	4.657482E-01	2.000E-01	4.090561E+00	7.645779E-02	3.571648E-03	1.529156E+00

ξ	θ	θ'	θ^{n+1}	$\xi\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi\theta'$
2.500E+01	4.440572E+00	6.433663E-02	2.571842E-03	1.608416E+00	2.600E+00	1.791970E+00	3.531210E-01	1.737832E-01	9.181145E-01
3.000E+01	4.739911E+00	5.585513E-02	1.981158E-03	1.675654E+00	2.800E+00	1.861807E+00	3.452114E-01	1.549515E-01	9.665919E-01
3.500E+01	5.002740E+00	4.956417E-02	1.596498E-03	1.734746E+00	3.000E+00	1.930045E+00	3.371599E-01	1.390905E-01	1.011480E+00
4.000E+01	5.237911E+00	4.469797E-02	1.328518E-03	1.787919E+00	3.200E+00	1.996673E+00	3.291258E-01	1.256260E-01	1.053203E+00
4.500E+01	5.451346E+00	4.081275E-02	1.132362E-03	1.836574E+00	3.400E+00	2.061704E+00	3.212141E-01	1.141093E-01	1.092128E+00
5.000E+01	5.647209E+00	3.763304E-02	9.832512E-04	1.881652E+00	3.600E+00	2.125171E+00	3.134923E-01	1.041882E-01	1.128572E+00
6.000E+01	5.997662E+00	3.272592E-02	7.728087E-04	1.963555E+00	3.800E+00	2.187116E+00	3.060025E-01	9.558381E-02	1.162809E+00
7.000E+01	6.305963E+00	2.910203E-02	6.324037E-04	2.037142E+00	4.000E+00	2.247589E+00	2.987685E-01	8.807431E-02	1.195074E+00
8.000E+01	6.582443E+00	2.630626E-02	5.326618E-04	2.104501E+00	4.200E+00	2.306642E+00	2.918024E-01	8.148161E-02	1.225570E+00
9.000E+01	6.833968E+00	2.407758E-02	4.584672E-04	2.166983E+00	4.400E+00	2.364328E+00	2.851078E-01	7.566181E-02	1.254474E+00
1.000E+02	7.065343E+00	2.225516E-02	4.012980E-04	2.225516E+00	4.600E+00	2.420703E+00	2.786827E-01	7.049781E-02	1.281940E+00
2.000E+02	8.759368E+00	1.343186E-02	1.698670E-04	2.686373E+00	4.800E+00	2.475819E+00	2.725213E-01	6.589363E-02	1.308102E+00
3.000E+02	9.916601E+00	1.010933E-02	1.034067E-04	3.032799E+00	5.000E+00	2.529728E+00	2.666155E-01	6.177011E-02	1.333078E+00
4.000E+02	1.082973E+01	8.304700E-03	7.269920E-05	3.321880E+00	5.500E+00	2.659548E+00	2.529045E-01	5.315898E-02	1.390975E+00
5.000E+02	1.159867E+01	7.148806E-03	5.525436E-05	3.574403E+00	6.000E+00	2.782861E+00	2.405646E-01	4.640078E-02	1.443387E+00
1.000E+03	1.439466E+01	4.542467E-03	2.329133E-05	4.542467E+00	6.500E+00	2.900313E+00	2.294310E-01	4.098881E-02	1.491301E+00
∞	∞	0.000000E+00	0.000000E+00	∞	7.000E+00	3.012468E+00	2.193526E-01	3.657907E-02	1.535468E+00
					7.500E+00	3.119819E+00	2.101961E-01	3.293151E-02	1.576471E+00
					8.000E+00	3.222799E+00	2.018465E-01	2.987449E-02	1.614772E+00
					8.500E+00	3.321784E+00	1.942046E-01	2.728261E-02	1.650739E+00
					9.000E+00	3.417107E+00	1.871860E-01	2.506249E-02	1.684674E+00
					9.500E+00	3.509082E+00	1.807183E-01	2.314339E-02	1.716824E+00
					1.000E+01	3.597907E+00	1.747392E-01	2.147090E-02	1.747392E+00
					1.100E+01	3.767167E+00	1.640409E-01	1.870490E-02	1.804450E+00
					1.200E+01	3.926455E+00	1.547446E-01	1.651955E-02	1.856936E+00
					1.300E+01	4.077036E+00	1.465888E-01	1.475592E-02	1.905654E+00
					1.400E+01	4.219945E+00	1.393724E-01	1.330698E-02	1.951214E+00
					1.500E+01	4.356041E+00	1.329391E-01	1.209829E-02	1.994087E+00
					1.600E+01	4.486043E+00	1.271653E-01	1.107668E-02	2.034645E+00
					1.700E+01	4.610559E+00	1.219523E-01	1.020327E-02	2.073189E+00
					1.800E+01	4.730108E+00	1.172201E-01	9.449023E-03	2.109962E+00
					1.900E+01	4.845137E+00	1.129034E-01	8.791880E-03	2.145165E+00
					2.000E+01	4.956035E+00	1.089483E-01	8.214799E-03	2.178967E+00
					2.500E+01	5.459081E+00	9.325072E-02	6.146691E-03	2.331268E+00
					3.000E+01	5.896035E+00	8.210670E-02	4.878877E-03	2.463201E+00
					3.500E+01	6.284742E+00	7.37434E-02	4.028446E-03	2.581052E+00
					4.000E+01	6.636509E+00	6.721246E-02	3.421290E-03	2.688499E+00
					4.500E+01	6.958977E+00	6.195310E-02	2.967316E-03	2.787889E+00

Cylinder of polytropic index $n = -4$

0.000E+00	1.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
1.000E-01	1.002494E+00	4.975145E-02	9.925538E-01	4.975145E-03
2.000E-01	1.009902E+00	9.804534E-02	9.708741E-01	1.960911E-02
3.000E-01	1.020211E+00	1.435857E-01	9.367713E-01	4.307571E-02
4.000E-01	1.038493E+00	1.853591E-01	8.928729E-01	7.414365E-02
5.000E-01	1.058934E+00	2.226989E-01	8.421565E-01	1.113495E-01
6.000E-01	1.082874E+00	2.552839E-01	7.875291E-01	1.531703E-01
7.000E-01	1.109832E+00	2.830913E-01	7.315241E-01	1.981639E-01
8.000E-01	1.139339E+00	3.063245E-01	6.761465E-01	2.450596E-01
9.000E-01	1.170956E+00	3.253365E-01	6.228429E-01	2.928029E-01
1.000E+00	1.204280E+00	3.405623E-01	5.725505E-01	3.405623E-01
1.200E+00	1.274678E+00	3.615060E-01	4.828346E-01	4.338072E-01
1.400E+00	1.348232E+00	3.726632E-01	4.080434E-01	5.217284E-01
1.600E+00	1.423286E+00	3.769378E-01	3.468353E-01	6.031004E-01
1.800E+00	1.498696E+00	3.765218E-01	2.970706E-01	6.777393E-01
2.000E+00	1.573688E+00	3.729907E-01	2.565924E-01	7.459815E-01
2.200E+00	1.647758E+00	3.674469E-01	2.235215E-01	8.083832E-01
2.400E+00	1.720584E+00	3.606505E-01	1.963235E-01	8.655611E-01

ξ	θ	θ'	θ^{n+1}	$\xi\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi\theta'$
5.000E+01	7.257571E+00	5.761642E-02	2.615930E-03	2.880821E+00	3.600E+00	2.327314E+00	4.079755E-01	1.846248E-01	1.468712E+00
6.000E+01	7.798172E+00	5.085945E-02	2.108733E-03	3.051567E+00	3.800E+00	2.408229E+00	4.011728E-01	1.724267E-01	1.524457E+00
7.000E+01	8.280405E+00	4.581203E-02	1.761349E-03	3.206842E+00	4.000E+00	2.487785E+00	3.94060E-01	1.615750E-01	1.577624E+00
8.000E+01	8.718102E+00	4.187920E-02	1.509156E-03	3.350336E+00	4.200E+00	2.565997E+00	3.877249E-01	1.518755E-01	1.628445E+00
9.000E+01	9.120541E+00	3.871636E-02	1.318069E-03	3.484472E+00	4.400E+00	2.642863E+00	3.811644E-01	1.431674E-01	1.677124E+00
1.000E+02	9.494274E+00	3.610939E-02	1.168462E-03	3.610939E+00	4.600E+00	2.718472E+00	3.747484E-01	1.353163E-01	1.723643E+00
2.000E+02	1.232912E+01	2.31213E-02	5.35857E-04	4.626426E+00	4.800E+00	2.792793E+00	3.684942E-01	1.282101E-01	1.768763E+00
3.000E+02	1.435737E+01	1.801426E-02	3.378903E-04	5.404279E+00	5.000E+00	2.865880E+00	3.634061E-01	1.217542E-01	1.812030E+00
4.000E+02	1.600402E+01	1.514705E-02	2.349588E-04	6.058820E+00	5.500E+00	3.043426E+00	3.479583E-01	1.079629E-01	1.913774E+00
5.000E+02	1.741897E+01	1.326691E-02	1.892050E-04	6.633455E+00	6.000E+00	3.214022E+00	3.346040E-01	9.68603E-02	2.007624E+00
1.000E+03	2.274599E+01	8.854906E-03	8.497387E-05	8.854906E+00	6.500E+00	3.378203E+00	3.222878E-01	8.762510E-02	2.094870E+00
"	"	0.000000E+00	0.000000E+00	"	7.000E+00	3.536469E+00	3.109315E-01	7.995768E-02	2.176521E+00
					7.500E+00	3.689280E+00	3.004499E-01	7.347113E-02	2.253374E+00
					8.000E+00	3.837031E+00	2.907592E-01	6.792112E-02	2.326074E+00
					8.500E+00	3.980158E+00	2.817815E-01	6.312470E-02	2.395143E+00
					9.000E+00	4.118940E+00	2.734459E-01	5.894258E-02	2.461013E+00
					9.500E+00	4.253700E+00	2.656888E-01	5.526704E-02	2.524043E+00
					1.000E+01	4.384715E+00	2.584536E-01	5.201363E-02	2.584536E+00
					1.100E+01	4.636477E+00	2.453541E-01	4.651830E-02	2.698895E+00
					1.200E+01	4.875942E+00	2.338111E-01	4.206133E-02	2.805733E+00
					1.300E+01	5.104529E+00	2.235603E-01	3.837856E-02	2.906283E+00
					1.400E+01	5.323423E+00	2.143924E-01	3.528727E-02	3.001493E+00
					1.500E+01	5.533619E+00	2.061404E-01	3.265739E-02	3.092107E+00
					1.600E+01	5.735964E+00	1.986700E-01	3.039395E-02	3.178720E+00
					1.700E+01	5.931182E+00	1.918715E-01	2.842611E-02	3.261815E+00
					1.800E+01	6.119901E+00	1.856550E-01	2.670000E-02	3.341790E+00
					1.900E+01	6.302662E+00	1.799462E-01	2.517399E-02	3.418977E+00
					2.000E+01	6.479941E+00	1.746827E-01	2.381540E-02	3.493655E+00
					2.500E+01	7.297222E+00	1.534659E-01	1.877954E-02	3.836647E+00
					3.000E+01	8.024172E+00	1.380593E-01	1.553101E-02	4.14178E+00
					3.500E+01	8.63821E+00	1.252866E-01	1.326106E-02	4.420032E+00
					4.000E+01	9.291072E+00	1.169496E-01	1.158426E-02	4.677985E+00
					4.500E+01	9.856169E+00	1.093316E-01	1.029399E-02	4.919921E+00
					5.000E+01	1.038648E+01	1.029760E-01	9.269640E-03	5.148801E+00
					6.000E+01	1.136350E+01	9.292385E-02	7.744189E-03	5.575431E+00
					7.000E+01	1.225292E+01	8.527935E-02	6.660719E-03	5.969554E+00

Cylinder of polytropic index $n = -3$

0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	0.000000E+00
1.000E-01	1.002495E+00	4.981336E-02	9.950280E-01	4.981336E-03	0.000000E+00
2.000E-01	1.009926E+00	9.852697E-02	9.804399E-01	1.970539E-02	0.000000E+00
3.000E-01	1.022130E+00	1.451374E-01	9.571662E-01	4.354123E-02	0.000000E+00
4.000E-01	1.038855E+00	1.888157E-01	9.265947E-01	7.552627E-02	0.000000E+00
5.000E-01	1.059775E+00	2.289539E-01	8.903739E-01	1.144769E-01	0.000000E+00
6.000E-01	1.084515E+00	2.651785E-01	8.502145E-01	1.591071E-01	0.000000E+00
7.000E-01	1.112675E+00	2.973340E-01	8.077244E-01	2.081338E-01	0.000000E+00
8.000E-01	1.143847E+00	3.254470E-01	7.643000E-01	2.603576E-01	0.000000E+00
9.000E-01	1.177635E+00	3.496823E-01	7.210717E-01	3.147141E-01	0.000000E+00
1.000E+00	1.213663E+00	3.702984E-01	6.788967E-01	3.702984E-01	0.000000E+00
1.200E+00	1.291086E+00	4.019495E-01	5.999151E-01	4.823394E-01	0.000000E+00
1.400E+00	1.373741E+00	4.230645E-01	5.298956E-01	5.922903E-01	0.000000E+00
1.600E+00	1.459776E+00	4.361345E-01	4.692750E-01	6.978152E-01	0.000000E+00
1.800E+00	1.547798E+00	4.432330E-01	4.174184E-01	7.978194E-01	0.000000E+00
2.000E+00	1.636780E+00	4.459819E-01	3.732666E-01	8.919638E-01	0.000000E+00
2.200E+00	1.725993E+00	4.456065E-01	3.356812E-01	9.803344E-01	0.000000E+00
2.400E+00	1.814876E+00	4.430150E-01	3.036031E-01	1.063236E+00	0.000000E+00
2.600E+00	1.903086E+00	4.388734E-01	2.761107E-01	1.141071E+00	0.000000E+00
2.800E+00	1.990354E+00	4.336675E-01	2.524289E-01	1.214269E+00	0.000000E+00
3.000E+00	2.076506E+00	4.277511E-01	2.319176E-01	1.283253E+00	0.000000E+00
3.200E+00	2.161425E+00	4.213811E-01	2.140522E-01	1.348420E+00	0.000000E+00
3.400E+00	2.245041E+00	4.147442E-01	1.984045E-01	1.410130E+00	0.000000E+00

ξ	θ	θ'	θ^{n+1}	$\xi\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi\theta'$
8.000E+01	1.307437E+01	7.922986E-02	5.850037E-03	6.338397E+00	4.200E+00	2.971914E+00	5.602329E-01	3.364835E-01	2.352940E+00
9.000E+01	1.384122E+01	7.429773E-02	5.219769E-03	6.686796E+00	4.400E+00	3.083548E+00	5.560805E-01	3.243017E-01	2.446754E+00
1.000E+02	1.456303E+01	7.018178E-02	4.715160E-03	7.018178E+00	4.600E+00	3.194335E+00	5.517604E-01	3.130542E-01	2.538098E+00
2.000E+02	2.031618E+01	4.879312E-02	2.422790E-03	9.758623E+00	4.800E+00	3.304244E+00	5.473189E-01	3.026411E-01	2.627131E+00
3.000E+02	2.469636E+01	3.976155E-02	1.639585E-03	1.492846E+01	5.000E+00	3.413357E+00	5.428001E-01	2.929753E-01	2.714001E+00
4.000E+02	2.838877E+01	3.447594E-02	1.240815E-03	1.379038E+01	5.500E+00	3.681804E+00	5.313773E-01	2.716060E-01	2.922575E+00
5.000E+02	3.164733E+01	3.089770E-02	9.984489E-04	1.544885E+01	6.000E+00	3.944651E+00	5.200474E-01	2.535079E-01	3.120284E+00
1.000E+03	4.450521E+01	2.204988E-02	5.048885E-04	2.204988E+01	7.500E+00	4.201902E+00	5.091847E-01	2.379875E-01	3.308619E+00
∞	∞	0.000000E+00	0.000000E+00	∞	7.000E+00	4.453738E+00	4.983397E-01	2.245305E-01	3.488798E+00
Cylinder of polytropic index $n = -2$									
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	9.000E+00	5.411499E+00	4.606217E-01	1.847917E-01	4.145595E+00
1.000E-01	1.002497E+00	4.987542E-02	9.975093E-01	4.987542E-03	9.500E+00	5.639718E+00	4.523224E-01	1.773138E-01	4.297063E+00
2.000E-01	1.009950E+00	9.901314E-02	9.901476E-01	1.980263E-02	1.000E+01	5.863692E+00	4.444409E-01	1.705352E-01	4.444039E+00
3.000E-01	1.022232E+00	1.467230E-01	9.782326E-01	4.401690E-02	1.100E+01	6.300905E+00	4.298323E-01	1.587074E-01	4.728155E+00
4.000E-01	1.039227E+00	1.924029E-01	9.622535E-01	7.696116E-02	1.200E+01	6.724016E+00	4.166036E-01	1.487206E-01	4.999243E+00
5.000E-01	1.060648E+00	2.35674E-01	9.428195E-01	1.177837E-01	1.300E+01	7.134514E+00	4.045790E-01	1.401637E-01	5.259528E+00
6.000E-01	1.086245E+00	2.758616E-01	9.206027E-01	1.655170E-01	1.400E+01	7.533524E+00	3.936049E-01	1.327400E-01	5.510468E+00
7.000E-01	1.115717E+00	3.130618E-01	8.962843E-01	2.191433E-01	1.500E+01	7.922029E+00	3.835490E-01	1.262303E-01	5.753234E+00
8.000E-01	1.148751E+00	3.470644E-01	8.705110E-01	2.776515E-01	1.600E+01	8.300890E+00	3.742886E-01	1.204690E-01	5.988777E+00
9.000E-01	1.185024E+00	3.778682E-01	8.438650E-01	3.400814E-01	1.700E+01	8.670863E+00	3.657577E-01	1.153288E-01	6.217881E+00
1.000E+00	1.224220E+00	4.055534E-01	8.168465E-01	4.055534E-01	1.800E+01	9.032615E+00	3.578445E-01	1.107099E-01	6.441203E+00
1.200E+00	1.310179E+00	4.521675E-01	7.632546E-01	5.426010E-01	1.900E+01	9.386738E+00	3.504890E-01	1.065333E-01	6.659291E+00
1.400E+00	1.404396E+00	4.884134E-01	7.120501E-01	6.837787E-01	2.000E+01	9.733758E+00	3.436311E-01	1.027352E-01	6.872622E+00
1.600E+00	1.504967E+00	5.159939E-01	6.644665E-01	8.255902E-01	2.500E+01	1.137726E+01	3.152142E-01	8.789464E-02	7.880555E+00
1.800E+00	1.610324E+00	5.365331E-01	6.209302E-01	9.657596E-01	3.000E+01	1.289738E+01	2.93749E-01	7.753514E-02	8.812474E+00
2.000E+00	1.719206E+00	5.514512E-01	5.816640E-01	1.102927E+01	3.500E+01	1.423227E+01	2.768261E-01	6.982135E-02	9.688914E+00
2.200E+00	1.830609E+00	5.619309E-01	5.462662E-01	1.236248E+00	4.000E+01	1.567086E+01	2.603503E-01	6.381272E-02	1.052201E+01
2.400E+00	1.943746E+00	5.689287E-01	5.144704E-01	1.365429E+00	4.500E+01	1.685656E+01	2.515557E-01	5.897424E-02	1.132001E+01
2.600E+00	2.058000E+00	5.732060E-01	4.859087E-01	1.490335E+00	5.000E+01	1.818926E+01	2.417748E-01	5.497750E-02	1.208837E+01
2.800E+00	2.172888E+00	5.753626E-01	4.602171E-01	1.611015E+00	5.500E+01	1.88926E+01	2.259103E-01	4.872300E-02	1.355462E+01
3.000E+00	2.289035E+00	5.758695E-01	4.370562E-01	1.727609E+00	7.000E+01	2.052419E+01	2.134807E-01	4.401636E-02	1.494365E+01
3.200E+00	2.403150E+00	5.750956E-01	4.161205E-01	1.840306E+00	8.000E+01	2.271683E+01	2.033897E-01	4.032006E-02	1.627117E+01
3.400E+00	2.518007E+00	5.733300E-01	3.971394E-01	1.949322E+00	9.000E+01	2.679218E+01	1.949747E-01	3.732432E-02	1.754773E+01
3.600E+00	2.632431E+00	5.707997E-01	3.798769E-01	2.054879E+00	1.000E+02	2.870520E+01	1.878095E-01	3.483690E-02	1.878095E+01
3.800E+00	2.746288E+00	5.676830E-01	3.641278E-01	2.157195E+00					
4.000E+00	2.859475E+00	5.641207E-01	3.497145E-01	2.256483E+00					

ξ	θ	θ'	θ^{n+1}	$\xi\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi\theta'$
2.000E+02	4.516671E-01	1.478857E-01	2.214020E-02	2.957713E-01	4.800E+00	3.699620E+00	7.046132E-01	5.199020E-01	3.382143E+00
3.000E+02	5.893218E-01	1.291606E-01	1.696866E-02	3.874818E-01	5.000E+00	3.840407E+00	7.032049E-01	5.102833E-01	3.516023E+00
4.000E+02	7.122560E-01	1.174667E-01	1.403990E-02	4.698670E-01	5.500E+00	4.190912E+00	6.985924E-01	4.884788E-01	3.842258E+00
5.000E+02	8.253684E-01	1.091720E-01	1.211580E-02	5.458398E-01	6.000E+00	4.538324E+00	6.929235E-01	4.693840E-01	4.157541E+00
1.000E+03	1.307404E-02	8.701058E-02	7.648744E-03	8.701058E-01	6.500E+00	4.883733E+00	6.866369E-01	4.525057E-01	4.463140E+00
∞	∞	0.000000E+00	0.000000E+00	∞	7.000E+00	5.225407E+00	6.800193E-01	4.374616E-01	4.760133E+00
Cylinder of polytropic index $n = -1.5$									
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	7.500E+00	5.563729E+00	6.732601E-01	4.239523E-01	5.049450E+00
1.000E-01	1.002498E+00	4.990650E-02	9.987535E-01	4.990650E-03	8.000E+00	5.898664E+00	6.664841E-01	4.117401E-01	5.331873E+00
2.000E-01	1.009963E+00	9.925803E-02	9.950555E-01	1.985161E-02	8.500E+00	6.230224E+00	6.597737E-01	4.006343E-01	5.608077E+00
3.000E-01	1.022313E+00	1.475288E-01	9.890267E-01	4.425864E-02	9.000E+00	6.558458E+00	6.531827E-01	3.904803E-01	5.878645E+00
4.000E-01	1.039417E+00	1.942478E-01	9.808558E-01	7.769914E-02	9.500E+00	6.883433E+00	6.467451E-01	3.811513E-01	6.144079E+00
5.000E-01	1.061098E+00	2.390183E-01	9.707833E-01	1.195091E-01	1.000E+01	7.205232E+00	6.404817E-01	3.725427E-01	6.404817E+00
6.000E-01	1.087145E+00	2.815287E-01	9.590832E-01	1.689172E-01	1.000E+01	7.839668E+00	6.285174E-01	3.571504E-01	6.913692E+00
7.000E-01	1.117321E+00	3.215566E-01	9.460435E-01	2.250896E-01	1.000E+01	8.462522E+00	6.173179E-01	3.437558E-01	7.407814E+00
8.000E-01	1.151369E+00	3.589647E-01	9.319503E-01	2.871717E-01	1.000E+01	9.074551E+00	6.068605E-01	3.319613E-01	7.889188E+00
9.000E-01	1.189024E+00	3.936926E-01	9.170746E-01	3.543234E-01	1.000E+01	9.676476E+00	5.971009E-01	3.214707E-01	8.359413E+00
1.000E+00	1.230018E+00	4.257453E-01	9.016630E-01	4.257453E-01	1.500E+01	1.026897E+01	5.879861E-01	3.120589E-01	8.819791E+00
1.200E+00	1.320970E+00	4.820924E-01	8.700687E-01	5.785109E-01	1.600E+01	1.085265E+01	5.794621E-01	3.035514E-01	9.271394E+00
1.400E+00	1.422216E+00	5.288681E-01	8.385272E-01	7.404153E-01	1.800E+01	1.199376E+01	5.639832E-01	2.887261E-01	1.015170E+01
1.600E+00	1.531956E+00	5.672342E-01	8.079357E-01	9.075748E-01	1.900E+01	1.255819E+01	5.569365E-01	2.822092E-01	1.058179E+01
1.800E+00	1.648631E+00	5.984188E-01	7.788220E-01	1.077154E+00	2.000E+01	1.310977E+01	5.502972E-01	2.761863E-01	1.100594E+01
2.000E+00	1.770923E+00	6.235815E-01	7.514502E-01	1.247163E+00	2.500E+01	1.578762E+01	5.220906E-01	2.516759E-01	1.305226E+01
2.200E+00	1.897732E+00	6.437518E-01	7.259097E-01	1.416254E+00	3.000E+01	1.834088E+01	4.800498E-01	2.335016E-01	1.500149E+01
2.400E+00	2.028150E+00	6.598103E-01	7.021825E-01	1.583545E+00	3.500E+01	2.079510E+01	4.822232E-01	2.192903E-01	1.687781E+01
2.600E+00	2.161431E+00	6.724937E-01	6.801885E-01	1.748484E+00	4.000E+01	2.316811E+01	4.674110E-01	2.077563E-01	1.869642E+01
2.800E+00	2.296963E+00	6.824103E-01	6.598162E-01	1.910749E+00	4.500E+01	2.547292E+01	4.548356E-01	1.981347E-01	2.046760E+01
3.000E+00	2.432424E+00	6.900586E-01	6.409408E-01	2.070176E+00	5.000E+01	2.771933E+01	4.439744E-01	1.899366E-01	2.219872E+01
3.200E+00	2.572863E+00	6.958453E-01	6.234358E-01	2.226703E+00	6.000E+01	3.206571E+01	4.260241E-01	1.768595E-01	2.556145E+01
3.400E+00	2.712480E+00	7.001023E-01	6.071789E-01	2.380348E+00	7.000E+01	3.625164E+01	4.116504E-01	1.660872E-01	2.881553E+01
3.600E+00	2.852820E+00	7.031002E-01	5.920561E-01	2.531161E+00	8.000E+01	4.030696E+01	3.997628E-01	1.575107E-01	3.198102E+01
3.800E+00	2.993651E+00	7.050600E-01	5.779621E-01	2.679228E+00	9.000E+01	4.425291E+01	3.896879E-01	1.503243E-01	3.507191E+01
4.000E+00	3.134787E+00	7.061629E-01	5.648017E-01	2.824651E+00	1.000E+02	4.810527E+01	3.809847E-01	1.441793E-01	3.809847E+01
4.200E+00	3.276069E+00	7.065574E-01	5.524888E-01	2.967541E+00	2.000E+02	8.327873E+01	3.297772E-01	1.095804E-01	6.599544E+01
4.400E+00	3.417371E+00	7.063662E-01	5.409461E-01	3.108011E+00	3.000E+02	1.148797E+02	3.042004E-01	9.329928E-02	9.126013E+01
4.600E+00	3.558583E+00	7.056902E-01	5.301044E-01	3.246175E+00	4.000E+02	1.444068E+02	2.873282E-01	8.321586E-02	1.149313E+02
					5.000E+02	1.724925E+02	2.749415E-01	7.614035E-02	1.374707E+02

ξ	θ	θ'	θ^{n-1}	$\xi\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi\theta'$
1.000E+03	2.999796E+02	2.398180E-01	5.773699E-02	2.398180E+02	6.500E+00	5.912741E+00	9.899539E-01	9.823859E-01	6.434700E+00
∞	∞	0.000000E+00	0.000000E+00	∞	7.000E+00	6.409316E+00	9.980965E-01	9.815940E-01	6.972676E+00
Cylinder of polytropic Index $n = -1.01$									
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	7.500E+00	6.908618E+00	1.000916E+00	9.808579E-01	7.506868E+00
1.000E-01	1.002498E+00	4.993701E-02	9.999750E-01	4.993701E-03	8.000E+00	7.410060E+00	1.004700E+00	9.801709E-01	8.037600E+00
2.000E-01	1.009975E+00	9.94919E-02	9.999008E-01	9.99984E-02	8.500E+00	7.913182E+00	1.007667E+00	9.795272E-01	8.565170E+00
3.000E-01	1.02374E+00	1.483271E-01	9.997788E-01	4.449812E-02	9.000E+00	8.417619E+00	1.009984E+00	9.789220E-01	9.089853E+00
4.000E-01	1.039605E+00	1.960903E-01	9.996117E-01	7.84361E-02	9.500E+00	8.923079E+00	1.011779E+00	9.783514E-01	9.611898E+00
5.000E-01	1.061547E+00	2.424989E-01	9.994029E-01	1.212494E-01	1.000E+01	9.429328E+00	1.013153E+00	9.778116E-01	1.013153E+01
6.000E-01	1.088051E+00	2.873103E-01	9.991565E-01	1.723862E-01	1.100E+01	1.044347E+01	1.014940E+00	9.768133E-01	1.116434E+01
7.000E-01	1.118949E+00	3.303336E-01	9.988767E-01	2.312336E-01	1.200E+01	1.145890E+01	1.015807E+00	9.759073E-01	1.218968E+01
8.000E-01	1.154054E+00	3.714296E-01	9.985682E-01	2.971437E-01	1.300E+01	1.247487E+01	1.016051E+00	9.750786E-01	1.320866E+01
9.000E-01	1.193168E+00	4.105079E-01	9.982334E-01	3.694571E-01	1.400E+01	1.349086E+01	1.015872E+00	9.743155E-01	1.422220E+01
1.000E+00	1.236086E+00	4.475223E-01	9.978827E-01	4.475223E-01	1.500E+01	1.450652E+01	1.015404E+00	9.736085E-01	1.523108E+01
1.200E+00	1.332511E+00	5.153592E-01	9.971335E-01	6.184310E-01	1.600E+01	1.552161E+01	1.014741E+00	9.729502E-01	1.623585E+01
1.400E+00	1.441697E+00	5.752162E-01	9.963485E-01	8.053027E-01	1.700E+01	1.653596E+01	1.013947E+00	9.723345E-01	1.723709E+01
1.600E+00	1.562104E+00	6.276781E-01	9.955496E-01	1.004285E+00	1.800E+01	1.754947E+01	1.013068E+00	9.717563E-01	1.823522E+01
1.800E+00	1.692325E+00	6.734792E-01	9.947528E-01	1.212282E+00	1.900E+01	1.856208E+01	1.012137E+00	9.712113E-01	1.923061E+01
2.000E+00	1.831104E+00	7.133941E-01	9.939691E-01	1.426788E+00	2.000E+01	1.957374E+01	1.011178E+00	9.706960E-01	2.022357E+01
2.200E+00	1.977340E+00	7.481711E-01	9.932057E-01	1.645977E+00	3.000E+01	2.461758E+01	1.006405E+00	9.684731E-01	2.516012E+01
2.400E+00	2.130075E+00	7.784974E-01	9.924669E-01	1.868394E+00	3.500E+01	3.464001E+01	9.985055E-01	9.651709E-01	3.494769E+01
2.600E+00	2.288483E+00	8.049845E-01	9.917553E-01	2.092960E+00	4.000E+01	3.962467E+01	9.954465E-01	9.638742E-01	3.981786E+01
2.800E+00	2.451849E+00	8.281661E-01	9.910717E-01	2.318865E+00	4.500E+01	4.459527E+01	9.928629E-01	9.627358E-01	4.467883E+01
3.000E+00	2.619560E+00	8.485025E-01	9.904162E-01	2.54507E+00	5.000E+01	4.953395E+01	9.906663E-01	9.617213E-01	4.953331E+01
3.200E+00	2.791087E+00	8.668375E-01	9.897882E-01	2.772440E+00	6.000E+01	5.944221E+01	9.871571E-01	9.599731E-01	5.922943E+01
3.400E+00	2.965974E+00	8.821570E-01	9.891868E-01	2.999340E+00	7.000E+01	6.929991E+01	9.844986E-01	9.585012E-01	6.891490E+01
3.600E+00	3.143828E+00	8.960966E-01	9.886109E-01	3.225948E+00	8.000E+01	7.913413E+01	9.824269E-01	9.572301E-01	7.859415E+01
3.800E+00	3.324307E+00	9.084496E-01	9.880593E-01	3.452109E+00	9.000E+01	8.894983E+01	9.807729E-01	9.561115E-01	8.826956E+01
4.000E+00	3.507116E+00	9.194230E-01	9.875305E-01	3.677692E+00	1.000E+02	9.875060E+01	9.794248E-01	9.551127E-01	9.794248E+01
4.200E+00	3.691999E+00	9.291933E-01	9.870233E-01	3.902612E+00	2.000E+02	1.963081E+02	9.730839E-01	9.485727E-01	1.946168E+02
4.400E+00	3.878723E+00	9.379114E-01	9.865364E-01	4.126810E+00	3.000E+02	2.934863E+02	9.707494E-01	9.447657E-01	2.912248E+02
4.600E+00	4.067099E+00	9.457068E-01	9.860687E-01	4.350251E+00	4.000E+02	3.904894E+02	9.694058E-01	9.420716E-01	3.877623E+02
4.800E+00	4.256951E+00	9.526904E-01	9.856189E-01	4.572914E+00	5.000E+02	4.873803E+02	9.684580E-01	9.399859E-01	4.842290E+02
5.000E+00	4.448128E+00	9.589581E-01	9.851860E-01	4.794791E+00	1.000E+03	5.800E+02	9.656582E-01	9.335306E-01	5.800E+03
5.200E+00	4.640012E+00	9.621012E-01	9.841712E-01	5.046064E+00	∞	∞	∞	∞	∞
5.400E+00	4.831012E+00	9.720116E-01	9.832417E-01	5.346064E+00	∞	∞	∞	∞	∞
5.600E+00	5.019647E+00	9.822101E-01	9.822101E-01	5.692608E+00	∞	∞	∞	∞	∞

Cylinder of polytropic index $n = -0.9$												
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
1.000E-01	9.974986E-01	-5.005636E-02	9.997495E-01	9.997495E-01	-5.005636E-02	9.997495E-01	9.997495E-01	-5.005636E-02	9.997495E-01	9.997495E-01	-5.005636E-02	9.997495E-01
2.000E-01	9.899774E-01	-1.004536E-01	9.989932E-01	9.989932E-01	-1.004536E-01	9.989932E-01	9.989932E-01	-1.004536E-01	9.989932E-01	9.989932E-01	-1.004536E-01	9.989932E-01
3.000E-01	9.773847E-01	-1.51548E-01	9.977151E-01	9.977151E-01	-1.51548E-01	9.977151E-01	9.977151E-01	-1.51548E-01	9.977151E-01	9.977151E-01	-1.51548E-01	9.977151E-01
4.000E-01	9.596323E-01	-2.037172E-01	9.958880E-01	9.958880E-01	-2.037172E-01	9.958880E-01	9.958880E-01	-2.037172E-01	9.958880E-01	9.958880E-01	-2.037172E-01	9.958880E-01
5.000E-01	9.365911E-01	-2.573970E-01	9.934705E-01	9.934705E-01	-2.573970E-01	9.934705E-01	9.934705E-01	-2.573970E-01	9.934705E-01	9.934705E-01	-2.573970E-01	9.934705E-01
6.000E-01	9.080860E-01	-3.130859E-01	9.90447E-01	9.90447E-01	-3.130859E-01	9.90447E-01	9.90447E-01	-3.130859E-01	9.90447E-01	9.90447E-01	-3.130859E-01	9.90447E-01
7.000E-01	8.738870E-01	-3.718376E-01	9.866100E-01	9.866100E-01	-3.718376E-01	9.866100E-01	9.866100E-01	-3.718376E-01	9.866100E-01	9.866100E-01	-3.718376E-01	9.866100E-01
8.000E-01	8.336965E-01	-4.330538E-01	9.819758E-01	9.819758E-01	-4.330538E-01	9.819758E-01	9.819758E-01	-4.330538E-01	9.819758E-01	9.819758E-01	-4.330538E-01	9.819758E-01
9.000E-01	7.871314E-01	-4.990678E-01	9.763482E-01	9.763482E-01	-4.990678E-01	9.763482E-01	9.763482E-01	-4.990678E-01	9.763482E-01	9.763482E-01	-4.990678E-01	9.763482E-01
1.000E+00	7.336948E-01	-5.707421E-01	9.695083E-01	9.695083E-01	-5.707421E-01	9.695083E-01	9.695083E-01	-5.707421E-01	9.695083E-01	9.695083E-01	-5.707421E-01	9.695083E-01
1.100E+00	6.727331E-01	-6.499414E-01	9.611348E-01	9.611348E-01	-6.499414E-01	9.611348E-01	9.611348E-01	-6.499414E-01	9.611348E-01	9.611348E-01	-6.499414E-01	9.611348E-01
1.200E+00	6.032647E-01	-7.394493E-01	9.507332E-01	9.507332E-01	-7.394493E-01	9.507332E-01	9.507332E-01	-7.394493E-01	9.507332E-01	9.507332E-01	-7.394493E-01	9.507332E-01
1.300E+00	5.243541E-01	-8.437388E-01	9.374810E-01	9.374810E-01	-8.437388E-01	9.374810E-01	9.374810E-01	-8.437388E-01	9.374810E-01	9.374810E-01	-8.437388E-01	9.374810E-01
1.400E+00	4.338674E-01	-9.707613E-01	9.198896E-01	9.198896E-01	-9.707613E-01	9.198896E-01	9.198896E-01	-9.707613E-01	9.198896E-01	9.198896E-01	-9.707613E-01	9.198896E-01
1.500E+00	3.289180E-01	-1.137038E+00	8.947645E-01	8.947645E-01	-1.137038E+00	8.947645E-01	8.947645E-01	-1.137038E+00	8.947645E-01	8.947645E-01	-1.137038E+00	8.947645E-01
1.600E+00	2.037465E-01	-1.388253E+00	8.529214E-01	8.529214E-01	-1.388253E+00	8.529214E-01	8.529214E-01	-1.388253E+00	8.529214E-01	8.529214E-01	-1.388253E+00	8.529214E-01
1.660E+00	1.133136E-01	-1.652815E+00	8.04188E-01	8.04188E-01	-1.652815E+00	8.04188E-01	8.04188E-01	-1.652815E+00	8.04188E-01	8.04188E-01	-1.652815E+00	8.04188E-01
1.680E+00	7.889263E-02	-1.798011E+00	7.757176E-01	7.757176E-01	-1.798011E+00	7.757176E-01	7.757176E-01	-1.798011E+00	7.757176E-01	7.757176E-01	-1.798011E+00	7.757176E-01
1.700E+00	4.082342E-02	-2.033763E+00	7.262580E-01	7.262580E-01	-2.033763E+00	7.262580E-01	7.262580E-01	-2.033763E+00	7.262580E-01	7.262580E-01	-2.033763E+00	7.262580E-01
1.710E+00	1.947331E-02	-2.262523E+00	6.744410E-01	6.744410E-01	-2.262523E+00	6.744410E-01	6.744410E-01	-2.262523E+00	6.744410E-01	6.744410E-01	-2.262523E+00	6.744410E-01
1.712E+00	1.487616E-02	-2.337745E+00	6.565217E-01	6.565217E-01	-2.337745E+00	6.565217E-01	6.565217E-01	-2.337745E+00	6.565217E-01	6.565217E-01	-2.337745E+00	6.565217E-01
1.714E+00	1.010520E-02	-2.439228E+00	6.316180E-01	6.316180E-01	-2.439228E+00	6.316180E-01	6.316180E-01	-2.439228E+00	6.316180E-01	6.316180E-01	-2.439228E+00	6.316180E-01
1.716E+00	5.080142E-03	-2.602816E+00	5.896409E-01	5.896409E-01	-2.602816E+00	5.896409E-01	5.896409E-01	-2.602816E+00	5.896409E-01	5.896409E-01	-2.602816E+00	5.896409E-01
1.717E+00	2.407749E-03	-2.759486E+00	5.472189E-01	5.472189E-01	-2.759486E+00	5.472189E-01	5.472189E-01	-2.759486E+00	5.472189E-01	5.472189E-01	-2.759486E+00	5.472189E-01
1.718E+00	0.000000E+00	-4.307079E+00	0.000000E+00	0.000000E+00	-4.307079E+00	0.000000E+00	0.000000E+00	-4.307079E+00	0.000000E+00	0.000000E+00	-4.307079E+00	0.000000E+00
Cylinder of polytropic index $n = -0.5$												
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
1.000E-01	9.974992E-01	-5.003130E-02	9.987488E-01	9.987488E-01	-5.003130E-02	9.987488E-01	9.987488E-01	-5.003130E-02	9.987488E-01	9.987488E-01	-5.003130E-02	9.987488E-01
2.000E-01	9.899873E-01	-1.002515E-01	9.949811E-01	9.949811E-01	-1.002515E-01	9.949811E-01	9.949811E-01	-1.002515E-01	9.949811E-01	9.949811E-01	-1.002515E-01	9.949811E-01
3.000E-01	9.774362E-01	-1.508550E-01	9.886537E-01	9.886537E-01	-1.508550E-01	9.886537E-01	9.886537E-01	-1.508550E-01	9.886537E-01	9.886537E-01	-1.508550E-01	9.886537E-01
4.000E-01	9.597968E-01	-2.020481E-01	9.796927E-01	9.796927E-01	-2.020481E-01	9.796927E-01	9.796927E-01	-2.020481E-01	9.796927E-01	9.796927E-01	-2.020481E-01	9.796927E-01
5.000E-01	9.369994E-01	-2.540556E-01	9.679873E-01	9.679873E-01	-2.540556E-01	9.679873E-01	9.679873E-01	-2.540556E-01	9.679873E-01	9.679873E-01	-2.540556E-01	9.679873E-01
6.000E-01	9.089502E-01	-3.071298E-01	9.533888E-01	9.533888E-01	-3.071298E-01	9.533888E-01	9.533888E-01	-3.071298E-01	9.533888E-01	9.533888E-01	-3.071298E-01	9.533888E-01
7.000E-01	8.755283E-01	-3.615619E-01	9.356967E-01	9.356967E-01	-3.615619E-01	9.356967E-01	9.356967E-01	-3.615619E-01	9.356967E-01	9.356967E-01	-3.615619E-01	9.356967E-01
8.000E-01	8.365811E-01	-4.176970E-01	9.146481E-01	9.146481E-01	-4.176970E-01	9.146481E-01	9.146481E-01	-4.176970E-01	9.146481E-01	9.146481E-01	-4.176970E-01	9.146481E-01
9.000E-01	7.919181E-01	-4.759371E-01	8.898978E-01	8.898978E-01	-4.759371E-01	8.898978E-01	8.898978E-01	-4.759371E-01	8.898978E-01	8.898978E-01	-4.759371E-01	8.898978E-01
1.000E+00	7.413013E-01	-5.368733E-01	8.609886E-01	8.609886E-01	-5.368733E-01	8.609886E-01	8.609886E-01	-5.368733E-01	8.609886E-01	8.609886E-01	-5.368733E-01	8.609886E-01
1.100E+00	6.844321E-01	-6.011382E-01	8.273041E-01	8.273041E-01	-6.011382E-01	8.273041E-01	8.273041E-01	-6.011382E-01	8.273041E-01	8.273041E-01	-6.011382E-01	8.273041E-01
1.200E+00	6.209312E-01	-6.696906E-01	7.879919E-01	7.879919E-01	-6.696906E-01	7.879919E-01	7.879919E-01	-6.696906E-01	7.879919E-01	7.879919E-01	-6.696906E-01	7.879919E-01
1.300E+00	5.503073E-01	-7.438663E-01	7.418270E-01	7.418270E-01	-7.438663E-01	7.418270E-01	7.418270E-01	-7.438663E-01	7.418270E-01	7.418270E-01	-7.438663E-01	7.418270E-01
1.400E+00	4.719045E-01	-8.255687E-01	6.869309E-01	6.869309E-01	-8.255687E-01	6.869309E-01	6.869309E-01	-8.255687E-01	6.869309E-01	6.869309E-01	-8.255687E-01	6.869309E-01
1.500E+00	3.843069E-01	-9.184839E-01	6.203280E-01	6.203280E-01	-9.184839E-01	6.203280E-01	6.203280E-01	-9.184839E-01	6.203280E-01	6.203280E-01	-9.184839E-01	6.203280E-01
1.600E+00	2.876416E-01	-1.028490E+00	5.363223E-01	5.363223E-01	-1.028490E+00	5.363223E-01	5.363223E-01	-1.028490E+00	5.363223E-01	5.363223E-01	-1.028490E+00	5.363223E-01
1.700E+00	1.780784E-01	-1.170276E+00	4.219933E-01	4.219933E-01	-1.170276E+00	4.219933E-01	4.219933E-01	-1.170276E+00	4.219933E-01	4.219933E-01	-1.170276E+00	4.219933E-01
1.750E+00	1.172855E-01	-1.285671E+00	3.424698E-01	3.424698E-01	-1.285671E+00	3.424698E-01	3.424698E-01	-1.285671E+00	3.424698E-01	3.424698E-01	-1.285671E+00	3.424698E-01
1.800E+00	5.083987E-02	-1.403693E+00	2.547709E-01	2.547709E-01	-1.403693E+00	2.547709E-01	2.547709E-01	-1.403693E+00	2.547709E-01	2.547709E-01	-1.403693E+00	2.547709E-01
1.810E+00	3.660893E-02	-1.443774E+00	1.913346E-01	1.913346E-01	-1.443774E+00	1.913346E-01	1.913346E-01	-1.443774E+00	1.913346E-01	1.913346E-01	-1.443774E+00	1.913346E-01
1.820E+00	2.192952E-02	-1.494567E+00	1.480862E-01	1.480862E-01	-1.494567E+00	1.480862E-01	1.480862E-01	-1.494567E+00	1.480862E-01	1.480862E-01	-1.494567E+00	1.480862E-01
1.830E+00	6.632797E-03	-1.573103E+00	8.144199E-02	8.144199E-02	-1.573103E+00	8.144199E-02	8.144199E-02	-1.573103E+00	8.144199E-02	8.144199E-02	-1.573103E+00	8.144199E-02
1.832E+00	3.461342E-03	-1.599876E+00	5.883317E-02	5.883317E-02	-1.599876E+00	5.883317E-02	5.883317E-02	-1.599876E+00	5.883317E-02	5.883317E-02	-1.599876E+00	5.883317E-02
1.834E+00	2.199659E-04	-1.652234E+00	1.483125E-02	1.483125E-02	-1.652234E+00	1.483125E-02	1.483125E-02	-1.652234E+00	1.483125E-02	1.483125E-02	-1.652234E+00	1.483125E-02
1.83413266E+00	0.000000E+00	-1.669972E+00	0.000000E+00	0.000000E+00	-1.669972E+00	0.000000E+00	0.000000E+00	-1.669972E+00	0.000000E+00	0.000000E+00	-1.669972E+00	0.000000E+00
Cylinder of polytropic index $n = 0$												
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
1.000E-01	9.975000E-01	-5.000000E-02	9.975000E-01	9.975000E-01	-5.000000E-02	9.975000E-01	9.975000E-01	-5.000000E-02	9.975000E-01	9.975000E-01	-5.000000E-02	9.975000E-01
2.000E-01	9.900000E-01	-1.000000E-01	9.900000E-01	9.900000E-01	-1.000000E-01	9.900000E-01	9.900000E-01	-1.000000E-01	9.900000E-01	9.900000E-01	-1.000000E-01	9.900000E-01
3.000E-01	9.775000E-01	-1.500000E-01	9.775000E-01	9.775000E-01	-1.500000E-0							

ξ	θ	θ'	θ^{n+1}	$\xi\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi\theta'$
Cylinder of polytropic index $n = 1$									
1.300E+00	5.775000E-01	-6.500000E-01	5.775000E-01	-8.450000E-01	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
1.400E+00	5.100000E-01	-7.000000E-01	5.100000E-01	-9.800000E-01	1.000E-01	9.975016E-01	-4.993753E-02	9.950094E-01	-4.993753E-03
1.500E+00	4.375000E-01	-7.500000E-01	4.375000E-01	-1.125000E+00	2.000E-01	9.900250E-01	-9.950083E-02	9.801494E-01	-1.950017E-02
1.600E+00	3.600000E-01	-8.000000E-01	3.600000E-01	-1.280000E+00	3.000E-01	9.776262E-01	-1.483188E-01	9.557531E-01	-4.449564E-02
1.700E+00	2.775000E-01	-8.500000E-01	2.775000E-01	-1.445000E+00	4.000E-01	9.603982E-01	-1.960266E-01	9.223648E-01	-7.841063E-02
1.800E+00	1.900000E-01	-9.000000E-01	1.900000E-01	-1.620000E+00	5.000E-01	9.384698E-01	-2.422685E-01	8.807256E-01	-1.211342E-01
1.900E+00	9.750000E-02	-9.500000E-01	9.750000E-02	-2.000000E+00	6.000E-01	9.120049E-01	-2.867010E-01	8.317529E-01	-1.720206E-01
2.000000000E+00	0.000000E+00	-1.000000E+00	0.000000E+00	-2.000000E+00	7.000E-01	8.812009E-01	-3.289957E-01	7.765150E-01	-2.302970E-01
Cylinder of polytropic index $n = 0.5$									
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	8.000E-01	8.462874E-01	-3.688420E-01	7.162023E-01	-2.950736E-01
1.000E-01	9.975008E-01	-4.996874E-02	9.962355E-01	-4.996874E-03	9.000E-01	8.0752238E-01	-4.059495E-01	6.520947E-01	-3.653546E-01
2.000E-01	9.900125E-01	-9.974979E-02	9.850562E-01	-1.994996E-02	1.000E+00	7.651977E-01	-4.400506E-01	5.855275E-01	-4.400506E-01
3.000E-01	9.775634E-01	-1.491547E-01	9.665345E-01	-4.474640E-02	1.100E+00	7.186220E-01	-4.709024E-01	5.178558E-01	-5.179926E-01
4.000E-01	9.602004E-01	-1.979932E-01	9.408987E-01	-7.919730E-02	1.200E+00	6.711327E-01	-4.982891E-01	4.504192E-01	-5.979469E-01
5.000E-01	9.379900E-01	-2.460730E-01	9.084422E-01	-1.230365E-01	1.300E+00	6.200860E-01	-5.220232E-01	3.845066E-01	-6.786302E-01
6.000E-01	9.110177E-01	-2.931978E-01	8.695413E-01	-1.759187E-01	1.400E+00	5.668551E-01	-5.419477E-01	3.213247E-01	-7.587268E-01
7.000E-01	8.793890E-01	-3.391671E-01	8.246535E-01	-2.374170E-01	1.500E+00	5.118277E-01	-5.579365E-01	2.619676E-01	-8.369048E-01
8.000E-01	8.432296E-01	-3.837744E-01	7.743170E-01	-3.070195E-01	1.600E+00	4.554022E-01	-5.698959E-01	2.073911E-01	-9.118335E-01
9.000E-01	8.026866E-01	-4.268058E-01	7.191492E-01	-3.841253E-01	1.700E+00	3.979849E-01	-5.777652E-01	1.583919E-01	-9.822009E-01
1.000E+00	7.579284E-01	-4.680380E-01	6.598456E-01	-4.680380E-01	1.800E+00	3.399884E-01	-5.815170E-01	1.155908E-01	-1.046731E+00
1.100E+00	7.091468E-01	-5.072357E-01	5.971786E-01	-5.579593E-01	1.900E+00	2.818186E-01	-5.811571E-01	7.942170E-02	-1.104198E+00
1.200E+00	6.565574E-01	-5.441487E-01	5.319969E-01	-6.529784E-01	2.000E+00	2.238908E-01	-5.767248E-01	5.012708E-02	-1.153450E+00
1.300E+00	6.004022E-01	-5.785070E-01	4.652254E-01	-7.520591E-01	2.100E+00	1.666070E-01	-5.682921E-01	2.775789E-02	-1.193413E+00
1.400E+00	5.409510E-01	-6.100154E-01	3.978660E-01	-8.540215E-01	2.200E+00	1.158623E-01	-5.559630E-01	1.217983E-02	-1.223119E+00
1.500E+00	4.785050E-01	-6.383440E-01	3.310014E-01	-9.575159E-01	2.300E+00	5.53978E-02	-5.398725E-01	3.084668E-03	-1.241707E+00
1.600E+00	4.134007E-01	-6.631155E-01	2.658011E-01	-1.060985E+00	2.400E+00	2.507683E-03	-5.201853E-01	6.288476E-06	-1.248445E+00
1.700E+00	3.460153E-01	-6.838840E-01	2.035368E-01	-1.162603E+00	2.4082556E+00	0.000000E+00	-5.191475E-01	0.000000E+00	-1.248445E+00
1.800E+00	2.767755E-01	-7.009982E-01	1.456101E-01	-1.260177E+00	Cylinder of polytropic index $n = 1.5$				
1.900E+00	2.061713E-01	-7.110324E-02	9.361432E-02	-1.350962E+00	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
2.000E+00	1.347795E-01	-7.156315E-01	4.948068E-02	-1.431263E+00	1.000E-01	9.975023E-01	-4.990635E-02	9.937675E-01	-4.990635E-03
2.100E+00	6.331513E-02	-7.120552E-01	1.593167E-02	-1.495316E+00	2.000E-01	9.900374E-01	-9.925312E-02	9.752793E-01	-1.985062E-02
2.150E+00	2.784930E-02	-7.059669E-01	4.647521E-03	-1.517829E+00	3.000E-01	9.776887E-01	-1.474923E-01	9.451515E-01	-4.424770E-02
2.170E+00	1.376498E-02	-7.023168E-01	1.614963E-03	-1.524027E+00	4.000E-01	9.605934E-01	-1.940988E-01	9.403759E-01	-7.763952E-02
2.180E+00	6.752620E-03	-7.001004E-01	5.548920E-04	-1.526219E+00	5.000E-01	9.389398E-01	-2.385808E-01	8.542684E-01	-1.192904E-01
2.18966219E+00	0.000000E+00	-6.975389E-01	0.000000E+00	-1.527374E+00					

ξ	θ	θ'	θ^{n+1}	$\xi\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi\theta'$
6.000E-01	9.129631E-01	-2.804893E-01	7.964033E-01	-1.682936E-01	1.000E+00	7.780992E-01	-3.933192E-01	4.710911E-01	-3.933192E-01
7.000E-01	8.829410E-01	-3.194266E-01	7.325364E-01	-2.235986E-01	1.100E+00	7.377782E-01	-4.123884E-01	4.015850E-01	-4.536272E-01
8.000E-01	8.491882E-01	-3.550524E-01	6.645228E-01	-2.840419E-01	1.200E+00	6.957619E-01	-4.272477E-01	3.368077E-01	-5.126967E-01
9.000E-01	8.120502E-01	-3.870897E-01	5.942336E-01	-3.483807E-01	1.300E+00	6.524635E-01	-4.380663E-01	2.777593E-01	-5.694863E-01
1.000E+00	7.718970E-01	-4.153277E-01	5.234779E-01	-4.153277E-01	1.400E+00	6.082755E-01	-4.450829E-01	2.250614E-01	-6.231161E-01
1.100E+00	7.291163E-01	-4.396219E-01	4.539332E-01	-4.835841E-01	1.500E+00	5.635840E-01	-4.485869E-01	1.789904E-01	-6.728803E-01
1.200E+00	6.841066E-01	-4.598980E-01	3.870878E-01	-5.518776E-01	1.600E+00	5.186643E-01	-4.489069E-01	1.395272E-01	-7.182510E-01
1.300E+00	6.372710E-01	-4.761457E-01	3.241981E-01	-6.189894E-01	1.700E+00	4.738770E-01	-4.463711E-01	1.064133E-01	-7.588751E-01
1.400E+00	5.890101E-01	-4.884184E-01	2.662605E-01	-6.837857E-01	1.800E+00	4.294669E-01	-4.414251E-01	7.921168E-02	-7.945652E-01
1.500E+00	5.397162E-01	-4.968277E-01	2.139999E-01	-7.452416E-01	1.900E+00	3.856617E-01	-4.343612E-01	5.736137E-02	-8.252862E-01
1.600E+00	4.897678E-01	-5.015390E-01	1.678710E-01	-8.024623E-01	2.000E+00	3.426523E-01	-4.255894E-01	4.023100E-02	-8.511388E-01
1.700E+00	4.395246E-01	-5.027648E-01	1.260731E-01	-8.547001E-01	2.100E+00	3.005937E-01	-4.154004E-01	2.716062E-02	-8.723408E-01
1.800E+00	3.893226E-01	-5.007590E-01	9.456745E-02	-9.013663E-01	2.200E+00	2.596070E-01	-4.041857E-01	1.749642E-02	-8.892086E-01
1.900E+00	3.394709E-01	-4.958104E-01	6.714389E-02	-9.420398E-01	2.300E+00	2.197811E-01	-3.922342E-01	1.061625E-02	-9.021386E-01
2.000E+00	2.902491E-01	-4.882365E-01	4.538616E-02	-9.764729E-01	2.400E+00	1.811753E-01	-3.798292E-01	5.946983E-03	-9.115902E-01
2.100E+00	2.418998E-01	-4.783778E-01	2.877987E-02	-1.004593E+00	2.500E+00	1.438218E-01	-3.672278E-01	2.974910E-03	-9.180699E-01
2.200E+00	1.946387E-01	-4.665943E-01	1.671329E-02	-1.026507E+00	2.600E+00	1.077285E-01	-3.546604E-01	1.250237E-03	-9.221170E-01
2.300E+00	1.486326E-01	-4.532621E-01	8.516972E-03	-1.042503E+00	2.700E+00	7.288172E-02	-3.423314E-01	3.871291E-04	-9.242947E-01
2.400E+00	1.040228E-01	-4.387753E-01	3.489968E-03	-1.053061E+00	2.800E+00	3.924828E-02	-3.304210E-01	6.045913E-05	-9.251783E-01
2.500E+00	6.090210E-02	-4.235533E-01	9.153363E-04	-1.058883E+00	2.880E+00	1.318212E-02	-3.213008E-01	2.290632E-06	-9.253464E-01
2.550E+00	3.991766E-02	-4.158106E-01	3.183597E-04	-1.060317E+00	2.900E+00	6.778285E-03	-3.190870E-01	3.114293E-07	-9.253523E-01
2.600E+00	1.932102E-02	-4.080697E-01	5.188898E-05	-1.060981E+00	2.920E+00	4.184451E-04	-3.169018E-01	7.326820E-11	-9.253533E-01
2.620E+00	1.119043E-02	-4.04925E-01	1.324697E-05	-1.061080E+00	2.92132072E+00	0.000000E+00	-3.167383E-01	0.000000E+00	-9.253533E-01
2.640E+00	3.121177E-03	-4.019369E-01	5.442471E-07	-1.061113E+00	Cylinder of polytropic index $n = 3$				
2.6477677E-00	0.000000E+00	-4.007569E-01	0.000000E+00	-1.061115E+00	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	1.000E-01	9.975004E-01	-4.981305E-02	9.900560E-01	4.981305E-02
1.000E-01	9.975031E-01	-4.987521E-02	9.925281E-01	-4.987521E-03	2.000E-01	9.900744E-01	-9.851732E-02	9.608849E-01	-1.970346E-03
2.000E-01	9.800498E-01	-9.900663E-02	9.704454E-01	-1.980133E-02	3.000E-01	9.778732E-01	-1.450874E-01	9.143871E-01	-4.352021E-02
3.000E-01	9.777506E-01	-1.466750E-01	9.347259E-01	-4.400250E-02	4.000E-01	9.611638E-01	-1.885377E-01	8.534726E-01	-7.541507E-02
4.000E-01	9.607860E-01	-1.922088E-01	8.869109E-01	-7.688352E-02	5.000E-01	9.402339E-01	-2.281672E-01	7.817257E-01	-1.140836E-01
5.000E-01	9.394002E-01	-2.350047E-01	8.289951E-01	-1.175023E-01	6.000E-01	9.156775E-01	-2.633881E-01	7.030241E-01	-1.580328E-01
6.000E-01	9.138937E-01	-2.745443E-01	7.632856E-01	-1.647266E-01	7.000E-01	8.877755E-01	-2.938371E-01	6.211732E-01	-2.056860E-01
7.000E-01	8.846140E-01	-3.104072E-01	6.922476E-01	-2.172850E-01	8.000E-01	8.570748E-01	-3.193516E-01	5.396037E-01	-2.554813E-01
8.000E-01	8.519455E-01	-3.422778E-01	6.183515E-01	-2.738222E-01	9.000E-01	8.240692E-01	-3.399513E-01	4.611633E-01	-3.059562E-01
9.000E-01	8.162986E-01	-3.699488E-01	5.439351E-01	-3.329539E-01	1.000E+00	7.892427E-01	-3.558114E-01	3.880094E-01	-3.558114E-01
					1.100E+00	7.530551E-01	-3.672298E-01	3.215933E-01	-4.039528E-01

Cylinder of polytropic index $n = 2$

ξ	θ	θ'	θ^{n+1}	$\xi\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi\theta'$
1.200E+00	7.159320E-01	-3.745928E-01	2.627163E-01	-4.495114E-01	7.000E-01	8.907154E-01	-2.789712E-01	5.606538E-01	-1.952798E-01
1.300E+00	6.782570E-01	-3.783427E-01	2.116300E-01	-4.918455E-01	8.000E-01	8.617543E-01	-2.993570E-01	4.752446E-01	-2.394856E-01
1.400E+00	6.403626E-01	-3.789486E-01	1.681586E-01	-5.305280E-01	9.000E-01	8.310171E-01	-3.145544E-01	3.963235E-01	-2.830990E-01
1.500E+00	6.025563E-01	-3.768823E-01	1.318228E-01	-5.653235E-01	1.000E+00	7.990017E-01	-3.25069E-01	3.256405E-01	-3.250069E-01
1.600E+00	5.650656E-01	-3.726000E-01	1.019519E-01	-5.961601E-01	1.200E+00	7.328709E-01	-3.338896E-01	2.114158E-01	-4.006676E-01
1.700E+00	5.280959E-01	-3.665287E-01	7.777701E-02	-6.230988E-01	1.400E+00	6.662537E-01	-3.306399E-01	1.312798E-01	-4.628959E-01
1.800E+00	4.918065E-01	-3.590574E-01	5.850284E-02	-6.463034E-01	1.600E+00	6.011368E-01	-3.195318E-01	7.849943E-02	-5.112509E-01
1.900E+00	4.563195E-01	-3.505326E-01	4.335869E-02	-6.660119E-01	1.800E+00	5.367366E-01	-3.039559E-01	4.538187E-02	-5.471207E-01
2.000E+00	4.217249E-01	-3.412562E-01	3.163131E-02	-6.825125E-01	2.000E+00	4.796886E-01	-2.863394E-01	2.539785E-02	-5.726788E-01
2.100E+00	3.88046E-01	-3.314869E-01	2.268328E-02	-6.961225E-01	2.200E+00	4.242288E-01	-2.682819E-01	1.374045E-02	-5.902201E-01
2.200E+00	3.554367E-01	-3.214417E-01	1.596060E-02	-7.071717E-01	2.400E+00	3.723394E-01	-2.507566E-01	7.156407E-03	-6.018158E-01
2.300E+00	3.237995E-01	-3.112995E-01	1.099271E-02	-7.159887E-01	2.600E+00	3.238544E-01	-2.342992E-01	3.562451E-03	-6.091778E-01
2.400E+00	2.931752E-01	-3.012049E-01	7.387695E-03	-7.228917E-01	2.800E+00	2.785316E-01	-2.191574E-01	1.676381E-03	-6.136407E-01
2.500E+00	2.635531E-01	-2.912723E-01	4.824724E-03	-7.281807E-01	3.000E+00	2.360989E-01	-2.053997E-01	7.336181E-04	-6.161992E-01
2.600E+00	2.349124E-01	-2.815897E-01	3.045255E-03	-7.321333E-01	3.200E+00	1.962818E-01	-1.929849E-01	2.913401E-04	-6.175660E-01
2.700E+00	2.072246E-01	-2.722226E-01	1.844019E-03	-7.350011E-01	3.400E+00	1.588195E-01	-1.818326E-01	1.010462E-04	-6.182309E-01
2.800E+00	1.804538E-01	-2.632174E-01	1.060433E-03	-7.370088E-01	3.600E+00	1.234732E-01	-1.718097E-01	2.869873E-05	-6.185148E-01
2.900E+00	1.545680E-01	-2.546044E-01	5.707933E-04	-7.383528E-01	3.800E+00	9.002858E-02	-1.627933E-01	5.914283E-06	-6.186145E-01
3.000E+00	1.295212E-01	-2.464008E-01	2.814258E-04	-7.392024E-01	4.000E+00	5.829699E-02	-1.546600E-01	6.733341E-07	-6.186400E-01
3.100E+00	1.052740E-01	-2.386127E-01	1.228245E-04	-7.396995E-01	4.100E+00	4.302111E-02	-1.508885E-01	1.473697E-07	-6.186427E-01
3.200E+00	8.178492E-02	-2.312376E-01	4.473968E-05	-7.399603E-01	4.200E+00	2.811332E-02	-1.472961E-01	1.756148E-08	-6.186435E-01
3.300E+00	5.901307E-02	-2.242655E-01	1.212810E-05	-7.400760E-01	4.300E+00	1.355633E-02	-1.438706E-01	4.578375E-10	-6.186435E-01
3.400E+00	3.691890E-02	-2.176808E-01	1.857784E-06	-7.401147E-01	4.350E+00	6.404306E-03	-1.422169E-01	1.077359E-11	-6.186436E-01
3.500E+00	1.546466E-02	-2.114634E-01	5.719546E-08	-7.401219E-01	4.370E+00	3.566487E-03	-1.415660E-01	5.770364E-13	-6.186436E-01
3.540E+00	7.054092E-03	-2.090740E-01	2.476079E-09	-7.401221E-01	4.390E+00	7.416256E-04	-1.409211E-01	2.243487E-16	-6.186436E-01
3.560E+00	2.884379E-03	-2.079955E-01	6.921645E-11	-7.401221E-01	4.39526586E+00	0.000000E+00	-1.407522E-01	0.000000E+00	-6.186436E-01
3.570E+00	8.082988E-04	-2.073171E-01	4.268622E-13	-7.401221E-01	Cylinder of polytropic Index $n = 5$				
3.57390098E+00	0.000000E+00	-2.070908E-01	0.000000E+00	-7.401221E-01	Cylinder of polytropic Index $n = 5$				
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
1.000E-01	9.975062E-01	-4.975104E-02	9.875932E-01	-4.975104E-03	1.000E-01	9.970708E-01	-4.968918E-02	9.851396E-01	-4.968918E-03
2.000E-01	9.900989E-01	-9.803283E-02	9.514652E-01	-1.960657E-02	2.000E-01	9.901232E-01	-9.755309E-02	9.421835E-01	-1.951062E-02
3.000E-01	9.779939E-01	-1.434947E-01	8.947068E-01	-4.304840E-02	3.000E-01	9.781129E-01	-1.419558E-01	8.756569E-01	-4.258675E-02
4.000E-01	9.615320E-01	-1.850049E-01	8.218995E-01	-7.400195E-02	4.000E-01	9.618911E-01	-1.816027E-01	7.920550E-01	-7.264109E-02
5.000E-01	9.411532E-01	-2.217195E-01	7.384167E-01	-1.108597E-01	5.000E-01	9.419802E-01	-2.156290E-01	6.986357E-01	-1.078145E-01
6.000E-01	9.173658E-01	-2.531128E-01	6.496999E-01	-1.518677E-01	6.000E-01	9.189670E-01	-2.436177E-01	6.022814E-01	-1.461700E-01
					7.000E-01	8.934587E-01	-2.655575E-01	5.086826E-01	-1.858902E-01
					8.000E-01	8.660470E-01	-2.817605E-01	4.219380E-01	-2.254084E-01

ξ	θ	θ'	θ^{n+1}	$\xi\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi\theta'$
9.000E-01	8.372801E-01	-2.927639E-01	3.445280E-01	-2.634873E-01	7.000E-01	8.960267E-01	-2.533915E-01	4.637103E-01	-1.773741E-01
1.000E+00	8.076454E-01	-2.992337E-01	2.775391E-01	-2.992337E-01	8.000E-01	8.700041E-01	-2.661513E-01	3.772672E-01	-2.129210E-01
1.000E+00	7.473723E-01	-3.014194E-01	1.742697E-01	-3.617033E-01	9.000E-01	8.429865E-01	-2.738544E-01	3.024594E-01	-2.464689E-01
1.400E+00	6.877373E-01	-2.936705E-01	1.058121E-01	-4.111387E-01	1.000E+00	8.153739E-01	-2.773374E-01	2.396059E-01	-2.773374E-01
1.600E+00	6.302784E-01	-2.802620E-01	6.268947E-02	-4.484191E-01	1.200E+00	7.599766E-01	-2.748756E-01	1.464204E-01	-3.298508E-01
1.800E+00	5.758078E-01	-2.641881E-01	3.644728E-02	-4.755387E-01	1.400E+00	7.059544E-01	-2.643953E-01	8.738492E-02	-3.701534E-01
2.000E+00	5.246519E-01	-2.473572E-01	2.085583E-02	-4.947144E-01	1.600E+00	6.544816E-01	-2.499187E-01	5.143774E-02	-3.998700E-01
2.200E+00	4.722397E-01	-2.308871E-01	1.175532E-02	-5.079516E-01	1.800E+00	6.060810E-01	-2.339954E-01	3.004101E-02	-4.211913E-01
2.400E+00	4.232372E-01	-2.153712E-01	6.520876E-03	-5.168909E-01	2.000E+00	5.608012E-01	-2.180985E-01	1.746185E-02	-4.361970E-01
2.600E+00	3.806094E-01	-2.010762E-01	3.551862E-03	-5.227981E-01	2.200E+00	5.187873E-01	-2.030013E-01	1.011406E-02	-4.466029E-01
2.800E+00	3.517157E-01	-1.880761E-01	1.893000E-03	-5.266132E-01	2.400E+00	4.796023E-01	-1.890549E-01	5.836723E-03	-4.537318E-01
3.000E+00	3.152947E-01	-1.763385E-01	9.824258E-04	-5.290154E-01	2.600E+00	4.430810E-01	-1.763699E-01	3.352601E-03	-4.585617E-01
3.200E+00	2.811019E-01	-1.657764E-01	4.933812E-04	-5.304843E-01	2.800E+00	4.089714E-01	-1.649278E-01	1.913597E-03	-4.617979E-01
3.400E+00	2.489130E-01	-1.562802E-01	2.378403E-04	-5.313526E-01	3.000E+00	3.770324E-01	-1.546470E-01	1.083055E-03	-4.639409E-01
3.600E+00	2.185265E-01	-1.477346E-01	1.088991E-04	-5.318446E-01	3.200E+00	3.470424E-01	-1.454193E-01	6.062860E-04	-4.653417E-01
3.800E+00	1.897632E-01	-1.400289E-01	4.669521E-05	-5.321097E-01	3.400E+00	3.188021E-01	-1.371306E-01	3.348942E-04	-4.662439E-01
4.000E+00	1.624658E-01	-1.330609E-01	1.838954E-05	-5.322436E-01	3.600E+00	2.921349E-01	-1.296709E-01	1.815867E-04	-4.668152E-01
4.200E+00	1.364958E-01	-1.267395E-01	6.467188E-06	-5.323059E-01	3.800E+00	2.668852E-01	-1.229393E-01	9.844320E-05	-4.671700E-01
4.400E+00	1.117322E-01	-1.209845E-01	1.945676E-06	-5.323319E-01	4.000E+00	2.429166E-01	-1.168463E-01	4.991143E-05	-4.673853E-01
4.600E+00	8.806888E-02	-1.157263E-01	4.665894E-07	-5.323412E-01	4.200E+00	2.201094E-01	-1.113123E-01	2.503055E-05	-4.675125E-01
4.800E+00	6.541258E-02	-1.109050E-01	7.833712E-08	-5.323438E-01	4.400E+00	1.983589E-01	-1.062694E-01	1.208264E-05	-4.675852E-01
5.000E+00	4.368123E-02	-1.064689E-01	6.946548E-09	-5.323444E-01	4.600E+00	1.775729E-01	-1.016577E-01	5.567207E-06	-4.676254E-01
5.100E+00	3.313941E-02	-1.043813E-01	1.324551E-09	-5.323444E-01	4.800E+00	1.576703E-01	-9.742633E-02	2.422438E-06	-4.676463E-01
5.200E+00	2.280230E-02	-1.023739E-01	1.405634E-10	-5.323444E-01	5.000E+00	1.385800E-01	-9.352140E-02	9.815398E-07	-4.676570E-01
5.300E+00	1.266210E-02	-1.004423E-01	4.121305E-12	-5.323444E-01	5.500E+00	9.400704E-02	-8.502990E-02	6.488177E-08	-4.676644E-01
5.400E+00	7.711449E-03	-9.858230E-02	3.973825E-16	-5.323444E-01	6.000E+00	5.331488E-02	-7.794418E-02	1.224444E-09	-4.676651E-01
5.420E+00	7.424453E-04	-9.821853E-02	1.688473E-19	-5.323444E-01	6.200E+00	3.798022E-02	-7.542983E-02	1.139994E-10	-4.676651E-01
5.42757459E+00	0.000000E+00	-9.808145E-02	0.000000E+00	-5.323444E-01	6.400E+00	2.313248E-02	-7.307267E-02	3.544486E-12	-4.676651E-01
					6.600E+00	8.741634E-03	-7.085834E-02	3.900753E-15	-4.676651E-01
					6.700E+00	1.708944E-03	-6.890076E-02	4.256922E-20	-4.676651E-01
					6.72452797E+00	0.000000E+00	-6.954616E-02	0.000000E+00	-4.676651E-01

Cylinder of polytropic index $n = 10$

0.000E+00	1.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000E+00	1.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
1.000E-01	9.975093E-01	-4.962748E-02	9.826951E-01	-4.962748E-03	1.000E-01	9.975153E-01	-4.938221E-02	9.730075E-01	-4.938221E-03
2.000E-01	9.901474E-01	-9.707802E-02	9.330371E-01	-1.941560E-02	2.000E-01	9.902425E-01	-9.522322E-02	8.977534E-01	-1.904464E-02
3.000E-01	9.782302E-01	-1.404497E-01	8.572107E-01	-4.213492E-02	3.000E-01	9.786833E-01	-1.347328E-01	7.889763E-01	-4.041983E-02
4.000E-01	9.622413E-01	-1.763241E-01	7.638149E-01	-7.132962E-02					
5.000E-01	9.427771E-01	-2.098666E-01	6.620081E-01	-1.049333E-01					
6.000E-01	9.204883E-01	-2.348165E-01	5.599226E-01	-1.408899E-01					

ξ	θ	θ'	θ^{n+1}	$\xi\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi\theta'$
4.000E-01	9.635615E-01	-1.663188E-01	6.647733E-01	-6.652750E-02	1.200E-01	9.527408E-02	-2.633271E-02	5.871138E-12	-3.159923E-01
5.000E-01	9.456968E-01	-1.896185E-01	5.410922E-01	-9.480926E-02	1.300E-01	6.998118E-02	-2.430712E-02	1.971486E-13	-3.159923E-01
6.000E-01	9.258950E-01	-2.052175E-01	4.287255E-01	-1.231305E-01	1.400E-01	4.656361E-02	-2.257090E-02	2.231070E-15	-3.159923E-01
7.000E-01	9.048723E-01	-2.214252E-01	3.330124E-01	-1.499766E-01	1.500E-01	2.476238E-02	-2.106617E-02	2.146426E-18	-3.159923E-01
8.000E-01	8.832189E-01	-2.180557E-01	2.551243E-01	-1.744445E-01	1.600E-01	4.368887E-03	-1.974953E-02	1.106268E-26	-3.159923E-01
9.000E-01	8.613924E-01	-2.179158E-01	1.937366E-01	-1.961242E-01	1.610E-01	2.399880E-03	-1.962687E-02	1.520844E-29	-3.159923E-01
1.000E+00	8.397297E-01	-2.149493E-01	1.463979E-01	-2.149493E-01	1.620E-01	4.432635E-04	-1.950371E-02	1.298020E-37	-3.159923E-01
1.200E+00	7.97589E-01	-2.039355E-01	8.328915E-01	-2.447226E-01	1.62227407E-01	0.000000E+00	-1.947837E-02	0.000000E+00	-3.159923E-01
1.400E+00	7.583517E-01	-1.899024E-01	4.770644E-02	-2.658634E-01	Cylinder of polytropic index $n = 20$				
1.600E+00	7.218255E-01	-1.754116E-01	2.771753E-02	-2.806586E-01	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
1.800E+00	6.881367E-01	-1.616327E-01	1.638407E-02	-2.909749E-01	1.000E-01	9.975308E-01	-4.877951E-02	9.494065E-01	-4.877951E-03
2.000E+00	6.570839E-01	-1.490911E-01	9.858910E-03	-2.981822E-01	2.000E-01	9.904699E-01	-9.088252E-02	8.178376E-01	-1.817650E-02
2.200E+00	6.284127E-01	-1.378371E-01	6.035242E-03	-3.032416E-01	3.000E-01	9.797147E-01	-1.222931E-01	6.502875E-01	-3.668793E-02
2.400E+00	6.018652E-01	-1.278392E-01	3.753974E-03	-3.068142E-01	4.000E-01	9.663913E-01	-1.423819E-01	4.877680E-01	-5.695278E-02
2.600E+00	5.772010E-01	-1.189819E-01	2.369177E-03	-3.093528E-01	5.000E-01	9.515602E-01	-1.528401E-01	3.525019E-01	-7.642003E-02
2.800E+00	5.542053E-01	-1.111314E-01	1.514878E-03	-3.111678E-01	6.000E-01	9.360598E-01	-1.561970E-01	2.496750E-01	-9.371813E-02
3.000E+00	5.326900E-01	-1.041576E-01	9.799862E-04	-3.124729E-01	7.000E-01	9.204774E-01	-1.548361E-01	1.754994E-01	-1.083853E-01
3.200E+00	5.124917E-01	-9.794258E-02	6.405533E-02	-3.141014E-01	8.000E-01	9.031874E-01	-1.506091E-01	1.234550E-01	-1.204873E-01
3.400E+00	4.934693E-01	-9.238276E-02	4.225340E-04	-3.141014E-01	9.000E-01	8.904076E-01	-1.448077E-01	8.737065E-02	-1.303289E-01
3.600E+00	4.755009E-01	-8.738916E-02	2.809730E-04	-3.146010E-01	1.000E+00	8.762500E-01	-1.382735E-01	6.240053E-02	-1.382735E-01
3.800E+00	4.584810E-01	-8.238596E-02	1.881632E-04	-3.149666E-01	1.200E+00	8.499411E-01	-1.248781E-01	3.289767E-02	-1.498537E-01
4.000E+00	4.423181E-01	-7.880877E-02	1.267892E-04	-3.152351E-01	1.400E+00	8.282300E-01	-1.124685E-01	1.816017E-02	-1.574560E-01
4.200E+00	4.269327E-01	-7.510301E-02	8.589225E-05	-3.154327E-01	1.600E+00	8.048506E-01	-1.015881E-01	1.047180E-02	-1.625409E-01
4.400E+00	4.122553E-01	-7.172237E-02	5.845589E-05	-3.155784E-01	1.800E+00	7.854928E-01	-9.223125E-02	6.280394E-03	-1.660163E-01
4.600E+00	3.982248E-01	-6.862742E-02	3.994025E-05	-3.156861E-01	2.000E+00	7.678683E-01	-8.422129E-02	3.899613E-03	-1.684426E-01
4.800E+00	3.847875E-01	-6.578454E-02	2.737982E-05	-3.157658E-01	2.500E+00	7.298569E-01	-6.876796E-02	1.342742E-03	-1.719199E-01
5.000E+00	3.718961E-01	-6.316496E-02	1.882077E-05	-3.158248E-01	3.000E+00	6.963496E-01	-5.785420E-02	5.315340E-04	-1.735628E-01
5.500E+00	3.417900E-01	-5.743888E-02	7.436240E-06	-3.159139E-01	3.500E+00	6.715247E-01	-4.983016E-02	2.335084E-04	-1.744055E-01
6.000E+00	3.142999E-01	-5.265932E-02	2.956560E-06	-3.159559E-01	4.000E+00	6.427031E-01	-4.371669E-02	1.111560E-04	-1.748668E-01
6.500E+00	2.890091E-01	-4.861165E-02	1.174968E-06	-3.159757E-01	4.500E+00	6.252590E-01	-3.891828E-02	5.638935E-05	-1.751323E-01
7.000E+00	2.655924E-01	-4.514071E-02	4.638432E-07	-3.159850E-01	5.000E+00	6.091291E-01	-3.505830E-02	3.012286E-05	-1.752913E-01
7.500E+00	2.437915E-01	-4.213189E-02	1.808022E-07	-3.159892E-01	6.000E+00	5.771532E-01	-2.924224E-02	9.707624E-06	-1.754534E-01
8.000E+00	2.233979E-01	-3.949889E-02	6.916250E-08	-3.159911E-01	7.000E+00	5.501010E-01	-2.507461E-02	3.542352E-06	-1.755223E-01
8.500E+00	2.042411E-01	-3.717552E-02	2.579727E-08	-3.159919E-01	8.000E+00	5.266610E-01	-2.194425E-02	1.419561E-06	-1.755540E-01
9.000E+00	1.861795E-01	-3.511026E-02	9.316425E-09	-3.159923E-01	9.000E+00	5.059827E-01	-1.950773E-02	6.121374E-07	-1.755776E-01
9.500E+00	1.690946E-01	-3.326236E-02	3.231676E-09	-3.159925E-01	1.000E+01	4.874841E-01	-1.755776E-02	2.800078E-07	-1.755776E-01
1.000E+01	1.528863E-01	-3.159925E-02	1.066732E-09	-3.159925E-01					
1.100E+01	1.227690E-01	-2.872659E-02	9.549417E-11	-3.159925E-01					

ξ	θ	θ'	θ^{n+1}	$\xi\theta'$	ξ	θ	θ'	$\exp(-\theta)$	$\xi\theta'$
1.500E+01	4.162910E-01	-1.170579E-02	1.017039E-08	-1.755869E-01	2.400E+00	1.084649E+00	6.976744E-01	3.380206E-01	1.674413E+00
2.000E+01	3.657777E-01	-8.779379E-03	6.722968E-10	-1.755876E-01	2.600E+00	1.224959E+00	7.046070E-01	2.937699E-01	1.831978E+00
3.000E+01	2.945830E-01	-5.852923E-03	7.134463E-12	-1.755877E-01	2.800E+00	1.366194E+00	7.070707E-01	2.550760E-01	1.979798E+00
4.000E+01	2.406866E-01	-4.399692E-03	1.373367E-13	-1.755877E-01	3.000E+00	1.507544E+00	7.058824E-01	2.214533E-01	2.117647E+00
5.000E+01	2.048883E-01	-3.511754E-03	3.482259E-15	-1.755877E-01	3.200E+00	1.648351E+00	7.017554E-01	1.923669E-01	2.245614E+00
6.000E+01	1.728749E-01	-2.926461E-03	9.825861E-17	-1.755877E-01	3.400E+00	1.789090E+00	6.952963E-01	1.672793E-01	2.364008E+00
8.000E+01	1.223615E-01	-2.194846E-03	6.926891E-20	-1.755877E-01	3.600E+00	1.926349E+00	6.870229E-01	1.456792E-01	2.473282E+00
1.000E+02	8.318021E-02	-1.755877E-03	2.091320E-23	-1.755877E-01	3.800E+00	2.062807E+00	6.776319E-01	1.270967E-01	2.573973E+00
1.250E+02	4.399895E-02	-1.404701E-03	3.253040E-29	-1.755877E-01	4.000E+00	2.197225E+00	6.666667E-01	1.111111E-01	2.666667E+00
1.500E+02	1.198554E-02	-1.170585E-03	4.485451E-41	-1.755877E-01	4.200E+00	2.329424E+00	6.552262E-01	9.735179E-02	2.751950E+00
1.550E+02	6.228046E-03	-1.132824E-03	4.801629E-47	-1.755877E-01	4.400E+00	2.459281E+00	6.432749E-01	8.549639E-02	2.830409E+00
1.580E+02	2.862045E-03	-1.111314E-03	3.892303E-54	-1.755877E-01	4.600E+00	2.586713E+00	6.310014E-01	7.526706E-02	2.902606E+00
1.600E+02	6.533662E-04	-1.097423E-03	1.313073E-67	-1.755877E-01	4.800E+00	2.711670E+00	6.185567E-01	6.642576E-02	2.969072E+00
1.60596473E-02	0.000000E+00	-1.093347E-03	0.000000E+00	-1.755877E-01	5.000E+00	2.834132E+00	6.060606E-01	5.876951E-02	3.030303E+00
					5.500E+00	3.129404E+00	5.751634E-01	4.374386E-02	3.163399E+00
					6.000E+00	3.409496E+00	5.454545E-01	3.305785E-02	3.272727E+00
					6.500E+00	3.675138E+00	5.174129E-01	2.534591E-02	3.363184E+00
					7.000E+00	3.927219E+00	4.912281E-01	1.969837E-02	3.438596E+00
					7.500E+00	4.166880E+00	4.669261E-01	1.550364E-02	3.501946E+00
					8.000E+00	4.394449E+00	4.444444E-01	1.234568E-02	3.555556E+00
					8.500E+00	4.611410E+00	4.236760E-01	9.937792E-03	3.601246E+00
					9.000E+00	4.818390E+00	4.049444E-01	8.079788E-03	3.640449E+00
					9.500E+00	5.016147E+00	3.867684E-01	6.630020E-03	3.674300E+00
					1.000E+01	5.205379E+00	3.703704E-01	5.486968E-03	3.703704E+00
					1.100E+01	5.560742E+00	3.410853E-01	3.845923E-03	3.751938E+00
					1.200E+01	5.888787E+00	3.157895E-01	2.770083E-03	3.789474E+00
					1.300E+01	6.193416E+00	2.937853E-01	2.042838E-03	3.819209E+00
					1.400E+01	6.477357E+00	2.745098E-01	1.537870E-03	3.843137E+00
					1.500E+01	6.743194E+00	2.575107E-01	1.178876E-03	3.862661E+00
					1.600E+01	6.993015E+00	2.424242E-01	9.182736E-04	3.878789E+00
					1.700E+01	7.228581E+00	2.289562E-01	7.255493E-04	3.892256E+00
					1.800E+01	7.451387E+00	2.168675E-01	5.806358E-04	3.903614E+00
					1.900E+01	7.662710E+00	2.059621E-01	4.700318E-04	3.913279E+00
					2.000E+01	7.863951E+00	1.960784E-01	3.844673E-04	3.921569E+00
					2.500E+01	8.742058E+00	1.579779E-01	1.597249E-04	3.949447E+00
					3.000E+01	9.463606E+00	1.321586E-01	7.762619E-05	3.964758E+00
					3.500E+01	1.007553E+01	1.135442E-01	4.209726E-05	3.974047E+00
					4.000E+01	1.060661E+01	9.950249E-02	2.475186E-05	3.980100E+00

ξ	θ	θ'	$\exp(-\theta)$	$\xi\theta'$
0.000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
1.000E-01	2.498439E-03	4.993758E-02	9.975047E-01	4.993758E-03
2.000E-01	9.975083E-03	9.950249E-02	9.900745E-01	1.990050E-02
3.000E-01	2.274387E-02	1.483313E-01	9.778741E-01	4.449938E-02
4.000E-01	3.960325E-02	1.960784E-01	9.611688E-01	7.843137E-02
5.000E-01	6.154332E-02	2.424242E-01	9.403122E-01	1.212121E-01
6.000E-01	8.803377E-02	2.870813E-01	9.157300E-01	1.722488E-01
7.000E-01	1.188949E-01	3.297998E-01	8.879011E-01	2.308598E-01
8.000E-01	1.539221E-01	3.703704E-01	8.573388E-01	2.962963E-01
9.000E-01	1.928918E-01	4.086266E-01	8.245713E-01	3.677639E-01
1.000E+00	2.355661E-01	4.444444E-01	7.901235E-01	4.444444E-01
1.200E+00	3.102899E-01	5.084746E-01	7.181844E-01	6.101695E-01
1.400E+00	4.32711E-01	5.622490E-01	6.451509E-01	7.871486E-01
1.600E+00	5.552635E-01	6.060606E-01	5.739210E-01	9.696970E-01
1.800E+00	6.800746E-01	6.405694E-01	5.065792E-01	1.153025E+00
2.000E+00	8.109302E-01	6.666667E-01	4.444444E-01	1.333333E+00
2.200E+00	9.462475E-01	6.853583E-01	3.881950E-01	1.507788E+00

Cylinder of polytropic index $n = 1.5$

ξ	δ	ϑ'	$\exp(-\vartheta)$	$\xi\vartheta'$	ξ	δ	ϑ'	δ^{n+1}	$\xi^2\vartheta'$
4.500E+01	1.107565E+01	8.853910E-02	1.548479E-05	3.984260E+00	2.600E+00	1.341898E+00	1.112266E-01	7.088023E-02	7.519124E-01
5.000E+01	1.149560E+01	7.974482E-02	1.017478E-05	3.987241E+00	2.800E+00	1.363508E+00	1.049631E-01	6.138772E-02	8.229108E-01
6.000E+01	1.222293E+01	6.651885E-02	4.916397E-06	3.991131E+00	3.000E+00	1.383919E+00	9.932625E-02	5.370366E-02	8.931262E-01
7.000E+01	1.283836E+01	5.704971E-02	2.656873E-06	3.993480E+00	3.200E+00	1.402325E+00	9.400760E-02	4.740510E-02	9.626378E-01
8.000E+01	1.337172E+01	4.993758E-02	1.558601E-06	3.995006E+00	3.400E+00	1.421552E+00	8.932348E-02	4.218271E-02	1.031527E+00
9.000E+01	1.384233E+01	4.440059E-02	9.735371E-07	3.996053E+00	3.600E+00	1.438956E+00	8.486690E-02	3.780707E-02	1.098975E+00
1.000E+02	1.426340E+01	3.996803E-02	6.389772E-07	3.996803E+00	3.800E+00	1.455524E+00	8.038952E-02	3.410369E-02	1.167756E+00
2.000E+02	1.703479E+01	1.999600E-01	3.998400E-08	3.999200E+00	4.000E+00	1.471326E+00	7.720249E-02	3.094717E-02	1.235240E+00
3.000E+02	1.855642E+01	1.333215E-02	7.899830E-09	3.999644E+00	4.200E+00	1.486424E+00	7.383167E-02	2.823025E-02	1.369266E+00
4.000E+02	1.980708E+01	9.99500E-03	2.499750E-09	3.999800E+00	4.400E+00	1.500876E+00	7.072565E-02	2.587598E-02	1.469184E+00
5.000E+02	2.069961E+01	7.999744E-03	1.023934E-09	3.999872E+00	4.600E+00	1.514731E+00	6.786003E-02	2.382214E-02	1.435918E+00
1.000E+03	2.347215E+01	3.999968E-03	6.399898E-11	3.999968E+00	4.800E+00	1.528034E+00	6.520808E-02	2.201923E-02	1.502394E+00
∞	∞	0.000000E+00	0.000000E+00	4.000000E+00	5.000E+00	1.540827E+00	6.274943E-02	2.042750E-02	1.568736E+00
					5.500E+00	1.570807E+00	5.733029E-02	1.717494E-02	1.734241E+00
					6.000E+00	1.598300E+00	5.276741E-02	1.469184E-02	1.899627E+00
					6.500E+00	1.623687E+00	4.838210E-02	1.274910E-02	2.065269E+00
					7.000E+00	1.647272E+00	4.53991E-02	1.119727E-02	2.231455E+00
					7.500E+00	1.669300E+00	4.263838E-02	9.935513E-03	2.398409E+00
					8.000E+00	1.689971E+00	4.009843E-02	8.893806E-03	2.566300E+00
					8.500E+00	1.705448E+00	3.785827E-02	8.02247E-03	2.735260E+00
					9.000E+00	1.727871E+00	3.586901E-02	7.284487E-03	2.905390E+00
					9.500E+00	1.745353E+00	3.409161E-02	6.653516E-03	3.076768E+00
					1.000E+01	1.761992E+00	3.249452E-02	6.108914E-03	3.249452E+00
					1.100E+01	1.793067E+00	2.974297E-02	5.219542E-03	3.598900E+00
					1.200E+01	1.821633E+00	2.745800E-02	4.527428E-03	3.953952E+00
					1.300E+01	1.848101E+00	2.553077E-02	3.976196E-03	4.314700E+00
					1.400E+01	1.872788E+00	2.38843E-02	3.528603E-03	4.681159E+00
					1.500E+01	1.895942E+00	2.245894E-02	3.159175E-03	5.053262E+00
					1.600E+01	1.917766E+00	2.121464E-02	2.849969E-03	5.430947E+00
					1.700E+01	1.938421E+00	2.011800E-02	2.588019E-03	5.814101E+00
					1.800E+01	1.958042E+00	1.914383E-02	2.363748E-03	6.202602E+00
					1.900E+01	1.976743E+00	1.827236E-02	2.169943E-03	6.596322E+00
					2.000E+01	1.994616E+00	1.748781E-02	2.001088E-03	6.995124E+00
					2.500E+01	2.074032E+00	1.449699E-02	1.408187E-03	9.060621E+00
					3.000E+01	2.141184E+00	1.248166E-02	1.057094E-03	1.123349E+01
					3.500E+01	2.199761E+00	1.102007E-02	8.291258E-04	1.349959E+01
					4.000E+01	2.251957E+00	9.904495E-03	6.713660E-04	1.584719E+01
					4.500E+01	2.299190E+00	9.020596E-03	5.569633E-04	1.826671E+01

Sphere of polytropic index $n = -10$

ξ	δ	ϑ'	δ^{n+1}	$\xi^2\vartheta'$
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
1.000E-01	1.001658E+00	3.300334E-02	9.851975E-01	3.300334E-04
2.000E-01	1.006537E+00	6.410369E-02	9.430462E-01	2.564148E-03
3.000E-01	1.014363E+00	9.175053E-02	8.795450E-01	8.257548E-03
4.000E-01	1.024739E+00	1.149683E-01	8.025686E-01	1.839492E-02
5.000E-01	1.037197E+00	1.333955E-01	7.198827E-01	3.34887E-02
6.000E-01	1.051262E+00	1.471771E-01	6.376753E-01	5.298374E-02
7.000E-01	1.066493E+00	1.567896E-01	5.602448E-01	7.682688E-02
8.000E-01	1.082503E+00	1.628711E-01	4.899359E-01	1.042375E-01
9.000E-01	1.098972E+00	1.660930E-01	4.276815E-01	1.345353E-01
1.000E+00	1.115647E+00	1.670811E-01	3.734730E-01	1.670811E-01
1.200E+00	1.148881E+00	1.644291E-01	2.867640E-01	2.367779E-01
1.400E+00	1.181179E+00	1.581469E-01	2.234382E-01	3.099678E-01
1.600E+00	1.212032E+00	1.502191E-01	1.771632E-01	3.845610E-01
1.800E+00	1.241234E+00	1.417772E-01	1.429939E-01	4.593561E-01
2.000E+00	1.268750E+00	1.334293E-01	1.173860E-01	5.337171E-01
2.200E+00	1.294633E+00	1.254807E-01	9.787716E-02	6.073265E-01
2.400E+00	1.318978E+00	1.180667E-01	8.276803E-02	6.800640E-01

ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$
5.000E+01	2.342434E+00	8.300094E-03	4.709708E-04	2.075023E-01	3.600E+00	1.703385E+00	1.798113E-01	1.187816E-01	2.330355E+00
6.000E+01	2.419581E+00	7.190065E-03	3.518347E-04	2.588424E-01	3.800E+00	1.738754E+00	1.739331E-01	1.094076E-01	2.511594E+00
7.000E+01	2.487189E+00	6.369983E-03	2.745501E-04	3.120851E-01	4.000E+00	1.772978E+00	1.683615E-01	1.012014E-01	2.693785E+00
8.000E+01	2.547576E+00	5.733391E-03	2.212363E-04	3.669370E-01	4.200E+00	1.806118E+00	1.630894E-01	9.397559E-02	2.876898E+00
9.000E+01	2.602276E+00	5.224440E-03	1.827339E-04	4.231796E-01	4.400E+00	1.838233E+00	1.581053E-01	8.757850E-02	3.060918E+00
1.000E+02	2.652366E+00	4.806458E-03	1.539213E-04	4.806458E-01	4.600E+00	1.869379E+00	1.533953E-01	8.188619E-02	3.245844E+00
2.000E+02	3.009875E+00	2.756992E-03	4.932461E-05	1.102797E-02	4.800E+00	1.896609E+00	1.489443E-01	7.679686E-02	3.431673E+00
3.000E+02	3.242081E+00	1.980650E-03	2.526882E-05	1.782585E-02	5.000E+00	1.928973E+00	1.447372E-01	7.222633E-02	3.618429E+00
4.000E+02	3.417503E+00	1.563810E-03	1.572860E-05	2.502096E-02	5.500E+00	1.998899E+00	1.351870E-01	6.263779E-02	4.089408E+00
5.000E+02	3.559869E+00	1.301230E-03	1.089165E-05	3.253074E-02	6.000E+00	2.064361E+00	1.268451E-01	5.506268E-02	4.566425E+00
1.000E+03	4.039407E+00	7.348356E-04	3.492538E-06	7.348356E-02	6.500E+00	2.125914E+00	1.195197E-01	4.895732E-02	5.049708E+00
∞	∞	0.000000E+00	0.000000E+00	∞	7.000E+00	2.184023E+00	1.130501E-01	4.395122E-02	5.539455E+00
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	7.500E+00	2.239084E+00	1.073037E-01	3.978491E-02	6.035831E+00
1.000E-01	1.001663E+00	3.316755E-02	9.933775E-01	3.316755E-04	8.000E+00	2.291429E+00	1.027173E-01	3.627223E-02	6.538963E+00
2.000E-01	1.006601E+00	6.536129E-02	9.740263E-01	2.614451E-03	8.500E+00	2.341343E+00	9.756327E-02	3.327669E-02	7.048946E+00
3.000E-01	1.014673E+00	9.570668E-02	9.433995E-01	8.613601E-03	9.000E+00	2.389067E+00	9.340549E-02	3.069632E-02	7.565844E+00
4.000E-01	1.025657E+00	1.235067E-01	9.036310E-01	1.976107E-02	9.500E+00	2.434813E+00	8.963654E-02	2.845363E-02	8.089697E+00
5.000E-01	1.039273E+00	1.482851E-01	8.571975E-01	3.707128E-02	1.000E-01	2.478760E+00	8.620524E-02	2.648878E-02	8.620524E+00
6.000E-01	1.055205E+00	1.697926E-01	8.065898E-01	6.112535E-02	1.200E-01	2.561872E+00	8.019082E-02	2.321509E-02	9.703089E+00
7.000E-01	1.073121E+00	1.879828E-01	7.540586E-01	9.211155E-02	1.300E-01	2.639447E+00	7.509300E-02	2.060386E-02	1.081339E+01
8.000E-01	1.092695E+00	2.029694E-01	7.014631E-01	1.299004E-01	1.400E-01	2.781072E+00	6.691788E-02	1.671672E-02	1.311590E+01
9.000E-01	1.113616E+00	2.149788E-01	6.502176E-01	1.741328E-01	1.500E-01	2.846290E+00	6.358772E-02	1.523644E-02	1.430724E+01
1.000E+00	1.135601E+00	2.243041E-01	6.013083E-01	2.243041E-01	1.600E-01	2.908376E+00	6.064305E-02	1.397648E-02	1.552462E+01
1.200E+00	1.181787E+00	2.361963E-01	5.126769E-01	3.401227E-01	1.700E-01	2.967683E+00	5.801911E-02	1.289230E-02	1.676752E+01
1.400E+00	1.229618E+00	2.411574E-01	4.374040E-01	4.726685E-01	1.800E-01	3.024504E+00	5.566483E-02	1.195043E-02	1.803540E+01
1.600E+00	1.277931E+00	2.413212E-01	3.749475E-01	6.177824E-01	1.900E-01	3.079088E+00	5.353940E-02	1.112529E-02	1.932772E+01
1.800E+00	1.325937E+00	2.363192E-01	3.235258E-01	7.721542E-01	2.000E-01	3.131648E+00	5.160985E-02	1.039701E-02	2.064394E+01
2.000E+00	1.379127E+00	2.332515E-01	2.812922E-01	9.333005E-01	2.500E-01	3.369666E+00	4.410485E-02	7.756260E-03	2.756553E+01
2.200E+00	1.419189E+00	2.271527E-01	2.465127E-01	1.099419E+00	3.000E-01	3.576474E+00	3.890160E-02	6.111947E-03	3.501144E+01
2.400E+00	1.463946E+00	2.203520E-01	2.177206E-01	1.269228E+00	3.500E-01	3.760904E+00	3.504514E-02	4.998402E-03	4.293029E+01
2.600E+00	1.507312E+00	2.132877E-01	1.937256E-01	1.441825E+00	4.000E-01	3.928349E+00	3.204949E-02	4.199132E-03	5.127919E+01
2.800E+00	1.549260E+00	2.061969E-01	1.735814E-01	1.616583E+00	5.000E-01	4.225452E+00	2.964061E-02	3.600396E-03	6.002233E+01
3.000E+00	1.589799E+00	1.992298E-01	1.565419E-01	1.793068E+00	6.000E-01	4.485599E+00	2.765166E-02	3.136947E-03	6.912914E+01
3.200E+00	1.628966E+00	1.924787E-01	1.420208E-01	1.970982E+00	7.000E-01	4.718732E+00	2.453750E-02	2.470121E-03	8.833506E+01
3.400E+00	1.666809E+00	1.859967E-01	1.295558E-01	2.150122E+00	8.000E-01	4.931048E+00	2.218979E-02	2.016967E-03	1.087300E+02
					9.000E-01	5.126715E+00	2.034167E-02	1.691387E-03	1.301867E+02
							1.884015E-02	1.447583E-03	1.526052E+02

Sphere of polytropic index $n = -5$

ξ	θ	θ'	θ^{n-1}	$\xi^2\theta'$	ξ	θ	θ'	θ^{n-1}	$\xi^2\theta'$
1.000E+02	5.308688E+00	1.759050E-02	1.259074E-03	1.759050E+02	4.600E+00	2.015772E+00	1.966419E-01	1.220888E-01	4.160942E+00
2.000E+02	6.685533E+00	1.116722E-02	5.002604E-04	4.466889E+02	4.800E+00	2.054609E+00	1.917706E-01	1.152955E-01	4.418366E+00
3.000E+02	7.657142E+00	8.536242E-03	2.908930E-04	7.682618E+02	5.000E+00	2.092495E+00	1.871236E-01	1.091457E-01	4.678090E+00
4.000E+02	8.430443E+00	7.047618E-03	1.979695E-04	1.127619E+03	5.500E+00	2.183330E+00	1.764273E-01	9.608197E-02	5.336925E+00
5.000E+02	9.083411E+00	6.071909E-03	1.468940E-04	1.517977E+03	6.000E+00	2.269121E+00	1.669228E-01	8.559078E-02	6.009222E+00
1.000E+03	1.144890E+01	3.819149E-03	5.820287E-05	3.819149E+03	6.500E+00	2.350426E+00	1.584566E-01	7.701235E-02	6.694792E+00
∞	∞	0.000000E+00	0.000000E+00	∞	7.000E+00	2.427727E+00	1.508853E-01	6.988763E-02	7.393525E+00
					7.500E+00	2.501443E+00	1.440850E-01	6.388930E-02	8.105345E+00
					8.000E+00	2.571934E+00	1.379716E-01	5.877879E-02	8.830183E+00
					8.500E+00	2.639511E+00	1.324287E-01	5.437877E-02	9.567973E+00
					9.000E+00	2.704447E+00	1.273906E-01	5.055308E-02	1.031864E+01
					9.500E+00	2.766975E+00	1.227934E-01	4.720458E-02	1.108210E+01
					1.000E+01	2.827304E+00	1.185827E-01	4.424685E-02	1.185827E+01
					1.100E+01	2.942667E+00	1.111433E-01	3.926832E-02	1.344834E+01
					1.200E+01	3.049948E+00	1.047778E-01	3.524704E-02	1.508800E+01
					1.300E+01	3.151907E+00	9.926822E-02	3.193594E-02	1.677633E+01
					1.400E+01	3.248715E+00	9.445090E-02	2.916522E-02	1.851238E+01
					1.500E+01	3.340998E+00	9.020074E-02	2.681461E-02	2.029517E+01
					1.600E+01	3.429273E+00	8.642078E-02	2.479670E-02	2.212372E+01
					1.700E+01	3.513971E+00	8.303482E-02	2.304654E-02	2.399706E+01
					1.800E+01	3.595453E+00	7.998223E-02	2.151488E-02	2.591424E+01
					1.900E+01	3.674030E+00	7.721422E-02	2.016378E-02	2.787433E+01
					2.000E+01	3.749963E+00	7.469105E-02	1.896352E-02	2.987642E+01
					2.500E+01	4.097031E+00	6.477948E-02	1.454093E-02	4.048718E+01
					3.000E+01	4.402590E+00	5.780942E-02	1.171859E-02	5.202847E+01
					3.500E+01	4.678008E+00	5.258610E-02	9.768261E-03	6.441797E+01
					4.000E+01	4.930319E+00	4.934919E-02	8.344010E-03	7.758717E+01
					4.500E+01	5.194215E+00	4.517462E-02	7.260944E-03	9.147860E+01
					5.000E+01	5.382995E+00	4.241741E-02	6.411033E-03	1.060435E+02
					6.000E+01	5.784326E+00	3.806454E-02	5.167038E-03	1.370323E+02
					7.000E+01	6.147718E+00	3.475117E-02	4.303860E-03	1.702807E+02
					8.000E+01	6.481613E+00	3.212146E-02	3.672406E-03	2.055773E+02
					9.000E+01	6.791732E+00	2.996959E-02	3.191967E-03	2.427537E+02
					1.000E+02	7.032165E+00	2.816732E-02	2.815152E-03	2.816722E+02
					2.000E+02	9.340030E+00	1.869707E-02	1.227313E-03	7.478830E+02
					3.000E+02	1.098759E+01	1.468155E-02	7.538643E-04	1.321340E+03
					4.000E+02	1.233071E+01	1.233759E-02	5.333788E-04	1.977215E+03
					5.000E+02	1.348441E+01	1.080822E-02	4.078531E-04	2.702054E+03

Sphere of polytropic index $n = -4$

ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$
1.000E+03	1.779911E+01	7.124139E-03	1.773393E-04	7.124139E+03	6.500E+00	2.891566E+00	2.254659E-01	1.380352E-01	9.525933E+00
"	"	0.000000E+00	0.000000E+00	"	7.000E+00	2.802106E+00	2.168257E-01	1.273594E-01	1.062446E+01
Sphere of polytropic Index $n = -3$									
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	7.500E+00	2.908518E+00	2.089392E-01	1.182107E-01	1.175283E+01
1.000E-01	1.00164E+00	3.32368E-02	9.966799E-01	3.32368E-04	8.000E+00	3.011157E+00	2.017257E-01	1.102892E-01	1.291044E+01
2.000E-01	1.006627E+00	6.587756E-02	9.868765E-01	2.635102E-03	8.500E+00	3.110343E+00	1.951113E-01	1.033674E-01	1.409679E+01
3.000E-01	1.014802E+00	9.738122E-02	9.710413E-01	8.764310E-03	9.000E+00	3.206357E+00	1.890298E-01	9.726938E-02	1.531142E+01
4.000E-01	1.026049E+00	1.272674E-01	9.498687E-01	2.036278E-02	1.000E-01	3.389851E+00	1.782388E-01	8.702396E-02	1.782388E+01
5.000E-01	1.040188E+00	1.551554E-01	9.242214E-01	3.878886E-02	1.100E-01	3.563340E+00	1.689653E-01	7.875633E-02	2.044486E+01
6.000E-01	1.057005E+00	1.807735E-01	8.950479E-01	6.507847E-02	1.200E-01	3.728189E+00	1.609144E-01	7.194560E-02	2.317187E+01
7.000E-01	1.076262E+00	2.039584E-01	8.633047E-01	9.993960E-02	1.300E-01	3.885500E+00	1.538569E-01	6.623786E-02	2.600181E+01
8.000E-01	1.097713E+00	2.246464E-01	8.298937E-01	1.437737E-01	1.400E-01	4.036173E+00	1.476171E-01	6.138470E-02	2.892395E+01
1.000E+00	1.121109E+00	2.428585E-01	7.956182E-01	1.967154E-01	1.500E+01	4.180960E+00	1.420572E-01	5.720684E-02	3.196287E+01
1.200E+00	1.200385E+00	2.837318E-01	6.937673E-01	4.085738E-01	1.600E+01	4.320479E+00	1.370683E-01	5.357179E-02	3.508948E+01
1.400E+00	1.259190E+00	3.011667E-01	6.306924E-01	5.902867E-01	1.700E+01	4.455258E+00	1.325632E-01	5.037958E-02	3.831077E+01
1.600E+00	1.326444E+00	3.124714E-01	5.733616E-01	7.999268E-01	1.800E+01	4.585743E+00	1.284717E-01	4.755329E-02	4.162482E+01
1.800E+00	1.383862E+00	3.190176E-01	5.221733E-01	1.033617E+00	1.900E+01	4.712320E+00	1.247363E-01	4.503296E-02	4.502981E+01
2.000E+00	1.448012E+00	3.219669E-01	4.769315E-01	1.287867E+00	2.000E+01	4.835319E+00	1.213100E-01	4.277104E-02	4.852399E+01
2.200E+00	1.512471E+00	3.22537E-01	4.371452E-01	1.559708E+00	2.500E+01	5.405624E+00	1.076429E-01	3.422223E-02	6.727680E+01
2.400E+00	1.576784E+00	3.206059E-01	4.022123E-01	1.846690E+00	3.000E+01	5.918067E+00	9.781843E-02	2.855224E-02	8.803658E+01
2.600E+00	1.640622E+00	3.175786E-01	3.715206E-01	2.146831E+00	3.500E+01	6.387663E+00	9.032865E-02	2.450846E-02	1.106526E+02
2.800E+00	1.703752E+00	3.135895E-01	3.444984E-01	2.458542E+00	4.000E+01	6.823897E+00	8.437490E-02	2.147510E-02	1.349998E+02
3.000E+00	1.766015E+00	3.089504E-01	3.206353E-01	2.780554E+00	5.000E+01	7.620123E+00	7.339286E-02	1.911350E-02	1.609730E+02
3.200E+00	1.827304E+00	3.038919E-01	2.994872E-01	3.111853E+00	6.000E+01	8.339729E+00	6.883540E-02	1.722170E-02	1.884822E+02
3.400E+00	1.887555E+00	2.985838E-01	2.806731E-01	3.451629E+00	7.000E+01	9.001753E+00	6.376766E-02	1.437792E-02	2.478074E+02
3.600E+00	1.946730E+00	2.931506E-01	2.638692E-01	3.799232E+00	8.000E+01	9.618373E+00	5.969358E-02	1.080928E-02	3.820389E+02
3.800E+00	2.004813E+00	2.876826E-01	2.488011E-01	4.154137E+00	9.000E+01	1.019796E+01	5.632231E-02	9.615542E-03	4.562107E+02
4.000E+00	2.061805E+00	2.822448E-01	2.352366E-01	4.515917E+00	1.000E+02	1.074654E+01	5.347037E-02	8.658892E-03	5.347037E+02
4.200E+00	2.117716E+00	2.768835E-01	2.229793E-01	4.844223E+00	2.000E+02	1.186650E+01	3.796386E-02	4.335952E-03	1.518555E+03
4.400E+00	2.172565E+00	2.716309E-01	2.118628E-01	5.258774E+00	3.000E+02	1.860175E+01	3.103433E-02	2.889964E-03	2.793089E+03
4.600E+00	2.226377E+00	2.665088E-01	2.017449E-01	5.639326E+00	4.000E+02	2.148294E+01	2.688553E-02	2.166768E-03	4.301685E+03
4.800E+00	2.279179E+00	2.615315E-01	1.925056E-01	6.025683E+00	5.000E+02	2.402177E+01	2.404839E-02	1.732966E-03	6.012098E+03
5.000E+00	2.331000E+00	2.567073E-01	1.840414E-01	6.417684E+00	1.000E+03	3.398123E+01	1.698295E-02	8.660053E-04	1.699826E+04
5.500E+00	2.456470E+00	2.453380E-01	1.657208E-01	7.421475E+00	"	"	"	"	"
6.000E+00	2.576501E+00	2.349435E-01	1.506397E-01	8.457965E+00	"	"	"	"	"

ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	7.500E+00	3.621468E+00	3.429146E-01	2.761311E-01	1.928894E-01
1.000E-01	1.001665E+00	3.326683E-02	9.983378E-01	3.326683E-04	8.000E+00	3.791064E+00	3.335404E-01	2.637782E-01	2.147458E+01
2.000E-01	1.006640E+00	6.613861E-02	9.940368E-01	2.645544E-03	8.500E+00	3.957082E+00	3.386020E-01	2.527115E-01	2.374149E-01
3.000E-01	1.014867E+00	9.823953E-02	9.853508E-01	8.841558E-03	9.000E+00	4.119735E+00	3.220772E-01	2.427341E-01	2.608825E+01
4.000E-01	1.026251E+00	1.299302E-01	9.744205E-01	2.067684E-02	9.500E+00	4.279223E+00	3.159402E-01	2.336873E-01	2.851360E+01
5.000E-01	1.040666E+00	1.588211E-01	9.609228E-01	3.970529E-02	1.000E+01	4.435735E+00	3.101642E-01	2.254418E-01	3.101642E+01
6.000E-01	1.057961E+00	1.867810E-01	9.452146E-01	6.724117E-02	1.100E+01	4.740509E+00	2.995902E-01	2.109478E-01	3.625042E+01
7.000E-01	1.077962E+00	2.129405E-01	9.276761E-01	1.043408E-01	1.200E+01	5.321151E+00	2.817010E-01	1.879293E-01	4.178290E+01
8.000E-01	1.100485E+00	2.371909E-01	9.086901E-01	1.518022E-01	1.300E+01	5.598975E+00	2.740733E-01	1.786041E-01	5.371837E+01
9.000E-01	1.125335E+00	2.594797E-01	8.886241E-01	2.798032E-01	1.400E+01	5.869535E+00	2.671572E-01	1.703712E-01	6.011038E+01
1.000E+00	1.152316E+00	2.798032E-01	8.678178E-01	4.532144E-01	1.500E+01	6.391420E+00	2.550824E-01	1.564597E-01	6.677869E+01
1.400E+00	1.277737E+00	3.425990E-01	7.826336E-01	6.714940E-01	1.600E+01	6.434381E+00	2.608542E-01	1.630392E-01	6.677869E+01
1.600E+00	1.211893E+00	3.642824E-01	7.415535E-01	9.325630E-01	1.700E+01	6.394120E+00	2.50824E-01	1.564597E-01	7.371881E+01
1.800E+00	1.423101E+00	3.807344E-01	7.026906E-01	1.235380E-01	1.800E+01	6.643812E+00	2.497734E-01	1.505160E-01	8.092657E+01
2.000E+00	1.500526E+00	3.928667E-01	6.664328E-01	1.571467E-00	1.900E+01	6.891102E+00	2.448699E-01	1.451147E-01	8.839802E+01
2.200E+00	1.580015E+00	4.014929E-01	6.329055E-01	1.943226E-00	2.000E+01	7.133671E+00	2.403236E-01	1.401803E-01	9.612943E+01
2.400E+00	1.660936E+00	4.073088E-01	6.020700E-01	2.346099E-00	2.500E+01	8.2826250E+00	2.217092E-01	1.206819E-01	1.385682E+02
2.600E+00	1.742780E+00	4.108927E-01	5.737927E-01	2.777635E-00	3.000E+01	9.358555E+00	2.078228E-01	1.068541E-01	1.870406E+02
2.800E+00	1.825176E+00	4.127166E-01	5.478922E-01	3.235698E-00	3.500E+01	1.036942E+01	1.969257E-01	9.643740E-02	2.412340E+02
3.000E+00	1.907784E+00	4.136131E-01	5.241682E-01	3.718452E+00	4.000E+01	1.133117E+01	1.880550E-01	8.85215E-02	3.008880E+02
3.200E+00	1.990369E+00	4.125316E-01	5.024193E-01	4.224324E+00	4.500E+01	1.225237E+01	1.806324E-01	8.161684E-02	3.657806E+02
3.400E+00	2.072742E+00	4.110704E-01	4.824528E-01	4.751973E+00	5.000E+01	1.313928E+01	1.742879E-01	7.610765E-02	4.357198E+02
3.600E+00	2.154753E+00	4.089705E-01	4.640899E-01	5.300258E+00	5.500E+01	1.482810E+01	1.639122E-01	6.743953E-02	5.900839E+02
3.800E+00	2.236298E+00	4.063850E-01	4.471677E-01	5.868200E+00	6.000E+01	1.642462E+01	1.556826E-01	6.088420E-02	7.628445E+02
4.000E+00	2.317283E+00	4.034351E-01	4.315395E-01	6.454982E+00	7.000E+01	1.794661E+01	1.489189E-01	5.572084E-02	9.530807E+02
4.200E+00	2.397654E+00	4.002169E-01	4.021693E-01	7.059826E+00	8.000E+01	1.940651E+01	1.332131E-01	5.152910E-02	1.160026E+03
4.400E+00	2.477359E+00	3.968062E-01	3.932632E-01	7.682168E+00	9.000E+01	2.081351E+01	1.383030E-01	4.804571E-02	1.383030E+03
4.600E+00	2.556368E+00	3.926632E-01	3.896353E-01	8.321449E+00	1.000E+02	3.301158E+01	1.099706E-01	3.029240E-02	4.398825E+03
4.800E+00	2.634658E+00	3.896353E-01	3.795558E-01	8.977198E+00	2.000E+02	4.325512E+01	9.613470E-02	2.311865E-02	8.652123E+03
5.000E+00	2.712219E+00	3.859601E-01	3.687019E-01	9.649001E+00	3.000E+02	5.240237E+01	8.736697E-02	1.908310E-02	1.397872E+04
5.200E+00	2.790289E+00	3.767423E-01	3.444841E-01	1.139646E+01	4.000E+02	6.081090E+01	8.111227E-02	1.644442E-02	2.027807E+04
5.400E+00	2.868899E+00	3.677053E-01	3.237302E-01	1.323740E+01	5.000E+02	9.654558E+01	6.437779E-02	1.035780E-02	6.437779E+04
5.600E+00	2.947065E+00	3.590119E-01	3.057492E-01	1.516825E+01	1.000E+03	0.000000E+00	0.000000E+00	0.000000E+00	*
5.800E+00	3.025074E+00	3.507387E-01	2.900170E-01	1.718620E+01					
6.000E+00									
6.200E+00									
6.400E+00									
6.600E+00									
6.800E+00									
7.000E+00									
Sphere of polytropic index $n = -2$									
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	7.500E+00	3.621468E+00	3.429146E-01	2.761311E-01	1.928894E-01
1.000E-01	1.001665E+00	3.326683E-02	9.983378E-01	3.326683E-04	8.000E+00	3.791064E+00	3.335404E-01	2.637782E-01	2.147458E+01
2.000E-01	1.006640E+00	6.613861E-02	9.940368E-01	2.645544E-03	8.500E+00	3.957082E+00	3.386020E-01	2.527115E-01	2.374149E-01
3.000E-01	1.014867E+00	9.823953E-02	9.853508E-01	8.841558E-03	9.000E+00	4.119735E+00	3.220772E-01	2.427341E-01	2.608825E+01
4.000E-01	1.026251E+00	1.299302E-01	9.744205E-01	2.067684E-02	9.500E+00	4.279223E+00	3.159402E-01	2.336873E-01	2.851360E+01
5.000E-01	1.040666E+00	1.588211E-01	9.609228E-01	3.970529E-02	1.000E+01	4.435735E+00	3.101642E-01	2.254418E-01	3.101642E+01
6.000E-01	1.057961E+00	1.867810E-01	9.452146E-01	6.724117E-02	1.100E+01	4.740509E+00	2.995902E-01	2.109478E-01	3.625042E+01
7.000E-01	1.077962E+00	2.129405E-01	9.276761E-01	1.043408E-01	1.200E+01	5.321151E+00	2.817010E-01	1.879293E-01	4.178290E+01
8.000E-01	1.100485E+00	2.371909E-01	9.086901E-01	1.518022E-01	1.300E+01	5.598975E+00	2.740733E-01	1.786041E-01	5.371837E+01
9.000E-01	1.125335E+00	2.594797E-01	8.886241E-01	2.798032E-01	1.400E+01	5.869535E+00	2.671572E-01	1.703712E-01	6.011038E+01
1.000E+00	1.152316E+00	2.798032E-01	8.678178E-01	4.532144E-01	1.500E+01	6.391420E+00	2.550824E-01	1.564597E-01	6.677869E+01
1.400E+00	1.277737E+00	3.425990E-01	7.826336E-01	6.714940E-01	1.600E+01	6.434381E+00	2.608542E-01	1.630392E-01	6.677869E+01
1.600E+00	1.211893E+00	3.642824E-01	7.415535E-01	9.325630E-01	1.700E+01	6.394120E+00	2.50824E-01	1.564597E-01	7.371881E+01
1.800E+00	1.423101E+00	3.807344E-01	7.026906E-01	1.235380E-01	1.800E+01	6.643812E+00	2.497734E-01	1.505160E-01	8.092657E+01
2.000E+00	1.500526E+00	3.928667E-01	6.664328E-01	1.571467E-00	1.900E+01	6.891102E+00	2.448699E-01	1.451147E-01	8.839802E+01
2.200E+00	1.580015E+00	4.014929E-01	6.329055E-01	1.943226E-00	2.000E+01	7.133671E+00	2.403236E-01	1.401803E-01	9.612943E+01
2.400E+00	1.660936E+00	4.073088E-01	6.020700E-01	2.346099E-00	2.500E+01	8.2826250E+00	2.217092E-01	1.206819E-01	1.385682E+02
2.600E+00	1.742780E+00	4.108927E-01	5.737927E-01	2.777635E-00	3.000E+01	9.358555E+00	2.078228E-01	1.068541E-01	1.870406E+02
2.800E+00	1.825176E+00	4.127166E-01	5.478922E-01	3.235698E-00	3.500E+01	1.036942E+01	1.969257E-01	9.643740E-02	2.412340E+02
3.000E+00	1.907784E+00	4.136131E-01	5.241682E-01	3.718452E+00	4.000E+01	1.133117E+01	1.880550E-01	8.85215E-02	3.008880E+02
3.200E+00	1.990369E+00	4.125316E-01	5.024193E-01	4.224324E+00	4.500E+01	1.225237E+01	1.806324E-01	8.161684E-02	3.657806E+02
3.400E+00	2.072742E+00	4.110704E-01	4.824528E-01	4.751973E+00	5.000E+01	1.313928E+01	1.742879E-01	7.610765E-02	4.357198E+02
3.600E+00	2.154753E+00	4.089705E-01	4.640899E-01	5.300258E+00	5.500E+01	1.482810E+01	1.639122E-01	6.743953E-02	5.900839E+02
3.800E+00	2.236298E+00	4.063850E-01	4.471677E-01	5.868200E+00	6.000E+01	1.642462E+01	1.556826E-01	6.088420E-02	7.628445E+02
4.000E+00	2.317283E+00	4.034351E-01	4.315395E-01	6.454982E+00	7.000E+01	1.794661E+01	1.489189E-01	5.572084E-02	9.530807E+02
4.200E+00	2.397654E+00	4.002169E-01	4.021693E-01	7.059826E+00	8.000E+01	1.940651E+01	1.332131E-01	5.152910E-02	1.160026E+03
4.400E+00	2.477359E+00	3.968062E-01	3.932632E-01	7.682168E+00	9.000E+01	2.081351E+01	1.383030E-01	4.804571E-02	1.383030E+03
4.600E+00	2.556368E+00	3.926632E-01	3.896353E-01	8.321449E+00	1.000E+02	3.301158E+01	1.099706E-01	3.029240E-02	4.398825E+03
4.800E+00	2.634658E+00	3.896353E-01	3.795558E-01	8.977198E+00	2.000E+02	4.325512E+01	9.613470E-02	2.311865E-02	8.652123E+03
5.000E+00	2.712219E+00	3.859601E-01	3.687019E-01	9.649001E+00	3.000E+02	5.240237E+01	8.736697E-02	1.908310E-02	1.397872E+04
5.200E+00	2.790289E+00	3.767423E-01	3.444841E-01	1.139646E+01	4.000E+02	6.081090E+01	8.111227E-02	1.644442E-02	2.027807E+04
5.400E+00	2.868899E+00	3.677053E-01	3.237302E-01	1.323740E+01	5.000E+02	9.654558E+01	6.437779E-02	1.035780E-02	6.437779E+04
5.600E+00	2.947065E+00	3.590119E-01	3.057492E-01	1.516825E+01	1.000E+03	0.000000E+00	0.000000E+00	0.000000E+00	*
5.800E+00	3.025074E+00	3.507387E-01	2.900170E-01	1.718620E+01					
6.000E+00									
6.200E+00									
6.400E+00									
6.600E+00									
6.800E+00									
7.000E+00									

ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$
2.000E-01	1.006647E+00	6.626988E-02	9.966931E-01	2.650795E-03	1.000E+01	5.371255E+00	4.479722E-01	4.314814E-01	4.479722E+01
3.000E-01	1.014900E+00	9.867410E-02	9.926323E-01	8.880669E-03	1.100E+01	5.814722E+00	4.396864E-01	4.147014E-01	5.312945E+01
4.000E-01	1.026353E+00	1.302334E-01	9.870782E-01	2.083734E-02	1.200E+01	6.249670E+00	4.309261E-01	4.000106E-01	6.205335E+01
5.000E-01	1.040911E+00	1.607161E-01	9.801516E-01	4.017903E-02	1.300E+01	6.676797E+00	4.24331E-01	3.870044E-01	7.156019E+01
6.000E-01	1.058454E+00	1.899282E-01	9.719948E-01	6.837415E-02	1.400E+01	7.096737E+00	4.165423E-01	3.753796E-01	8.164229E+01
7.000E-01	1.078849E+00	2.177162E-01	9.627637E-01	1.066809E-01	1.500E+01	7.510061E+00	4.101898E-01	3.649037E-01	9.229271E+01
8.000E-01	1.101946E+00	2.439682E-01	9.526202E-01	1.561397E-01	1.600E+01	7.912777E+00	4.043170E-01	3.553956E-01	1.035052E+02
9.000E-01	1.127589E+00	2.686122E-01	9.417261E-01	2.175759E-01	1.700E+01	8.318837E+00	3.988713E-01	3.467119E-01	1.152738E+02
1.000E+00	1.155614E+00	2.916124E-01	9.302371E-01	2.916124E-01	1.800E+01	8.715146E+00	3.938063E-01	3.387370E-01	1.275933E+02
1.200E+00	1.218153E+00	3.326937E-01	9.060437E-01	4.790789E-01	1.900E+01	9.106563E+00	3.890814E-01	3.313773E-01	1.404584E+02
1.400E+00	1.28271E+00	3.674718E-01	8.810417E-01	7.202447E-01	2.000E+01	9.493410E+00	3.846611E-01	3.245554E-01	1.538644E+02
1.600E+00	1.364756E+00	3.964725E-01	8.559975E-01	1.014970E+00	2.500E+01	1.136827E+01	3.661713E-01	2.965875E-01	2.288571E+02
1.800E+00	1.446518E+00	4.203512E-01	8.314537E-01	1.361938E+00	3.000E+01	1.316221E+01	3.519702E-01	2.756358E-01	3.167732E+02
2.000E+00	1.532601E+00	4.397959E-01	8.077658E-01	1.759184E+00	3.500E+01	1.489261E+01	3.405769E-01	2.591281E-01	4.172067E+02
2.200E+00	1.622185E+00	4.554663E-01	7.851449E-01	2.204457E+00	4.000E+01	1.657121E+01	3.311341E-01	2.456535E-01	5.298146E+02
2.400E+00	1.714576E+00	4.679616E-01	7.636980E-01	2.695459E+00	4.500E+01	1.820631E+01	3.231113E-01	2.343630E-01	6.543003E+02
2.600E+00	1.809193E+00	4.778078E-01	7.434599E-01	3.229981E+00	5.000E+01	1.980410E+01	3.161615E-01	2.247100E-01	7.904037E+02
2.800E+00	1.905553E+00	4.854559E-01	7.244185E-01	3.805974E+00	6.000E+01	2.290569E+01	3.046006E-01	2.089432E-01	1.096562E+03
3.000E+00	2.003254E+00	4.912869E-01	7.065322E-01	4.421582E+00	7.000E+01	2.590341E+01	2.952453E-01	1.964814E-01	1.446702E+03
3.200E+00	2.101968E+00	4.956196E-01	6.897425E-01	5.075144E+00	8.000E+01	2.881567E+01	2.874227E-01	1.862883E-01	1.839505E+03
3.400E+00	2.201420E+00	4.987187E-01	6.739824E-01	5.765188E+00	9.000E+01	3.165559E+01	2.807228E-01	1.777357E-01	2.273854E+03
3.600E+00	2.301388E+00	5.008034E-01	6.591817E-01	6.490413E+00	1.000E+02	3.443297E+01	2.748781E-01	1.704169E-01	2.748781E+03
3.800E+00	2.401686E+00	5.020549E-01	6.452706E-01	7.249672E+00	2.000E-02	5.990676E+01	2.394765E-01	1.291999E-01	9.579061E+03
4.000E+00	2.502164E+00	5.026224E-01	6.321820E-01	8.041958E+00	3.000E-02	8.285142E+01	2.209103E-01	1.098626E-01	1.988193E+04
4.200E+00	2.602698E+00	5.026293E-01	6.198322E-01	8.866380E+00	4.000E-02	1.042915E+02	2.085949E-01	9.792094E-02	3.373519E+04
4.400E+00	2.703185E+00	5.021773E-01	6.082220E-01	9.722152E+00	5.000E-02	1.246764E+02	1.995063E-01	8.955874E-02	4.987657E+04
4.600E+00	2.803544E+00	5.013505E-01	5.972365E-01	1.060858E+01	1.000E+03	2.170989E+02	1.736905E-01	6.787038E-02	1.736905E+05
4.800E+00	2.903705E+00	5.002186E-01	5.868455E-01	1.152504E+01	∞	∞	0.000000E+00	0.000000E+00	∞
5.000E+00	3.003615E+00	4.988391E-01	5.770028E-01	1.247098E+01					
5.500E+00	3.252014E+00	4.946017E-01	5.545284E-01	1.496170E+01					
6.000E+00	3.498098E+00	4.896484E-01	5.346678E-01	1.762727E+01					
6.500E+00	3.741600E+00	4.843227E-01	5.169771E-01	2.046264E+01	0.00E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
7.000E+00	3.982396E+00	4.788520E-01	5.011039E-01	2.346375E+01	1.000E-01	1.001666E+00	3.329972E-02	9.99834E-01	3.329972E-04
7.500E+00	4.220451E+00	4.73741E-01	4.867664E-01	2.662729E+01	2.000E-01	1.006653E+00	6.639900E-02	9.99937E-01	2.655960E-03
8.000E+00	4.455784E+00	4.679766E-01	4.737377E-01	2.995050E+01	3.000E-01	1.014932E+00	9.910354E-02	9.998518E-01	8.91931E-03
8.500E+00	4.688450E+00	4.627131E-01	4.618334E-01	3.343102E+01	4.000E-01	1.026453E+00	1.312309E-01	9.997389E-01	2.09694E-02
9.000E+00	4.918525E+00	4.576150E-01	4.509024E-01	3.706682E+01	5.000E-01	1.041154E+00	1.626152E-01	9.995968E-01	4.065379E-02
9.500E+00	5.146095E+00	4.526990E-01	4.408198E-01	4.085608E+01	6.000E-01	1.058948E+00	1.831111E-01	9.994274E-01	6.951999E-02

Sphere of polytropic index $n = -1.01$

ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$
7.000E-01	1.079742E+00	2.225961E-01	9.992331E-01	1.090721E-01	1.500E-01	1.043347E+01	7.061796E-01	9.768226E-01	1.588904E+02
8.000E-01	1.103430E+00	2.509719E-01	9.990162E-01	1.606220E-01	1.600E-01	1.113938E+01	7.056257E-01	9.761833E-01	1.806402E+02
9.000E-01	1.129897E+00	2.781640E-01	9.987795E-01	2.253128E-01	1.700E-01	1.184711E+01	7.050482E-01	9.755842E-01	2.037589E+02
1.000E+00	1.159022E+00	3.041212E-01	9.985253E-01	3.041212E-01	1.800E-01	1.254947E+01	7.044661E-01	9.750205E-01	2.282470E+02
1.200E+00	1.224742E+00	3.522304E-01	9.979748E-01	5.072117E-01	1.900E-01	1.325365E+01	7.038916E-01	9.744883E-01	2.541049E+02
1.400E+00	1.299576E+00	3.952723E-01	9.973831E-01	7.747337E-01	2.000E-01	1.395726E+01	7.03324E-01	9.739844E-01	2.813330E+02
1.600E+00	1.382526E+00	4.334406E-01	9.967661E-01	1.109608E+00	2.500E-01	1.748751E+01	7.008666E-01	9.718018E-01	4.380417E+02
1.800E+00	1.472649E+00	4.670695E-01	9.961369E-01	1.513305E+00	3.000E-01	2.096687E+01	6.989602E-01	9.700289E-01	6.290642E+02
2.000E+00	1.569078E+00	4.965674E-01	9.955052E-01	1.986270E+00	3.500E-01	2.445785E+01	6.974972E-01	9.685361E-01	8.544300E+02
2.200E+00	1.671030E+00	5.223689E-01	9.948788E-01	2.528265E+00	4.000E-01	2.794238E+01	6.963574E-01	9.672469E-01	1.114172E+03
2.400E+00	1.777808E+00	5.449019E-01	9.942627E-01	3.138635E+00	4.500E-01	3.142182E+01	6.954513E-01	9.661125E-01	1.408289E+03
2.600E+00	1.888800E+00	5.645688E-01	9.936608E-01	3.816485E+00	5.000E-01	3.489717E+01	6.947161E-01	9.650995E-01	1.736774E+03
2.800E+00	2.003469E+00	5.817364E-01	9.930753E-01	4.560813E+00	6.000E-01	4.183841E+01	6.935960E-01	9.633503E-01	2.496946E+03
3.000E+00	2.121349E+00	5.967321E-01	9.925077E-01	5.370589E+00	7.000E-01	4.877009E+01	6.927782E-01	9.618746E-01	3.394613E+03
3.200E+00	2.242036E+00	6.098441E-01	9.919587E-01	6.244803E+00	8.000E-01	5.569460E+01	6.921478E-01	9.605984E-01	4.429746E+03
3.400E+00	2.365178E+00	6.213236E-01	9.914284E-01	7.182501E+00	9.000E+01	6.261345E+01	6.916404E-01	9.594742E-01	5.602287E+03
3.600E+00	2.490471E+00	6.313884E-01	9.909168E-01	8.182794E+00	1.000E-02	6.92768E+01	6.912177E-01	9.584698E-01	6.912177E+03
3.800E+00	2.617652E+00	6.402264E-01	9.904234E-01	9.244869E+00	2.000E-02	1.385128E+02	6.888298E-01	9.518864E-01	2.755319E+04
4.000E+00	2.746491E+00	6.479991E-01	9.899476E-01	1.036799E+01	3.000E-02	2.073283E+02	6.875273E-01	9.480548E-01	6.187746E+04
4.200E+00	2.876790E+00	6.548458E-01	9.894889E-01	1.155148E+01	4.000E-02	2.760302E+02	6.865933E-01	9.453452E-01	1.098549E+05
4.400E+00	3.008376E+00	6.608859E-01	9.890464E-01	1.279475E+01	5.000E-02	3.446515E+02	6.858592E-01	9.432487E-01	1.714648E+05
4.600E+00	3.141097E+00	6.662222E-01	9.886195E-01	1.409726E+01	1.000E-03	6.869356E+02	6.835353E-01	9.367654E-01	6.835353E+05
4.800E+00	3.274823E+00	6.709433E-01	9.882074E-01	1.545853E+01	*	*	0.000000E+00	0.000000E+00	*
5.000E+00	3.409439E+00	6.751255E-01	9.878094E-01	1.687814E+01	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
5.500E+00	3.749234E+00	6.836345E-01	9.868714E-01	2.067994E+01	1.000E-01	9.983326E-01	-3.336338E-02	9.998331E-01	-3.336338E-04
6.000E+00	4.092713E+00	6.899752E-01	9.860067E-01	2.483911E+01	2.000E-01	9.932113E-01	-6.690807E-02	9.99301E-01	-2.67623E-03
6.500E+00	4.438944E+00	6.947196E-01	9.852064E-01	2.935190E+01	3.000E-01	9.849387E-01	-1.008208E-01	9.984836E-01	-9.073868E-03
7.000E+00	4.787233E+00	6.982758E-01	9.844624E-01	3.421552E+01	4.000E-01	9.731383E-01	-1.332992E-01	9.972808E-01	-2.164787E-02
7.500E+00	5.137071E+00	7.009387E-01	9.837683E-01	3.942780E+01	5.000E-01	9.61414E-01	-1.705588E-01	9.957031E-01	-4.263971E-02
8.000E+00	5.488061E+00	7.029239E-01	9.831184E-01	4.498713E+01	6.000E-01	9.398995E-01	-2.068408E-01	9.937250E-01	-7.446271E-02
8.500E+00	5.839909E+00	7.043910E-01	9.825076E-01	5.082223E+01	7.000E-01	9.164414E-01	-2.444222E-01	9.913122E-01	-1.197669E-01
9.000E+00	6.192386E+00	7.054594E-01	9.819320E-01	5.714221E+01	8.000E-01	8.900539E-01	-2.836292E-01	9.894202E-01	-1.815227E-01
9.500E+00	6.545317E+00	7.062191E-01	9.813879E-01	6.373628E+01	9.000E-01	8.596484E-01	-3.248561E-01	9.849906E-01	-2.631335E-01
1.000E-01	6.898565E+00	7.067390E-01	9.808722E-01	7.067390E+01	1.000E+00	8.248592E-01	-3.685921E-01	9.809466E-01	-3.685921E-01
1.100E-01	7.605613E+00	7.072580E-01	9.799156E-01	8.557822E+01	1.000E+00	7.858258E-01	-4.154599E-01	9.761861E-01	-5.027065E-01
1.200E-01	8.312927E+00	7.073119E-01	9.790446E-01	1.018529E+02	1.000E+00	7.858258E-01	-4.154599E-01	9.761861E-01	-5.027065E-01
1.300E-01	9.020142E+00	7.070816E-01	9.782453E-01	1.194968E+02	1.000E+00	7.858258E-01	-4.154599E-01	9.761861E-01	-5.027065E-01
1.400E-01	9.727034E+00	7.066802E-01	9.775077E-01	1.385093E+02	1.000E+00	7.858258E-01	-4.154599E-01	9.761861E-01	-5.027065E-01

Sphere of polytropic index $n = -0.9$

ξ	θ	θ'	θ^{n+1}	$\xi^2 \theta'$	ξ	θ	θ'	θ^{n+1}	$\xi^2 \theta'$
1.200E+00	7.41759E-01	-4.662774E-01	9.705709E-01	-6.714394E-01	1.900E+00	3.208364E-01	-8.399335E-01	5.664242E-01	-3.032232E+00
1.300E+00	6.924021E-01	-5.221549E-01	9.639086E-01	-8.824419E-01	2.000E+00	2.320758E-01	-9.394628E-01	4.817423E-01	-3.757851E+00
1.400E+00	6.371247E-01	-5.846624E-01	9.559220E-01	-1.145938E+00	2.100E+00	1.317026E-01	-1.077598E+00	3.629086E-01	-4.752208E+00
1.500E+00	5.751723E-01	-6.561319E-01	9.461932E-01	-1.476297E+00	2.150E+00	7.539397E-02	-1.181146E+00	2.745893E-01	-5.459846E+00
1.600E+00	5.054792E-01	-7.402601E-01	9.340504E-01	-1.895066E+00	2.160E+00	6.345600E-02	-1.208042E+00	2.519047E-01	-5.636242E+00
1.700E+00	4.264911E-01	-8.434511E-01	9.183136E-01	-2.437574E+00	2.170E+00	5.122654E-02	-1.238556E+00	2.263232E-01	-5.832233E+00
1.800E+00	3.357478E-01	-9.783814E-01	8.966053E-01	-3.169956E+00	2.180E+00	3.866768E-02	-1.274276E+00	1.966410E-01	-6.055870E+00
1.900E+00	2.287725E-01	-1.176586E+00	8.628601E-01	-4.247475E+00	2.190E+00	2.571344E-02	-1.318410E+00	1.603541E-01	-6.323223E+00
2.000E+00	9.450434E-02	-1.572580E+00	7.898512E-01	-6.290319E+00	2.200E+00	1.224430E-02	-1.379633E+00	1.106540E-01	-6.678394E+00
2.020E+00	6.149834E-02	-1.740359E+00	7.566355E-01	-7.101362E+00	2.202E+00	8.683283E-03	-1.396543E+00	9.730533E-02	-6.771563E+00
2.030E+00	4.350610E-02	-1.865392E+00	7.308950E-01	-7.687095E+00	2.204E+00	6.656089E-03	-1.416345E+00	8.158488E-02	-6.880062E+00
2.040E+00	2.396453E-02	-2.061745E+00	6.885839E-01	-8.580159E+00	2.206E+00	3.799360E-03	-1.441671E+00	6.163895E-02	-7.015800E+00
2.048E+00	1.100168E-02	-2.286352E+00	6.370095E-01	-9.570934E+00	2.207E+00	2.349827E-03	-1.458516E+00	4.847293E-02	-7.104212E+00
2.048E+00	6.299133E-03	-2.428340E+00	6.024598E-01	-1.018515E+01	2.208E+00	8.800112E-04	-1.482764E+00	2.966498E-02	-7.228864E+00
2.050E+00	1.175205E-03	-2.780906E+00	5.093442E-01	-1.168676E+01	2.208E+00	0.000000E+00	-1.521468E+00	0.000000E+00	-7.421510E+00
2.05040073E+00	0.000000E+00	-4.232444E+00	0.000000E+00	-1.779380E+01	2.20856842E+00				

Sphere of polytropic index $n = -0.5$ Sphere of polytropic index $n = 0$

0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
1.000E-01	9.983329E-01	-3.335002E-02	9.991661E-01	-3.335002E-04	1.000E-01	9.983333E-01	-3.333333E-02	9.983333E-01	-3.333333E-04
2.000E-01	9.932266E-01	-6.680057E-02	9.966577E-01	-2.672023E-03	2.000E-01	9.933333E-01	-6.666667E-02	9.933333E-01	-2.666667E-03
3.000E-01	9.849660E-01	-1.004544E-01	9.924545E-01	-9.040895E-03	3.000E-01	9.850000E-01	-1.000000E-01	9.850000E-01	-9.000000E-03
4.000E-01	9.732254E-01	-1.344187E-01	9.865219E-01	-2.150699E-02	4.000E-01	9.733333E-01	-1.333333E-01	9.733333E-01	-2.133333E-02
5.000E-01	9.580681E-01	-1.688077E-01	9.788096E-01	-4.220193E-02	5.000E-01	9.583333E-01	-1.666667E-01	9.583333E-01	-4.166667E-02
6.000E-01	9.394456E-01	-2.037458E-01	9.692500E-01	-7.334849E-02	6.000E-01	9.400000E-01	-2.000000E-01	9.400000E-01	-7.200000E-02
7.000E-01	9.172961E-01	-2.393710E-01	9.577558E-01	-1.172918E-01	7.000E-01	9.183333E-01	-2.333333E-01	9.183333E-01	-1.143333E-01
8.000E-01	8.915433E-01	-2.758396E-01	9.442157E-01	-1.765373E-01	8.000E-01	8.933333E-01	-2.666667E-01	8.933333E-01	-1.706667E-01
9.000E-01	8.620941E-01	-3.133322E-01	9.284902E-01	-2.537991E-01	9.000E-01	8.650000E-01	-3.000000E-01	8.650000E-01	-2.430000E-01
1.000E+00	8.288357E-01	-3.520616E-01	9.104041E-01	-3.520616E-01	1.000E+00	8.333333E-01	-3.333333E-01	8.333333E-01	-3.333333E-01
1.100E+00	7.916321E-01	-3.922842E-01	8.897371E-01	-4.746639E-01	1.100E+00	7.983333E-01	-3.666667E-01	7.983333E-01	-4.436667E-01
1.200E+00	7.503187E-01	-4.343153E-01	8.662094E-01	-6.254143E-01	1.200E+00	7.600000E-01	-4.000000E-01	7.600000E-01	-5.760000E-01
1.300E+00	7.046956E-01	-4.785529E-01	8.394615E-01	-8.087541E-01	1.300E+00	7.183333E-01	-4.333333E-01	7.183333E-01	-7.323333E-01
1.400E+00	6.545175E-01	-5.255126E-01	8.090226E-01	-1.030005E+00	1.400E+00	6.733333E-01	-4.666667E-01	6.733333E-01	-9.146667E-01
1.500E+00	5.994796E-01	-5.758855E-01	7.742607E-01	-1.295742E+00	1.500E+00	6.250000E-01	-5.000000E-01	6.250000E-01	-1.125000E+00
1.600E+00	5.391951E-01	-6.306337E-01	7.342990E-01	-1.614422E+00	1.600E+00	5.733333E-01	-5.333333E-01	5.733333E-01	-1.365333E+00
1.700E+00	4.731609E-01	-6.911653E-01	6.878669E-01	-1.997468E+00	1.700E+00	5.183333E-01	-5.666667E-01	5.183333E-01	-1.637667E+00
1.800E+00	4.006975E-01	-7.596807E-01	6.330067E-01	-2.461366E+00	1.800E+00	4.600000E-01	-6.000000E-01	4.600000E-01	-1.944000E+00
					1.900E+00	3.983333E-01	-6.333333E-01	3.983333E-01	-2.286333E+00

ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$
2.000E+00	3.333333E-01	-6.666667E-01	3.333333E-01	-2.666667E+00	2.600E+00	7.904162E-02	-5.306457E-01	2.222203E-02	-3.587165E+00
2.100E+00	2.650000E-01	-7.000000E-01	2.650000E-01	-3.087000E+00	2.700E+00	2.674118E-02	-5.138861E-01	4.372913E-03	-3.746230E+00
2.200E+00	1.933333E-01	-7.333333E-01	1.933333E-01	-3.549333E+00	2.720E+00	1.650868E-02	-5.092768E-01	2.121136E-03	-3.767768E+00
2.300E+00	1.183333E-01	-7.666667E-01	1.183333E-01	-4.055667E+00	2.730E+00	1.42859E-02	-5.087196E-01	1.221769E-03	-3.776531E+00
2.400E+00	4.000000E-02	-8.000000E-01	4.000000E-02	-4.608000E+00	2.740E+00	6.374975E-03	-5.039642E-01	5.089999E-04	-3.783362E+00
2.420E+00	2.393333E-02	-8.066667E-01	2.393333E-02	-4.724163E+00	2.750E+00	1.350271E-03	-5.009125E-01	4.961712E-05	-3.788151E+00
2.430E+00	1.585000E-02	-8.100000E-01	1.585000E-02	-4.782969E+00	2.752E+00	3.491111E-04	-5.002421E-01	6.522971E-06	-3.788585E+00
2.440E+00	7.733333E-03	-8.133333E-01	7.733333E-03	-4.842261E+00	2.75269805E+00	0.000000E+00	-4.999971E-01	0.000000E+00	-3.788651E+00
2.44948974E+00	0.000000E+00	-8.164966E-01	0.000000E+00	-4.898979E+00	Sphere of polytropic index $n = 1$				
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
1.000E-01	9.983338E-01	-3.331666E-02	9.975017E-01	-3.331666E-04	1.000E-01	9.983342E-01	-3.330001E-02	9.986711E-01	-3.330001E-04
2.000E-01	9.933400E-01	-6.653327E-02	9.900267E-01	-2.661331E-03	2.000E-01	9.933467E-01	-6.640038E-02	9.867376E-01	-2.656015E-03
3.000E-01	9.850338E-01	-9.954952E-02	9.776349E-01	-8.959456E-03	3.000E-01	9.850674E-01	-9.910289E-02	9.703577E-01	-8.919260E-03
4.000E-01	9.734401E-01	-1.322646E-01	9.604239E-01	-2.116234E-02	4.000E-01	9.735459E-01	-1.312122E-01	9.477915E-01	-2.099394E-02
5.000E-01	9.585943E-01	-1.645770E-01	9.385388E-01	-4.114426E-02	5.000E-01	9.588511E-01	-1.625370E-01	9.193954E-01	-4.063426E-02
6.000E-01	9.405416E-01	-1.963843E-01	9.121515E-01	-7.069833E-02	6.000E-01	9.410708E-01	-1.928920E-01	8.856142E-01	-6.94410E-02
7.000E-01	9.193377E-01	-2.275824E-01	8.814803E-01	-1.115154E-01	7.000E-01	9.203110E-01	-2.220983E-01	8.469723E-01	-1.088282E-01
8.000E-01	8.950489E-01	-2.580659E-01	8.467791E-01	-1.651622E-01	8.000E-01	8.96951E-01	-2.499855E-01	8.040621E-01	-1.599907E-01
9.000E-01	8.677519E-01	-2.877272E-01	8.083387E-01	-2.330591E-01	9.000E-01	8.703632E-01	-2.763925E-01	7.575322E-01	-2.238779E-01
1.000E+00	8.375345E-01	-3.164564E-01	7.664857E-01	-3.164564E-01	1.000E+00	8.414710E-01	-3.011687E-01	7.080734E-01	-3.011687E-01
1.100E+00	8.044955E-01	-3.441403E-01	7.215816E-01	-4.164097E-01	1.200E+00	8.101885E-01	-3.241749E-01	6.564054E-01	-3.922516E-01
1.200E+00	7.687452E-01	-3.706623E-01	6.740213E-01	-5.337536E-01	1.300E+00	7.766992E-01	-3.432846E-01	6.032617E-01	-4.972098E-01
1.300E+00	7.304058E-01	-3.959013E-01	6.242324E-01	-6.690731E-01	1.400E+00	7.411986E-01	-3.643844E-01	5.493754E-01	-6.158097E-01
1.400E+00	6.896119E-01	-4.197309E-01	5.726736E-01	-8.226726E-01	1.500E+00	7.038927E-01	-3.813754E-01	4.954649E-01	-7.474957E-01
1.500E+00	6.465110E-01	-4.420184E-01	5.198331E-01	-9.945413E-01	1.600E+00	6.649967E-01	-3.961730E-01	4.422206E-01	-8.913892E-01
1.600E+00	6.012643E-01	-4.626227E-01	4.662278E-01	-1.184314E+00	1.700E+00	6.247335E-01	-4.087081E-01	3.902919E-01	-1.046293E+00
1.700E+00	5.540476E-01	-4.813930E-01	4.124019E-01	-1.391226E+00	1.800E+00	5.833322E-01	-4.189275E-01	3.402765E-01	-1.210700E+00
1.800E+00	5.050523E-01	-4.981655E-01	3.589257E-01	-1.614056E+00	1.900E+00	5.410265E-01	-4.267936E-01	2.927096E-01	-1.382811E+00
1.900E+00	4.544871E-01	-5.127604E-01	3.063934E-01	-1.851065E+00	2.000E+00	4.946487E-01	-4.33978E-01	2.480565E-01	-1.560550E+00
2.000E+00	4.025795E-01	-5.249758E-01	2.554333E-01	-2.099903E+00	2.100E+00	4.510521E-01	-4.361420E-01	1.689638E-01	-1.741591E+00
2.100E+00	3.495789E-01	-5.345805E-01	2.068892E-01	-2.357500E+00	2.200E+00	4.074984E-01	-4.345452E-01	1.350550E-01	-2.103199E+00
2.200E+00	2.957595E-01	-5.413018E-01	1.608452E-01	-2.619901E+00	2.300E+00	3.674984E-01	-4.306503E-01	1.051184E-01	-2.278140E+00
2.300E+00	2.412457E-01	-5.448049E-01	1.186247E-01	-2.882018E+00	2.400E+00	3.242197E-01	-4.245153E-01	7.921016E-02	-2.445208E+00
2.400E+00	1.869200E-01	-5.446574E-01	8.081346E-02	-3.137227E+00	2.500E+00	2.8144430E-01	-4.162130E-01	5.730703E-02	-2.601331E+00
2.500E+00	1.326357E-01	-5.405256E-01	4.830484E-02	-3.376597E+00	2.600E+00	2.393889E-01	-4.058302E-01	3.931090E-02	-2.743412E+00

ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$
2.700E+00	1.582888E-01	-3.934670E-01	2.505536E-02	-2.868375E+00	2.500E+00	3.158926E-01	-3.413414E-01	5.608524E-02	-2.133383E+00
2.800E+00	1.196386E-01	-3.792361E-01	1.431340E-02	-2.973211E+00	2.600E+00	2.822524E-01	-3.313061E-01	4.232472E-02	-2.239629E+00
2.900E+00	8.249977E-02	-3.632614E-01	6.806212E-03	-3.05028E+00	2.700E+00	2.496598E-01	-3.204174E-01	3.114380E-02	-2.335843E+00
3.000E+00	4.704000E-02	-3.456775E-01	2.212762E-03	-3.111097E+00	2.800E+00	2.181919E-01	-3.08400E-01	2.223804E-02	-2.421306E+00
3.100E+00	1.341312E-02	-3.266285E-01	1.799117E-04	-3.138900E+00	2.900E+00	1.879094E-01	-2.967337E-01	1.530634E-02	-2.495531E+00
3.120E+00	6.920185E-03	-3.226561E-01	4.788895E-05	-3.140864E+00	3.000E+00	1.588576E-01	-2.842527E-01	1.005820E-02	-2.558275E+00
3.130E+00	3.703640E-03	-3.206506E-01	1.371695E-05	-3.141382E+00	3.100E+00	1.310664E-01	-2.715447E-01	6.219118E-03	-2.609544E+00
3.140E+00	5.072143E-04	-3.186325E-01	2.572663E-07	-3.141589E+00	3.200E+00	1.045515E-01	-2.575707E-01	3.534484E-03	-2.649608E+00
3.141E+00	1.886831E-04	-3.184300E-01	3.560130E-08	-3.141592E+00	3.300E+00	7.931464E-02	-2.450063E-01	1.771672E-03	-2.679009E+00
3.14159265E+00	0.000000E+00	-3.183099E-01	0.000000E+00	-3.141593E+00	3.400E+00	5.534424E-02	-2.334426E-01	7.205782E-04	-2.698597E+00
Sphere of polytropic index $n = 1.5$									
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	3.500E+00	3.261573E-02	-2.211909E-01	1.921179E-04	-2.709588E+00
1.000E-01	9.983346E-01	-3.328337E-02	9.958417E-01	-3.328337E-04	3.600E+00	1.109099E-02	-2.093927E-01	1.295467E-05	-2.713729E+00
2.000E-01	9.935333E-01	-6.626800E-02	9.834660E-01	-2.650720E-03	3.620E+00	9.626088E-03	-2.071024E-01	3.992270E-06	-2.713953E+00
3.000E-01	9.851007E-01	-9.866007E-02	9.631671E-01	-8.879406E-03	3.640E+00	4.860744E-03	-2.059675E-01	1.647340E-06	-2.714013E+00
4.000E-01	9.736505E-01	-1.301756E-01	9.354223E-01	-2.082809E-02	3.650E+00	7.639242E-04	-2.037196E-01	1.612968E-08	-2.714055E+00
5.000E-01	9.591039E-01	-1.605449E-01	9.008741E-01	-4.013622E-02	3.652E+00	3.567080E-04	-2.034966E-01	2.403157E-09	-2.714055E+00
6.000E-01	9.415881E-01	-1.895169E-01	8.603050E-01	-6.822610E-02	3.653E+00	1.532672E-04	-2.038852E-01	2.908190E-10	-2.714055E+00
7.000E-01	9.212547E-01	-2.168630E-01	8.146092E-01	-1.062629E-01	3.65375374E+00	0.000000E+00	-2.033013E-01	0.000000E+00	-2.714055E+00
8.000E-01	8.982765E-01	-2.423798E-01	7.647600E-01	-1.551231E-01	Sphere of polytropic index $n = 2$				
9.000E-01	8.728456E-01	-2.658923E-01	7.117763E-01	-2.153728E-01	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
1.000E+00	8.451698E-01	-2.872555E-01	6.566892E-01	-2.872555E-01	1.000E-01	9.983350E-01	-3.326675E-02	9.950133E-01	-3.326675E-04
1.100E+00	8.154699E-01	-3.063557E-01	6.005095E-01	-3.706904E-01	2.000E-01	9.933599E-01	-6.613611E-02	9.802117E-01	-2.645445E-03
1.200E+00	7.839768E-01	-3.231109E-01	5.441994E-01	-4.652797E-01	3.000E-01	9.851339E-01	-9.822101E-02	9.560616E-01	-8.839891E-03
1.300E+00	7.509276E-01	-3.374711E-01	4.886470E-01	-5.703261E-01	4.000E-01	9.737541E-01	-1.291545E-01	9.233108E-01	-2.066473E-02
1.400E+00	7.165631E-01	-3.494173E-01	4.346464E-01	-6.848578E-01	5.000E-01	9.533527E-01	-1.585990E-01	8.829476E-01	-3.964974E-02
1.500E+00	6.811243E-01	-3.589602E-01	3.828829E-01	-8.076604E-01	6.000E-01	9.420940E-01	-1.862534E-01	8.361472E-01	-6.705123E-02
1.600E+00	6.448499E-01	-3.661387E-01	3.339232E-01	-9.373150E-01	7.000E-01	9.221703E-01	-2.118600E-01	7.842120E-01	-1.038114E-01
1.700E+00	6.079733E-01	-3.710173E-01	2.882115E-01	-1.072240E+00	8.000E-01	8.997974E-01	-2.352083E-01	7.285077E-01	-1.505333E-01
1.800E+00	5.707202E-01	-3.736839E-01	2.460697E-01	-1.2120736E+00	9.000E-01	8.752094E-01	-2.561379E-01	6.704029E-01	-2.074717E-01
1.900E+00	5.333066E-01	-3.742469E-01	2.077029E-01	-1.351031E+00	1.000E+00	8.486541E-01	-2.745394E-01	6.112124E-01	-2.745394E-01
2.000E+00	4.959368E-01	-3.728321E-01	1.732071E-01	-1.491329E+00	1.100E+00	8.203878E-01	-2.903532E-01	5.521506E-01	-3.513274E-01
2.100E+00	4.588015E-01	-3.695799E-01	1.425811E-01	-1.629847E+00	1.200E+00	7.906702E-01	-3.035673E-01	4.942948E-01	-4.371369E-01
2.200E+00	4.220770E-01	-3.646419E-01	1.157389E-01	-1.764867E+00	1.300E+00	7.597600E-01	-3.142135E-01	4.385602E-01	-5.310208E-01
2.300E+00	3.859239E-01	-3.581784E-01	9.252398E-02	-1.894764E+00	1.400E+00	7.279107E-01	-3.223630E-01	3.856865E-01	-6.318314E-01
2.400E+00	3.504866E-01	-3.503553E-01	7.272415E-02	-2.018046E+00	1.500E+00	6.953671E-01	-3.281207E-01	3.362347E-01	-7.382715E-01

ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$
Sphere of polytropic index $n = 2.5$									
1.600E+00	6.623619E-01	-3.316197E-01	2.905936E-01	-8.489463E-01	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
1.700E+00	6.291133E-01	-3.330153E-01	2.489927E-01	-9.624142E-01	1.000E-01	9.983354E-01	-3.325015E-02	9.941861E-01	-3.325015E-04
1.800E+00	5.95233E-01	-3.324793E-01	2.115205E-01	-1.077233E+00	2.000E-01	9.93665E-01	-6.00473E-02	9.769747E-01	-2.640189E-03
1.900E+00	5.625758E-01	-3.301943E-01	1.781454E-01	-1.192001E+00	3.000E-01	9.851670E-01	-9.778566E-02	9.490398E-01	-8.800710E-03
2.000E+00	5.298364E-01	-3.263489E-01	1.487392E-01	-1.416196E+00	4.000E-01	9.738567E-01	-1.281487E-01	9.114497E-01	-2.050379E-02
2.100E+00	4.974517E-01	-3.211330E-01	1.230985E-01	-1.416196E+00	5.000E-01	9.595978E-01	-1.566977E-01	8.655901E-01	-3.917443E-02
2.200E+00	4.656493E-01	-3.147340E-01	1.009664E-01	-1.523313E+00	6.000E-01	9.425889E-01	-1.830960E-01	8.130704E-01	-6.591456E-02
2.300E+00	4.345383E-01	-3.073338E-01	8.205104E-02	-1.625796E+00	7.000E-01	9.230593E-01	-2.070743E-01	7.556203E-01	-1.014664E-01
2.400E+00	4.042101E-01	-2.991056E-01	6.604219E-02	-1.722848E+00	8.000E-01	9.012614E-01	-2.284349E-01	6.949886E-01	-1.461984E-01
2.500E+00	3.747393E-01	-2.902125E-01	5.262447E-02	-1.813828E+00	9.000E-01	8.774638E-01	-2.470527E-01	6.328522E-01	-2.001127E-01
2.600E+00	3.461847E-01	-2.808059E-01	4.148811E-02	-1.898248E+00	1.000E+00	8.519442E-01	-2.628722E-01	5.707409E-01	-2.628722E-01
2.700E+00	3.185906E-01	-2.710244E-01	3.233694E-02	-1.975768E+00	1.100E+00	8.249824E-01	-2.759025E-01	5.099838E-01	-3.338420E-01
2.800E+00	2.919881E-01	-2.609938E-01	2.489405E-02	-2.046191E+00	1.200E+00	7.968545E-01	-2.862099E-01	4.516756E-01	-4.121423E-01
2.900E+00	2.663964E-01	-2.508263E-01	1.890536E-02	-2.109449E+00	1.300E+00	7.678274E-01	-2.939092E-01	3.966639E-01	-4.967066E-01
3.000E+00	2.418241E-01	-2.406215E-01	1.414160E-02	-2.165593E+00	1.400E+00	7.381545E-01	-2.991542E-01	3.45537E-01	-5.863422E-01
3.100E+00	2.182704E-01	-2.304662E-01	1.039884E-02	-2.214780E+00	1.500E+00	7.080723E-01	-3.021280E-01	2.987250E-01	-6.797880E-01
3.200E+00	1.957267E-01	-2.204354E-01	7.498081E-03	-2.257258E+00	1.600E+00	6.777979E-01	-3.030340E-01	2.563602E-01	-7.757671E-01
3.300E+00	1.741771E-01	-2.105930E-01	5.284124E-03	-2.293358E+00	1.700E+00	6.475273E-01	-3.020874E-01	2.184757E-01	-8.730327E-01
3.400E+00	1.536000E-01	-2.009927E-01	3.623879E-03	-2.323476E+00	1.800E+00	6.174348E-01	-2.995079E-01	1.849562E-01	-9.704055E-01
3.500E+00	1.339690E-01	-1.918787E-01	2.404433E-03	-2.348064E+00	1.900E+00	5.876729E-01	-2.955130E-01	1.555875E-01	-1.066802E+00
3.600E+00	1.152535E-01	-1.828867E-01	1.530956E-03	-2.367620E+00	2.000E+00	5.583723E-01	-2.903137E-01	1.300869E-01	-1.161255E+00
3.700E+00	9.741994E-02	-1.740451E-01	9.245782E-04	-2.382677E+00	2.100E+00	5.296436E-01	-2.841099E-01	1.081292E-01	-1.252925E+00
3.800E+00	8.043209E-02	-1.657754E-01	5.203411E-04	-2.393796E+00	2.200E+00	5.015776E-01	-2.770881E-01	8.936830E-02	-1.341107E+00
3.900E+00	6.425194E-02	-1.578933E-01	2.652521E-04	-2.401557E+00	2.300E+00	4.742475E-01	-2.740195E-01	7.345438E-02	-1.425229E+00
4.000E+00	4.884015E-02	-1.504097E-01	1.165014E-04	-2.406555E+00	2.400E+00	4.477101E-01	-2.612586E-01	6.004670E-02	-1.504849E+00
4.100E+00	3.415651E-02	-1.433309E-01	3.984929E-05	-2.409393E+00	2.500E+00	4.220076E-01	-2.527435E-01	4.882260E-02	-1.579847E+00
4.200E+00	2.016037E-02	-1.365599E-01	8.193995E-06	-2.410681E+00	2.600E+00	3.971691E-01	-2.439957E-01	3.948337E-02	-1.649411E+00
4.300E+00	6.810943E-03	-1.303965E-01	3.159525E-07	-2.411031E+00	2.700E+00	3.732127E-01	-2.351207E-01	3.175757E-02	-1.714030E+00
4.320E+00	4.215080E-03	-1.291923E-01	7.488892E-08	-2.411042E+00	2.800E+00	3.501462E-01	-2.262093E-01	2.540229E-02	-1.773481E+00
4.330E+00	2.926138E-03	-1.285966E-01	2.505443E-08	-2.411045E+00	2.900E+00	3.279695E-01	-2.173380E-01	2.020305E-02	-1.827813E+00
4.340E+00	1.643133E-03	-1.280047E-01	4.436287E-09	-2.411046E+00	3.000E+00	3.066751E-01	-2.085711E-01	1.597257E-02	-1.877139E+00
4.350E+00	3.660302E-04	-1.274169E-01	4.904003E-11	-2.411046E+00	3.100E+00	2.862500E-01	-1.999609E-01	1.254900E-02	-1.921624E+00
4.352E+00	1.113135E-04	-1.272998E-01	1.379252E-12	-2.411046E+00	3.200E+00	2.666763E-01	-1.915501E-01	9.793666E-03	-1.961473E+00
4.352E+00	0.000000E+00	-1.272487E-01	0.000000E+00	-2.411046E+00	3.300E+00	2.479322E-01	-1.833719E-01	7.588665E-03	-1.996920E+00
4.35287460E+00					3.400E+00	2.299933E-01	-1.754521E-01	5.834490E-03	-2.028226E+00

ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$
3.500E+00	2.12326E-01	-1.67809E-01	4.47677E-03	-2.05566E+00	9.000E-01	8.786172E-01	-2.385703E-01	5.986525E-01	-1.932419E-01
3.600E+00	1.964217E-01	-1.604568E-01	3.358638E-03	-2.079521E+00	1.000E+00	8.550576E-01	-2.521293E-01	5.345415E-01	-2.521293E-01
3.700E+00	1.807312E-01	-1.534026E-01	2.509669E-03	-2.100082E+00	1.100E+00	8.292881E-01	-2.67879E-01	4.729570E-01	-3.179734E-01
3.800E+00	1.657311E-01	-1.466507E-01	1.853165E-03	-2.117637E+00	1.200E+00	8.052919E-01	-2.706901E-01	4.149341E-01	-3.897937E-01
3.900E+00	1.513910E-01	-1.402017E-01	1.350051E-03	-2.132469E+00	1.300E+00	7.752356E-01	-2.760262E-01	3.611893E-01	-4.664843E-01
4.000E+00	1.376807E-01	-1.340534E-01	9.684029E-04	-2.144855E+00	1.400E+00	7.474648E-01	-2.790192E-01	3.121498E-01	-5.468776E-01
4.100E+00	1.245704E-01	-1.282013E-01	6.822611E-04	-2.155064E+00	1.500E+00	7.195018E-01	-2.799117E-01	2.679955E-01	-6.298013E-01
4.200E+00	1.120308E-01	-1.226390E-01	4.706317E-04	-2.163352E+00	1.600E+00	6.915441E-01	-2.789950E-01	2.287070E-01	-7.141247E-01
4.300E+00	1.000332E-01	-1.173588E-01	3.165958E-04	-2.169964E+00	1.700E+00	6.637641E-01	-2.763994E-01	1.941132E-01	-7.987943E-01
4.400E+00	8.854994E-02	-1.123518E-01	2.066137E-04	-2.175131E+00	1.800E+00	6.363094E-01	-2.724872E-01	1.639356E-01	-8.828583E-01
4.500E+00	7.755409E-02	-1.076083E-01	1.299022E-04	-2.179068E+00	1.900E+00	6.093043E-01	-2.674473E-01	1.378279E-01	-9.654842E-01
4.600E+00	6.701984E-02	-1.031181E-01	7.793128E-05	-2.181979E+00	2.000E+00	5.828505E-01	-2.614909E-01	1.154061E-01	-1.045964E+00
4.700E+00	5.692239E-02	-9.887048E-02	4.400387E-05	-2.184049E+00	2.100E+00	5.570301E-01	-2.548102E-01	9.627523E-02	-1.123713E+00
4.800E+00	4.723802E-02	-9.485453E-02	2.290978E-05	-2.185448E+00	2.200E+00	5.319069E-01	-2.475767E-01	8.004654E-02	-1.198271E+00
4.900E+00	3.794412E-02	-9.105924E-02	1.064157E-05	-2.186332E+00	2.300E+00	5.075282E-01	-2.399411E-01	6.634995E-02	-1.269289E+00
5.000E+00	2.901919E-02	-8.747353E-02	4.162922E-06	-2.186838E+00	2.400E+00	4.839277E-01	-2.320346E-01	5.484311E-02	-1.336519E+00
5.100E+00	2.042480E-02	-8.408638E-02	1.221494E-06	-2.187087E+00	2.500E+00	4.611266E-01	-2.239889E-01	4.521480E-02	-1.399806E+00
5.200E+00	1.219566E-02	-8.088683E-02	2.003172E-07	-2.187180E+00	2.600E+00	4.391359E-01	-2.158389E-01	3.718740E-02	-1.459071E+00
5.300E+00	4.259544E-03	-7.786397E-02	5.043949E-09	-2.187199E+00	2.700E+00	4.179583E-01	-2.077234E-01	3.051631E-02	-1.514303E+00
5.320E+00	2.708118E-03	-7.727965E-02	1.033562E-09	-2.187199E+00	2.800E+00	3.975887E-01	-1.996870E-01	2.498826E-02	-1.565546E+00
5.340E+00	1.168314E-03	-7.670186E-02	5.450780E-11	-2.187200E+00	2.900E+00	3.780165E-01	-1.917820E-01	2.041940E-02	-1.612887E+00
5.350E+00	4.027292E-04	-7.641539E-02	1.310828E-12	-2.187200E+00	3.000E+00	3.592265E-01	-1.840499E-01	1.665227E-02	-1.656449E+00
5.352E+00	2.499555E-04	-7.635829E-02	2.468990E-13	-2.187200E+00	3.100E+00	3.411997E-01	-1.765225E-01	1.355297E-02	-1.696381E+00
5.354E+00	9.729594E-05	-7.630125E-02	9.085139E-15	-2.187200E+00	3.200E+00	3.239144E-01	-1.692239E-01	1.100632E-02	-1.732855E+00
5.355E+00	2.100894E-05	-7.627276E-02	4.250249E-17	-2.187200E+00	3.300E+00	3.073467E-01	-1.621713E-01	8.923070E-03	-1.766045E+00
5.35527546E+00	0.000000E+00	-7.626491E-02	0.000000E+00	-2.187200E+00	3.400E+00	2.914715E-01	-1.553763E-01	7.217462E-03	-1.796150E+00
					3.500E+00	2.762626E-01	-1.488458E-01	5.824902E-03	-1.823361E+00
					3.600E+00	2.616934E-01	-1.425827E-01	4.689984E-03	-1.847877E+00
					3.700E+00	2.477372E-01	-1.365871E-01	3.766732E-03	-1.869877E+00
					3.800E+00	2.343672E-01	-1.308562E-01	3.017084E-03	-1.889564E+00
					3.900E+00	2.215573E-01	-1.253856E-01	2.409594E-03	-1.907115E+00
					4.000E+00	2.092818E-01	-1.201691E-01	1.918334E-03	-1.922705E+00
					4.100E+00	1.975152E-01	-1.151994E-01	1.521956E-03	-1.936501E+00
					4.200E+00	1.862338E-01	-1.104684E-01	1.202912E-03	-1.948662E+00
					4.300E+00	1.754139E-01	-1.059674E-01	9.467945E-04	-1.959337E+00
					4.400E+00	1.650329E-01	-1.016874E-01	7.417926E-04	-1.968667E+00
					4.500E+00	1.550693E-01	-9.761895E-02	5.782342E-04	-1.976784E+00
					4.600E+00	1.450524E-01	-9.375278E-02	4.482091E-04	-1.983809E+00
Sphere of polytropic index $n = 3$									
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00	0.000E+00	2.12326E-01	-1.67809E-01	4.47677E-03	-2.05566E+00
1.000E-01	9.983358E-01	-3.323356E-02	9.933599E-02	-3.323356E-04	1.000E-01	1.964217E-01	-1.604568E-01	3.358638E-03	-2.079521E+00
2.000E-01	9.937371E-01	-6.587385E-02	9.737547E-01	-2.634954E-03	2.000E-01	1.807312E-01	-1.534026E-01	2.509669E-03	-2.100082E+00
3.000E-01	9.851998E-01	-9.735399E-02	9.421005E-01	-8.761859E-03	3.000E-01	1.657311E-01	-1.466507E-01	1.853165E-03	-2.117637E+00
4.000E-01	9.739583E-01	-1.271577E-01	8.998319E-01	-2.034523E-02	3.900E+00	1.513910E-01	-1.402017E-01	1.350051E-03	-2.132469E+00
5.000E-01	9.598391E-01	-1.548396E-01	8.487772E-01	-3.870989E-02	4.800E+00	1.376807E-01	-1.340534E-01	9.684029E-04	-2.144855E+00
6.000E-01	9.430732E-01	-1.800396E-01	7.910093E-01	-6.481427E-02	5.700E+00	1.245704E-01	-1.282013E-01	6.822611E-04	-2.155064E+00
7.000E-01	9.239228E-01	-2.024921E-01	7.286900E-01	-9.922112E-02	6.600E+00	1.120308E-01	-1.226390E-01	4.706317E-04	-2.163352E+00
8.000E-01	9.025721E-01	-2.220277E-01	6.639266E-01	-1.420977E-01					

ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$
4.700E+00	1.363124E-01	-9.007952E-02	3.452558E-04	-1.989857E+00	6.000E-01	9.435472E-01	-1.770796E-01	7.699035E-01	-6.374865E-02
4.800E+00	1.274804E-01	-8.658996E-02	2.641030E-04	-1.995033E+00	7.000E-01	9.247621E-01	-1.981008E-01	7.032912E-01	-9.706937E-02
4.900E+00	1.189885E-01	-8.327506E-02	2.004567E-04	-1.999434E+00	8.000E-01	8.816262E-01	-2.159576E-01	6.350801E-01	-1.382129E-01
5.000E+00	1.108198E-01	-8.012604E-02	1.508238E-04	-2.003151E+00	9.000E-01	8.46768E-01	-2.306327E-01	5.674039E-01	-1.868125E-01
5.100E+00	1.029581E-01	-7.713439E-02	1.123678E-04	-2.006265E+00	1.000E+00	8.58096E-01	-2.422051E-01	5.020126E-01	-2.422051E-01
5.200E+00	9.53799E-02	-7.429188E-02	8.278940E-05	-2.008852E+00	1.200E+00	8.079342E-01	-2.567299E-01	3.829940E-01	-3.696910E-01
5.300E+00	8.809501E-02	-7.159064E-02	6.022893E-05	-2.010981E+00	1.400E+00	7.559767E-01	-2.613812E-01	2.839800E-01	-5.123072E-01
5.400E+00	8.106540E-02	-6.902309E-02	4.318592E-05	-2.014713E+00	1.600E+00	7.038915E-01	-2.583838E-01	2.059366E-01	-6.614625E-01
5.500E+00	7.428617E-02	-6.658201E-02	3.045313E-05	-2.021406E+00	1.800E+00	6.529871E-01	-2.499176E-01	1.469165E-01	-8.097329E-01
5.600E+00	6.774502E-02	-6.426050E-02	2.106248E-05	-2.015209E+00	2.000E+00	6.041619E-01	-2.378757E-01	1.035595E-01	-9.515027E-01
5.700E+00	6.143031E-02	-6.205202E-02	1.424068E-05	-2.016070E+00	2.200E+00	5.579736E-01	-2.237691E-01	7.240379E-02	-1.083042E+00
5.800E+00	5.533105E-02	-5.995032E-02	9.372938E-06	-2.016729E+00	2.400E+00	5.147160E-01	-2.087264E-01	5.035636E-02	-1.202264E+00
5.900E+00	4.943688E-02	-5.794947E-02	5.973161E-06	-2.017221E+00	2.600E+00	4.744918E-01	-1.935431E-01	3.491637E-02	-1.308351E+00
6.000E+00	4.373798E-02	-5.604388E-02	3.659612E-06	-2.017580E+00	2.800E+00	4.372724E-01	-1.78748E-01	2.417599E-02	-1.401385E+00
6.100E+00	3.822511E-02	-5.422823E-02	2.134985E-06	-2.017832E+00	3.000E+00	4.029445E-01	-1.646693E-01	1.673414E-02	-1.482024E+00
6.200E+00	3.288951E-02	-5.249747E-02	1.170118E-06	-2.018003E+00	3.200E+00	3.713445E-01	-1.514900E-01	1.158768E-02	-1.551258E+00
6.300E+00	2.772294E-02	-5.084687E-02	5.906867E-07	-2.018112E+00	3.400E+00	3.422828E-01	-1.392935E-01	8.030321E-03	-1.610233E+00
6.400E+00	2.271761E-02	-4.927190E-02	2.663489E-07	-2.018177E+00	3.600E+00	3.155603E-01	-1.280968E-01	5.570201E-03	-1.660134E+00
6.500E+00	1.786618E-02	-4.776834E-02	1.018889E-07	-2.018212E+00	3.800E+00	2.909791E-01	-1.178748E-01	3.867033E-03	-1.702112E+00
6.600E+00	1.316170E-02	-4.63215E-02	3.000876E-08	-2.018229E+00	4.000E+00	2.683488E-01	-1.085774E-01	2.686261E-03	-1.737238E+00
6.700E+00	8.597630E-03	-4.495956E-02	5.464054E-09	-2.018235E+00	4.200E+00	2.474907E-01	-1.001409E-01	1.866445E-03	-1.766485E+00
6.800E+00	4.167789E-03	-4.364697E-02	3.017332E-10	-2.018236E+00	4.400E+00	2.282396E-01	-9.249595E-02	1.296462E-03	-1.790722E+00
6.800E+00	1.571876E-03	-4.288681E-02	6.104827E-12	-2.018236E+00	4.600E+00	2.104443E-01	-8.557232E-02	8.997396E-04	-1.810710E+00
6.870E+00	1.143632E-03	-4.276204E-02	1.710590E-12	-2.018236E+00	4.800E+00	1.939672E-01	-7.930184E-02	6.234148E-04	-1.827114E+00
6.880E+00	7.166335E-04	-4.263783E-02	2.637476E-13	-2.018236E+00	5.000E+00	1.786843E-01	-7.362030E-02	4.309118E-04	-1.840508E+00
6.890E+00	2.908741E-04	-4.251415E-02	7.158470E-15	-2.018236E+00	5.500E+00	1.44958E-01	-6.161562E-02	1.683059E-04	-1.863873E+00
6.896E+00	3.601115E-05	-4.244020E-02	1.681697E-18	-2.018236E+00	6.000E+00	1.166474E-01	-5.214679E-02	6.323209E-05	-1.877209E+00
6.8984862E+00	0.000000E+00	-4.242976E-02	0.000000E+00	-2.018236E+00	6.500E+00	9.253061E-02	-4.241499E-02	2.229895E-05	-1.884426E+00
					7.000E+00	7.180075E-02	-3.853174E-02	7.121649E-06	-1.888055E+00
					7.500E+00	5.351051E-02	-3.59465E-02	1.944321E-06	-1.886999E+00
					8.000E+00	3.805997E-02	-2.953646E-02	4.093624E-07	-1.890334E+00
					8.500E+00	2.415962E-02	-2.616639E-02	5.295480E-08	-1.890522E+00
					9.000E+00	1.180312E-02	-2.334019E-02	2.108560E-09	-1.890555E+00
					9.200E+00	7.236560E-03	-2.233645E-02	2.332893E-10	-1.890557E+00
					9.300E+00	5.026933E-03	-2.185868E-02	4.527553E-11	-1.890557E+00
					9.400E+00	2.864319E-03	-2.139607E-02	3.602437E-12	-1.890557E+00
					9.500E+00	7.472341E-04	-2.094800E-02	8.522264E-15	-1.890557E+00
					9.510E+00	5.379743E-04	-2.090397E-02	1.942795E-15	-1.890557E+00

Sphere of polytropic index $n = 3.5$

ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$
9.520E+00	3.291542E-04	-2.086008E-02	2.129397E-16	-1.890557E+00
9.530E+00	1.207723E-04	-2.081632E-02	2.338050E-18	-1.890557E+00
9.532E+00	7.914844E-05	-2.080759E-02	3.491327E-19	-1.890557E+00
9.534E+00	3.754199E-05	-2.079886E-02	1.217104E-20	-1.890557E+00
9.535E+00	1.674531E-05	-2.079450E-02	3.217509E-22	-1.890557E+00
9.53580534E+00	0.000000E+00	-2.079098E-02	0.000000E+00	-1.890557E+00
Sphere of polytropic index $n = 4$				
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
1.000E-01	9.983367E-01	-3.320043E-02	9.917109E-01	-3.320043E-04
2.000E-01	9.933862E-01	-6.561355E-02	9.673656E-01	-2.624542E-03
3.000E-01	9.852649E-01	-9.650145E-02	9.294639E-01	-8.685130E-03
4.000E-01	9.741584E-01	-1.252190E-01	8.772966E-01	-2.003503E-02
5.000E-01	9.603109E-01	-1.512470E-01	8.166939E-01	-3.781176E-02
6.000E-01	9.440113E-01	-1.742114E-01	7.496973E-01	-6.271610E-02
7.000E-01	9.255784E-01	-1.938887E-01	6.793068E-01	-9.500546E-02
8.000E-01	9.053459E-01	-2.101991E-01	6.082369E-01	-1.345274E-01
9.000E-01	8.836493E-01	-2.271893E-01	5.387655E-01	-1.807833E-01
1.000E+00	8.608138E-01	-2.430096E-01	4.726570E-01	-2.330096E-01
1.200E+00	8.12949E-01	-2.441049E-01	3.550195E-01	-3.515111E-01
1.400E+00	7.637988E-01	-2.458005E-01	2.599331E-01	-4.817689E-01
1.600E+00	7.150639E-01	-2.405993E-01	1.869495E-01	-6.159343E-01
1.800E+00	6.678667E-01	-2.307699E-01	1.328767E-01	-7.476943E-01
2.000E+00	6.229408E-01	-2.181532E-01	9.380668E-02	-8.726129E-01
2.200E+00	5.806982E-01	-2.041302E-01	6.603189E-02	-9.879902E-01
2.400E+00	5.413170E-01	-1.896715E-01	4.647917E-02	-1.092508E+00
2.600E+00	5.048153E-01	-1.754198E-01	3.278404E-02	-1.185838E+00
2.800E+00	4.711084E-01	-1.617727E-01	2.320621E-02	-1.268298E+00
3.000E+00	4.400507E-01	-1.493544E-01	1.650112E-02	-1.340589E+00
3.200E+00	4.114642E-01	-1.370711E-01	1.179397E-02	-1.403603E+00
3.400E+00	3.851579E-01	-1.261522E-01	8.476056E-03	-1.458319E+00
3.600E+00	3.609402E-01	-1.161790E-01	6.125988E-03	-1.505679E+00
3.800E+00	3.386263E-01	-1.071048E-01	4.452495E-03	-1.546593E+00
4.000E+00	3.180424E-01	-9.886802E-02	3.254058E-03	-1.581888E+00
4.200E+00	2.990278E-01	-9.140069E-02	2.390880E-03	-1.612308E+00
4.400E+00	2.814355E-01	-8.463377E-02	1.765607E-03	-1.638310E+00
4.600E+00	2.651321E-01	-7.850046E-02	1.310121E-03	-1.661070E+00
ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$
4.800E+00	2.499972E-01	-7.293790E-02	9.765086E-04	-1.680489E+00
5.000E+00	2.359227E-01	-6.788810E-02	7.308848E-04	-1.697203E+00
5.500E+00	2.047620E-01	-5.717298E-02	3.599540E-04	-1.729483E+00
6.000E+00	1.783843E-01	-4.865644E-02	1.806270E-04	-1.751632E+00
6.500E+00	1.558267E-01	-4.181763E-02	9.187750E-05	-1.766799E+00
7.000E+00	1.363523E-01	-3.626805E-02	4.713163E-05	-1.777135E+00
7.500E+00	1.193924E-01	-3.171807E-02	2.425962E-05	-1.784142E+00
8.000E+00	1.045041E-01	-2.795075E-02	1.246423E-05	-1.788848E+00
8.500E+00	9.133874E-02	-2.480239E-02	6.357333E-06	-1.791973E+00
9.000E+00	7.961947E-02	-2.214833E-02	3.199605E-06	-1.794013E+00
9.500E+00	6.912410E-02	-1.989278E-02	1.578147E-06	-1.795324E+00
1.000E+01	5.967274E-02	-1.796142E-02	7.566237E-07	-1.796142E+00
1.100E+01	4.334010E-02	-1.485063E-02	1.529147E-07	-1.796926E+00
1.200E+01	2.972593E-02	-1.248033E-02	2.321012E-08	-1.797168E+00
1.300E+01	1.820540E-02	-1.063446E-02	1.999869E-09	-1.797223E+00
1.400E+01	8.330527E-03	-9.169339E-03	4.012013E-11	-1.797230E+00
1.420E+01	6.522448E-03	-8.913062E-03	1.180465E-11	-1.797230E+00
1.440E+01	4.764594E-03	-8.667197E-03	2.455442E-12	-1.797230E+00
1.460E+01	3.054901E-03	-8.431266E-03	2.660637E-13	-1.797230E+00
1.480E+01	1.391415E-03	-8.205031E-03	5.215353E-15	-1.797230E+00
1.490E+01	5.764189E-04	-8.095266E-03	6.363423E-17	-1.797230E+00
1.494E+01	2.534752E-04	-8.051973E-03	1.046351E-18	-1.797230E+00
1.496E+01	9.265099E-05	-8.030461E-03	6.827321E-21	-1.797230E+00
1.497E+01	1.240003E-05	-8.019735E-03	2.931658E-25	-1.797230E+00
1.49715463E+01	0.000000E+00	-8.018079E-03	0.000000E+00	-1.797230E+00
Sphere of polytropic index $n = 4.5$				
0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
1.000E-01	9.983371E-01	-3.318389E-02	9.908881E-01	-3.318389E-04
2.000E-01	9.933927E-01	-6.548414E-02	9.641962E-01	-2.619366E-03
3.000E-01	9.852972E-01	-9.608050E-02	9.217641E-01	-8.647245E-03
4.000E-01	9.742370E-01	-1.242707E-01	8.663722E-01	-1.988332E-02
5.000E-01	9.605416E-01	-1.495100E-01	8.013509E-01	-3.737749E-02
6.000E-01	9.444659E-01	-1.714309E-01	7.303389E-01	-6.171512E-02
7.000E-01	9.263725E-01	-1.898452E-01	6.566305E-01	-9.302415E-02
8.000E-01	9.066147E-01	-2.047287E-01	5.832101E-01	-1.310264E-01
9.000E-01	8.855408E-01	-2.161954E-01	5.124448E-01	-1.751183E-01

ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$
1.000E+00	8.634822E-01	-2.244656E-01	4.460604E-01	-2.244656E-01	1.800E+01	4.195731E-02	-5.362330E-03	2.663435E-08	-1.737395E+00
1.200E+00	8.176013E-01	-2.326322E-01	3.303532E-01	-3.349904E-01	1.900E+01	3.687696E-02	-4.813183E-03	1.309643E-08	-1.737559E+00
1.400E+00	7.710201E-01	-2.319352E-01	2.392552E-01	-4.545930E-01	2.000E+01	3.230429E-02	-4.344152E-03	6.323115E-09	-1.737661E+00
1.600E+00	7.252376E-01	-2.250671E-01	1.708605E-01	-5.761719E-01	2.200E+01	2.440558E-02	-3.590409E-03	1.352665E-09	-1.737758E+00
1.800E+00	6.812511E-01	-2.143129E-01	1.211127E-01	-6.943739E-01	2.400E+01	1.782310E-02	-3.016995E-03	2.401083E-10	-1.737789E+00
2.000E+00	6.396537E-01	-2.014225E-01	8.564403E-02	-8.056902E-01	2.600E+01	1.225324E-02	-2.570706E-03	3.057800E-11	-1.737797E+00
2.200E+00	6.007407E-01	-1.876381E-01	6.064276E-02	-9.081684E-01	2.800E+01	7.479072E-03	-2.2162580E-03	2.023776E-12	-1.737799E+00
2.400E+00	5.646023E-01	-1.737860E-01	4.311064E-02	-1.001007E+00	3.000E+01	3.341455E-03	-1.930888E-03	2.407943E-14	-1.737799E+00
2.600E+00	5.311960E-01	-1.603813E-01	3.082489E-02	-1.084178E+00	3.100E+01	1.472855E-03	-1.808323E-03	2.659985E-16	-1.737799E+00
2.800E+00	5.003999E-01	-1.477193E-01	2.219447E-02	-1.158119E+00	3.120E+01	1.113508E-03	-1.785214E-03	5.712338E-17	-1.737799E+00
3.000E+00	4.720487E-01	-1.359475E-01	1.610377E-02	-1.223528E+00	3.140E+01	7.587396E-04	-1.762545E-03	6.926438E-18	-1.737799E+00
3.200E+00	4.459580E-01	-1.251181E-01	1.177922E-02	-1.281209E+00	3.160E+01	4.084616E-04	-1.740305E-03	2.297904E-19	-1.737799E+00
3.400E+00	4.219391E-01	-1.152240E-01	8.687077E-03	-1.331990E+00	3.180E+01	6.258970E-05	-1.718483E-03	7.599165E-24	-1.737799E+00
3.600E+00	3.998087E-01	-1.062243E-01	6.459328E-03	-1.376667E+00	3.182E+01	2.824164E-05	-1.716324E-03	9.547612E-26	-1.737799E+00
3.800E+00	3.793937E-01	-9.805936E-02	4.841675E-03	-1.415977E+00	3.183E+01	1.108380E-05	-1.715245E-03	5.569133E-28	-1.737799E+00
4.000E+00	3.605338E-01	-9.066173E-02	3.657657E-03	-1.450588E+00	3.18364632E+01	0.000000E+00	-1.714549E-03	0.000000E+00	-1.737799E+00
4.200E+00	3.430825E-01	-8.396210E-02	2.784146E-03	-1.481091E+00	Sphere of polytropic index $n = 4.99$				
4.400E+00	3.269069E-01	-7.789310E-02	2.134679E-03	-1.508010E+00					
4.600E+00	3.118874E-01	-7.239135E-02	1.648118E-03	-1.531801E+00					
4.800E+00	2.979165E-01	-6.739845E-02	1.280914E-03	-1.552860E+00					
5.000E+00	2.848977E-01	-6.286131E-02	1.001818E-03	-1.571533E+00	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
5.500E+00	2.559722E-01	-5.321104E-02	5.59815E-04	-1.609634E+00	1.000E-01	9.983375E-01	-3.316769E-02	9.900827E-01	-3.316769E-04
6.000E+00	2.313631E-01	-4.550700E-02	3.188732E-04	-1.638252E+00	2.000E-01	9.933991E-01	-6.535779E-02	9.611063E-01	-2.614312E-03
6.500E+00	2.102182E-01	-3.928909E-02	1.882289E-04	-1.659964E+00	3.000E-01	9.853286E-01	-9.567135E-02	9.152734E-01	-8.610422E-03
7.000E+00	1.918837E-01	-3.421611E-02	1.139488E-04	-1.676589E+00	4.000E-01	9.743528E-01	-1.233546E-01	8.558746E-01	-1.973674E-02
7.500E+00	1.758539E-01	-3.003425E-02	7.052380E-05	-1.689426E+00	5.000E-01	9.607644E-01	-1.478442E-01	7.868198E-01	-3.686106E-02
8.000E+00	1.617329E-01	-2.655332E-02	4.450318E-05	-1.699412E+00	6.000E-01	9.449024E-01	-1.637873E-01	7.121439E-01	-6.076342E-02
8.500E+00	1.492079E-01	-2.362950E-02	2.856638E-05	-1.707231E+00	7.000E-01	9.271303E-01	-1.860366E-01	6.355843E-01	-9.115795E-02
9.000E+00	1.380290E-01	-2.115295E-02	1.861387E-05	-1.713389E+00	8.000E-01	9.078117E-01	-1.996271E-01	5.602868E-01	-1.277614E-01
9.500E+00	1.279946E-01	-1.903892E-02	1.229008E-05	-1.718263E+00	9.000E-01	8.873209E-01	-2.097397E-01	4.866563E-01	-1.698891E-01
1.000E+01	1.189407E-01	-1.722137E-02	8.209531E-06	-1.722137E+00	1.000E+00	8.659757E-01	-2.166602E-01	4.223371E-01	-2.166602E-01
1.100E+01	1.032586E-01	-1.427852E-02	3.772192E-06	-1.727701E+00	1.200E+00	8.219086E-01	-2.223613E-01	3.088846E-01	-2.202002E-01
1.200E+01	9.015578E-02	-1.202278E-02	1.789400E-06	-1.731281E+00	1.400E+00	7.775847E-01	-2.197517E-01	2.216045E-01	-4.307133E-01
1.300E+01	7.905006E-02	-1.025799E-02	8.678850E-07	-1.733600E+00	1.600E+00	7.343745E-01	-2.116399E-01	1.573426E-01	-5.417983E-01
1.400E+01	6.952045E-02	-8.852594E-03	4.281728E-07	-1.735108E+00	1.800E+00	6.931438E-01	-2.002797E-01	1.113094E-01	-6.489064E-01
1.500E+01	6.125558E-02	-7.715951E-03	2.134524E-07	-1.736089E+00	2.000E+00	6.548366E-01	-1.873117E-01	7.884541E-02	-7.492467E-01
1.600E+01	5.402048E-02	-6.784081E-03	1.069231E-07	-1.736725E+00	2.200E+00	6.182515E-01	-1.738445E-01	5.611508E-02	-8.414076E-01
1.700E+01	4.763467E-02	-6.010845E-03	5.352757E-08	-1.737134E+00	2.400E+00	5.848167E-01	-1.605793E-01	4.022057E-02	-9.249367E-01
					2.600E+00	5.539785E-01	-1.479271E-01	2.907520E-02	-9.999874E-01

ξ	θ	θ'	θ^{n-1}	$\xi^2\theta'$	ξ	θ	θ'	θ^{n-1}	$\xi^2\theta'$
2.800E+00	5.255903E-01	-1.361044E-01	2.121668E-02	-1.067058E+00	6.000E-01	2.785366E-02	-4.803129E-04	4.839990E-10	-1.729126E+00
3.000E+00	4.994753E-01	-1.252020E-01	1.563503E-02	-1.126818E+00	7.000E-01	2.373609E-02	-3.529833E-04	1.856524E-10	-1.729618E+00
3.200E+00	4.754471E-01	-1.152377E-01	1.163703E-02	-1.179993E+00	8.000E-01	2.084720E-02	-2.703013E-04	8.054119E-11	-1.729928E+00
3.400E+00	4.532151E-01	-1.061684E-01	8.747299E-03	-1.227306E+00	9.000E-01	1.824437E-02	-2.135969E-04	3.838506E-11	-1.730135E+00
3.600E+00	4.329232E-01	-9.795024E-02	6.638980E-03	-1.269435E+00	1.000E-02	1.632192E-02	-1.730279E-04	1.970150E-11	-1.730279E+00
3.800E+00	4.140894E-01	-9.051245E-02	5.086204E-03	-1.307000E+00	2.000E-02	7.689336E-03	-4.326725E-05	2.136481E-13	-1.730690E+00
4.000E+00	3.967099E-01	-8.378468E-02	3.931859E-03	-1.340555E+00	3.000E-02	4.784801E-03	-1.923050E-05	1.265858E-14	-1.730745E+00
4.200E+00	3.805328E-01	-7.769788E-02	2.410325E-03	-1.370591E+00	4.000E-02	3.342508E-03	-1.081724E-05	1.476366E-15	-1.730758E+00
4.400E+00	3.655534E-01	-7.218681E-02	2.403258E-03	-1.397537E+00	5.000E-02	2.477128E-03	-6.923051E-06	2.453300E-16	-1.730763E+00
4.600E+00	3.516237E-01	-6.719131E-02	1.909889E-03	-1.421768E+00	1.000E-03	7.46631E-04	-1.730765E-06	1.857682E-19	-1.730765E+00
4.800E+00	3.386462E-01	-6.265673E-02	1.524683E-03	-1.443611E+00	1.500E-03	1.694413E-04	-7.692290E-07	2.581224E-23	-1.730765E+00
5.000E+00	3.265336E-01	-5.853396E-02	1.225823E-03	-1.463349E+00	1.700E-03	3.369502E-05	-5.988807E-07	1.622253E-27	-1.730765E+00
5.500E+00	2.995453E-01	-4.975274E-02	7.311567E-04	-1.505020E+00	1.75818915E-03	0.000000E+00	-5.598955E-07	0.000000E+00	-1.730765E+00
6.000E+00	2.764897E-01	-4.272312E-02	4.525402E-04	-1.538032E+00	Sphere of polytropic index $n = 5$				
6.500E+00	2.565999E-01	-3.703069E-02	2.893653E-04	-1.564546E+00	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
7.000E+00	2.392876E-01	-3.236966E-02	1.904284E-04	-1.586113E+00	1.000E-01	9.983737E-01	-3.316736E-02	9.900663E-01	-3.316736E-04
7.500E+00	2.240965E-01	-2.851308E-02	1.285600E-04	-1.603861E+00	2.000E-01	9.933993E-01	-6.535321E-02	9.610434E-01	-2.614209E-03
8.000E+00	2.106690E-01	-2.529095E-02	8.879069E-05	-1.618621E+00	3.000E-01	9.853293E-01	-9.566304E-02	9.151417E-01	-8.609673E-03
8.500E+00	1.987214E-01	-2.257459E-02	6.258721E-05	-1.631014E+00	4.000E-01	9.743547E-01	-1.233360E-01	8.556625E-01	-1.973377E-02
9.000E+00	1.880266E-01	-2.026558E-02	4.493386E-05	-1.641512E+00	5.000E-01	9.607689E-01	-1.478106E-01	7.865271E-01	-3.695265E-02
9.500E+00	1.784007E-01	-1.826783E-02	3.279938E-05	-1.650476E+00	6.000E-01	9.449112E-01	-1.687341E-01	7.117802E-01	-6.074429E-02
1.000E+01	1.696936E-01	-1.658187E-02	2.430499E-05	-1.658187E+00	7.000E-01	9.271455E-01	-1.859604E-01	6.351664E-01	-9.112061E-02
1.100E+01	1.545615E-01	-1.380729E-02	1.389049E-05	-1.670682E+00	8.000E-01	9.078413E-01	-1.995256E-01	5.598344E-01	-1.276984E-01
1.200E+01	1.418680E-01	-1.166855E-02	8.313592E-06	-1.680271E+00	9.000E-01	8.873565E-01	-2.096118E-01	4.881900E-01	-1.697855E-01
1.300E+01	1.310727E-01	-9.986869E-03	5.174859E-06	-1.687781E+00	1.000E-01	8.660254E-01	-2.221606E-01	4.218750E-01	-2.165084E-01
1.400E+01	1.217828E-01	-8.641666E-03	3.331613E-06	-1.693767E+00	1.200E+00	8.219949E-01	-2.221608E-01	3.084714E-01	-3.199115E-01
1.500E+01	1.137053E-01	-7.549380E-03	2.208652E-06	-1.698611E+00	1.400E+00	7.777138E-01	-2.195160E-01	2.212681E-01	-4.302513E-01
1.600E+01	1.066194E-01	-6.650716E-03	1.502241E-06	-1.702583E+00	1.600E+00	7.345532E-01	-2.113822E-01	1.370867E-01	-5.411384E-01
1.700E+01	1.003538E-01	-5.902698E-03	1.045172E-06	-1.705880E+00	1.800E+00	6.946337E-01	-1.870439E-01	7.871720E-02	-7.481756E-01
1.800E+01	9.477448E-02	-5.273593E-03	7.419619E-07	-1.708644E+00	2.000E+00	6.546337E-01	-1.870439E-01	5.602935E-02	-8.401456E-01
1.900E+01	8.977499E-02	-4.739567E-03	5.362913E-07	-1.710984E+00	2.400E+00	6.185896E-01	-1.735838E-01	4.016534E-02	-9.235028E-01
2.000E+01	8.526976E-02	-4.282452E-03	3.939691E-07	-1.712981E+00	2.600E+00	5.852057E-01	-1.603303E-01	2.904118E-02	-9.984032E-01
2.500E+01	6.810549E-02	-2.751391E-03	1.025068E-07	-1.719620E+00	2.800E+00	5.544160E-01	-1.476928E-01	2.119707E-02	-1.065346E+00
3.000E+01	5.662905E-02	-1.914665E-03	3.393950E-08	-1.723198E+00	3.000E+00	5.260730E-01	-1.358860E-01	1.562500E-02	-1.125000E+00
3.500E+01	4.841812E-02	-1.408436E-03	1.327999E-08	-1.725334E+00	3.200E+00	4.760106E-01	-1.150479E-01	1.163322E-02	-1.178090E+00
4.000E+01	4.225371E-02	-1.079190E-03	5.873952E-09	-1.726705E+00					
4.500E+01	3.745599E-02	-8.531521E-04	2.853596E-09	-1.727633E+00					
5.000E+01	3.361606E-02	-6.913154E-04	1.492848E-09	-1.728288E+00					

ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$
3.400E+00	4.539206E-01	-1.059980E-01	8.747413E-03	-1.225336E+00	9.000E+01	1.924145E-02	-2.137147E-04	5.074885E-11	-1.731089E+00
3.600E+00	4.335550E-01	-9.779436E-02	6.641484E-03	-1.267415E+00	1.000E-02	1.731791E-02	-1.731272E-04	2.697571E-11	-1.731272E+00
3.800E+00	4.147509E-01	-9.037005E-02	5.090077E-03	-1.304943E+00	2.000E-02	8.659299E-03	-4.329640E-05	4.217801E-13	-1.731856E+00
4.000E+00	3.973597E-01	-8.365468E-02	3.936437E-03	-1.338475E+00	3.000E-02	5.773406E-03	-1.924405E-05	3.703333E-14	-1.731964E+00
4.200E+00	3.812464E-01	-7.757921E-02	3.070681E-03	-1.368497E+00	4.000E-02	4.330868E-03	-1.082501E-05	6.591426E-15	-1.732002E+00
4.400E+00	3.662897E-01	-7.207848E-02	2.415174E-03	-1.395439E+00	5.000E-02	3.464081E-03	-6.928079E-06	1.727938E-15	-1.732020E+00
4.600E+00	3.523807E-01	-6.709236E-02	1.914578E-03	-1.419674E+00	1.000E-03	1.732048E-03	-1.732043E-06	2.699976E-17	-1.732043E+00
4.800E+00	3.394221E-01	-6.256629E-02	1.529117E-03	-1.441527E+00	*	0.000000E+00	0.000000E+00	0.000000E+00	-1.732051E+00
5.000E+00	3.273268E-01	-5.845122E-02	1.229956E-03	-1.461281E+00					
5.500E+00	3.003757E-01	-4.968621E-02	7.344950E-04	-1.503008E+00					
6.000E+00	2.773501E-01	-4.266925E-02	4.551661E-04	-1.536093E+00					
6.500E+00	2.574846E-01	-3.698674E-02	2.914124E-04	-1.562890E+00	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
7.000E+00	2.401922E-01	-3.233375E-02	1.920232E-04	-1.584345E+00	1.000E-01	9.983383E-01	-3.313435E-02	9.884260E-01	-3.313435E-04
7.500E+00	2.250176E-01	-2.848324E-02	1.298072E-04	-1.602182E+00	2.000E-01	9.934123E-01	-6.509881E-02	9.547872E-01	-2.603952E-03
8.000E+00	2.116037E-01	-2.526611E-02	8.977168E-05	-1.617031E+00	3.000E-01	9.853930E-01	-9.483841E-02	9.021239E-01	-8.535457E-03
8.500E+00	1.996675E-01	-2.253380E-02	6.336424E-05	-1.629512E+00	4.000E-01	9.745473E-01	-1.215064E-01	8.348729E-01	-1.944102E-02
9.000E+00	1.898922E-01	-2.024810E-02	4.555394E-05	-1.640096E+00	5.000E-01	9.612138E-01	-1.445204E-01	7.581235E-01	-3.613010E-02
9.500E+00	1.793644E-01	-1.827305E-02	3.329793E-05	-1.649142E+00	6.000E-01	9.457754E-01	-1.635773E-01	6.768868E-01	-5.888784E-02
1.000E+01	1.706640E-01	-1.656932E-02	2.470882E-05	-1.656932E+00	7.000E-01	9.286320E-01	-1.786313E-01	5.95325E-01	-8.752933E-02
1.100E+01	1.555428E-01	-1.379815E-02	1.416116E-05	-1.669576E+00	8.000E-01	9.101772E-01	-1.898466E-01	5.174660E-01	-1.215018E-01
1.200E+01	1.428571E-01	-1.166181E-02	8.499860E-06	-1.679300E+00	9.000E-01	8.907802E-01	-1.975392E-01	4.450347E-01	-1.600067E-01
1.300E+01	1.320676E-01	-9.981856E-03	5.306137E-06	-1.686934E+00	1.000E+00	8.707732E-01	-2.021179E-01	3.796082E-01	-2.021179E-01
1.400E+01	1.227818E-01	-8.637917E-03	3.426136E-06	-1.693032E+00	1.200E+00	8.300401E-01	-2.037396E-01	2.714524E-01	-2.933850E-01
1.500E+01	1.147079E-01	-7.548570E-03	2.278029E-06	-1.697978E+00	1.400E+00	7.897537E-01	-1.981944E-01	1.916204E-01	-3.884610E-01
1.600E+01	1.076244E-01	-6.648612E-03	1.554048E-06	-1.702045E+00	1.600E+00	7.510458E-01	-1.883709E-01	1.347923E-01	-4.822295E-01
1.700E+01	1.013806E-01	-5.901131E-03	1.084464E-06	-1.705427E+00	1.800E+00	7.145489E-01	-1.763707E-01	9.510946E-02	-5.714410E-01
1.800E+01	9.578263E-02	-5.272438E-03	7.721835E-07	-1.708270E+00	2.000E+00	6.805484E-01	-1.635898E-01	6.761033E-02	-6.543591E-01
1.900E+01	9.078413E-02	-4.738732E-03	5.598344E-07	-1.710682E+00	2.200E+00	6.491069E-01	-1.508885E-01	4.855271E-02	-7.303005E-01
2.000E+01	8.627960E-02	-4.281866E-03	4.125234E-07	-1.712746E+00	2.400E+00	6.201545E-01	-1.387568E-01	3.527762E-02	-7.992389E-01
2.500E+01	6.911635E-02	-2.751447E-03	1.090146E-07	-1.719664E+00	2.600E+00	5.935493E-01	-1.274436E-01	2.595359E-02	-8.615188E-01
3.000E+01	5.763904E-02	-1.914918E-03	3.666912E-08	-1.723426E+00	2.800E+00	5.691156E-01	-1.170493E-01	1.933764E-02	-9.176867E-01
3.500E+01	4.942668E-02	-1.408741E-03	1.458037E-08	-1.725708E+00	3.000E+00	5.466674E-01	-1.075863E-01	1.459022E-02	-9.682764E-01
4.000E+01	4.326073E-02	-1.079494E-03	6.554857E-09	-1.727191E+00	3.200E+00	5.260214E-01	-9.901787E-02	1.114356E-02	-1.013943E+00
4.500E+01	3.846154E-02	-8.534385E-04	3.237128E-09	-1.728209E+00	3.400E+00	5.070046E-01	-9.128272E-02	8.611593E-03	-1.055228E+00
5.000E+01	3.462025E-02	-6.913751E-04	1.721794E-09	-1.728938E+00	3.600E+00	4.894575E-01	-8.430899E-02	6.729846E-03	-1.092644E+00
5.500E+01	3.285549E-02	-4.805244E-04	5.772594E-10	-1.729888E+00	3.800E+00	4.732352E-01	-7.802282E-02	5.315442E-03	-1.126650E+00
6.000E+01	2.473601E-02	-3.531554E-04	2.290752E-10	-1.730461E+00	4.000E+00	4.622073E-01	-7.235295E-02	4.240668E-03	-1.157647E+00
8.000E+01	2.164556E-02	-2.704428E-04	1.028521E-10	-1.730834E+00	4.200E+00	4.442574E-01	-6.723304E-02	3.415408E-03	-1.185991E+00

Sphere of polytropic index $n = 6$

ξ	θ	θ'	θ^{n+1}	$\xi^2 \theta'$	ξ	θ	θ'	θ^{n+1}	$\xi^2 \theta'$
4.400E+00	4.312815E-01	-6.260272E-02	2.775401E-03	-1.211989E+00	5.000E+02	6.833864E-02	-4.109422E-05	6.860883E-09	-1.027355E+01
4.600E+00	4.191873E-01	-5.840778E-02	2.274347E-03	-1.235909E+00	1.000E+03	5.313600E-02	-2.270810E-05	1.195974E-09	-2.270810E+01
4.800E+00	4.078926E-01	-5.459997E-02	1.878541E-03	-1.257983E+00	∞	0.000000E+00	0.000000E+00	0.000000E+00	∞
5.000E+00	3.973243E-01	-5.113662E-02	1.563206E-03	-1.278416E+00	Sphere of polytropic index $n = 10$				
5.500E+00	3.767222E-01	-4.374915E-02	1.017269E-03	-1.323412E+00	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
6.000E+00	3.533335E-01	-3.781617E-02	6.875330E-04	-1.361382E+00	1.000E-01	9.83416E-01	-3.300295E-02	9.819083E-01	-3.300295E-04
6.500E+00	3.306718E-01	-3.299193E-02	4.801799E-04	-1.393909E+00	2.000E-01	9.394636E-01	-6.409207E-02	9.304037E-01	-2.563683E-03
7.000E+00	3.201993E-01	-2.902347E-02	3.450981E-04	-1.422150E+00	3.000E-01	9.856408E-01	-9.167123E-02	8.529145E-01	-8.250411E-03
7.500E+00	3.065371E-01	-2.572384E-02	2.543205E-04	-1.446966E+00	4.000E-01	9.752824E-01	-1.146783E-01	7.593365E-01	-1.834858E-02
8.000E+00	2.943875E-01	-2.295323E-02	1.916175E-04	-1.469006E+00	5.000E-01	9.628735E-01	-1.326511E-01	6.595713E-01	-3.316277E-02
8.500E+00	2.835135E-01	-2.060580E-02	1.472367E-04	-1.488769E+00	6.000E-01	9.489183E-01	-1.456545E-01	5.617160E-01	-5.243562E-02
9.000E+00	2.732747E-01	-1.860052E-02	1.151326E-04	-1.506642E+00	7.000E-01	9.338937E-01	-1.541331E-01	4.712723E-01	-5.52521E-02
9.500E+00	2.648655E-01	-1.687456E-02	9.145115E-05	-1.522929E+00	8.000E-01	9.182206E-01	-1.587440E-01	3.912168E-01	-1.015962E-01
1.000E+01	2.568119E-01	-1.537875E-02	7.367255E-05	-1.537875E+00	9.000E-01	9.024953E-01	-1.602174E-01	3.225469E-01	-1.297761E-01
1.100E+01	2.427111E-01	-1.292967E-02	4.961661E-05	-1.564490E+00	1.000E+00	8.862581E-01	-1.592593E-01	2.649491E-01	-1.592593E-01
1.200E+01	2.307717E-01	-1.102546E-02	3.485607E-05	-1.587666E+00	1.200E+00	8.550013E-01	-1.524554E-01	1.784984E-01	-2.195358E-01
1.300E+01	2.205291E-01	-9.516059E-03	2.536652E-05	-1.608214E+00	1.400E+00	8.255076E-01	-1.421262E-01	1.213199E-01	-2.785673E-01
1.400E+01	2.116423E-01	-8.299523E-03	1.902032E-05	-1.626706E+00	1.600E+00	7.982286E-01	-1.305942E-01	8.383018E-02	-3.343213E-01
1.500E+01	2.038563E-01	-7.304721E-03	1.463084E-05	-1.643562E+00	1.800E+00	7.732641E-01	-1.191292E-01	5.910253E-02	-3.859787E-01
1.600E+01	1.969761E-01	-6.480832E-03	1.150321E-05	-1.659093E+00	2.000E+00	7.505303E-01	-1.083522E-01	4.256479E-02	-4.334087E-01
1.700E+01	1.908502E-01	-5.790785E-03	9.222478E-06	-1.673537E+00	2.200E+00	7.298600E-01	-9.851733E-02	3.130653E-02	-4.768239E-01
1.800E+01	1.853592E-01	-5.207031E-03	7.517929E-06	-1.687078E+00	2.400E+00	7.110564E-01	-8.968438E-02	2.349332E-02	-5.165820E-01
2.000E+01	1.759185E-01	-4.280013E-03	5.214114E-06	-1.712005E+00	2.600E+00	6.939218E-01	-8.181674E-02	1.796452E-02	-5.530812E-01
2.500E+01	1.585411E-01	-2.895420E-03	2.517634E-06	-1.765888E+00	2.800E+00	6.782706E-01	-7.483528E-02	1.397760E-02	-5.867086E-01
3.000E+01	1.466189E-01	-2.014213E-03	1.456566E-06	-1.812792E+00	3.000E+00	6.597348E-01	-6.864649E-02	1.105045E-02	-6.178184E-01
3.500E+01	1.378878E-01	-1.515147E-03	9.477234E-07	-1.856055E+00	3.200E+00	6.507654E-01	-6.315666E-02	8.864800E-03	-6.467242E-01
4.000E+01	1.311881E-01	-1.185883E-03	6.687461E-07	-1.897413E+00	3.400E+00	6.366314E-01	-5.827847E-02	7.206972E-03	-6.736991E-01
4.500E+01	1.258634E-01	-9.589763E-04	5.003757E-07	-1.937877E+00	3.600E+00	6.274185E-01	-5.393356E-02	5.931032E-03	-6.989790E-01
5.000E+01	1.215139E-01	-7.912354E-04	3.911844E-07	-1.978088E+00	3.800E+00	6.170270E-01	-5.005307E-02	4.935669E-03	-7.227664E-01
6.000E+01	1.147936E-01	-5.720406E-04	2.626778E-07	-2.059346E+00	4.000E+00	6.073703E-01	-4.657722E-02	4.149435E-03	-7.452355E-01
8.000E+01	1.097962E-01	-4.374204E-04	1.923582E-07	-2.143360E+00	4.200E+00	5.983726E-01	-4.345443E-02	3.521185E-03	-7.665361E-01
9.000E+01	1.058943E-01	-3.486476E-04	1.493171E-07	-2.231345E+00	4.400E+00	5.899679E-01	-4.064035E-02	3.013787E-03	-7.867972E-01
9.000E+01	1.027340E-01	-2.869187E-04	1.207811E-07	-2.324042E+00	4.600E+00	5.820984E-01	-3.809688E-02	2.599923E-03	-8.061300E-01
1.000E+02	1.000997E-01	-2.421930E-04	1.007003E-07	-2.421930E+00	4.800E+00	5.747133E-01	-3.579129E-02	2.259249E-03	-8.246313E-01
2.000E+02	8.571765E-02	-9.297530E-05	3.400101E-08	-3.719012E+00	5.000E+00	5.677680E-01	-3.369540E-02	1.976422E-03	-8.423851E-01
3.000E+02	7.825810E-02	-6.176759E-05	1.797650E-08	-5.559083E+00	5.500E+00	5.520803E-01	-2.922175E-02	1.452194E-03	-8.839580E-01
4.000E+02	7.280372E-02	-4.873645E-05	1.084114E-08	-7.797832E+00					

ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$
Sphere of polytropic index $n = 20$									
6.000E+00	5.384019E-01	-2.561709E-02	1.101976E-03	-9.222154E-01	0.000E+00	1.000000E+00	0.000000E+00	1.000000E+00	0.000000E+00
6.500E+00	5.263538E-01	-2.267064E-02	8.591235E-04	-9.578345E-01	1.000E-01	9.983498E-01	-3.267875E-02	9.659117E-01	-3.267875E-04
7.000E+00	5.156470E-01	-2.023108E-02	6.852941E-04	-9.913230E-01	2.000E-01	9.958575E-01	-6.170052E-02	8.736328E-01	-2.468021E-03
7.500E+00	5.060568E-01	-1.818788E-02	5.574324E-04	-1.023068E+00	3.000E-01	9.862166E-01	-8.457184E-02	7.471691E-01	-7.611465E-03
8.000E+00	4.974068E-01	-1.645894E-02	4.611359E-04	-1.053372E+00	4.000E-01	9.796902E-01	-1.004041E-01	6.122497E-01	-1.607041E-02
8.500E+00	4.895559E-01	-1.496231E-02	3.871020E-04	-1.082472E+00	5.000E-01	9.663460E-01	-1.098095E-01	4.872885E-01	-2.745239E-02
9.000E+00	4.823904E-01	-1.371060E-02	3.291432E-04	-1.110559E+00	6.000E-01	9.551219E-01	-1.139143E-01	3.812720E-01	-4.100914E-02
9.500E+00	4.758173E-01	-1.260705E-02	2.830368E-04	-1.137786E+00	7.000E-01	9.436916E-01	-1.141671E-01	2.969392E-01	-5.593921E-02
1.000E+01	4.697602E-01	-1.164281E-02	2.458318E-04	-1.164281E+00	8.000E-01	9.323760E-01	-1.118081E-01	2.298330E-01	-7.155717E-02
1.100E+01	4.649848E-01	-1.004527E-02	1.902860E-04	-1.215478E+00	9.000E-01	9.213835E-01	-1.078387E-01	1.791633E-01	-8.734937E-02
1.200E+01	4.595822E-01	-8.783332E-03	1.515813E-04	-1.264800E+00	1.000E+00	9.108381E-01	-1.029657E-01	1.406905E-01	-1.029657E-01
1.300E+01	4.413004E-01	-7.767563E-03	1.236168E-04	-1.312718E+00	1.200E+00	8.913063E-01	-9.228373E-02	8.924141E-02	-1.328886E-01
1.400E+01	4.339617E-01	-6.936633E-03	1.027934E-04	-1.359580E+00	1.400E+00	8.739014E-01	-8.190645E-02	5.898085E-02	-1.805862E-01
1.500E+01	4.273801E-01	-6.247326E-03	8.688742E-05	-1.405648E+00	1.600E+00	8.584714E-01	-7.259580E-02	4.057347E-02	-1.658452E-01
1.600E+01	4.214303E-01	-5.668458E-03	7.447038E-05	-1.451125E+00	1.800E+00	8.447807E-01	-6.451328E-02	2.894836E-02	-2.090230E-01
1.700E+01	4.160141E-01	-5.177058E-03	6.459339E-05	-1.496170E+00	2.000E+00	8.325888E-01	-5.758762E-02	2.133250E-02	-2.303513E-01
1.800E+01	4.110528E-01	-4.755886E-03	5.660752E-05	-1.540907E+00	2.200E+00	8.216781E-01	-5.167461E-02	1.617097E-02	-2.501051E-01
1.900E+01	4.064832E-01	-4.391796E-03	5.005756E-05	-1.585438E+00	2.400E+00	8.118618E-01	-4.661930E-02	1.256389E-02	-2.685272E-01
2.000E+01	4.022536E-01	-4.074611E-03	4.461694E-05	-1.629844E+00	2.600E+00	8.029826E-01	-4.228148E-02	9.973069E-03	-2.858228E-01
2.500E+01	3.849282E-01	-2.963129E-03	2.749018E-05	-1.851956E+00	2.800E+00	7.949094E-01	-3.854114E-02	8.066226E-03	-3.021625E-01
3.000E+01	3.718791E-01	-2.308006E-03	1.881133E-05	-2.077205E+00	3.000E+00	7.875330E-01	-3.529861E-02	6.631991E-03	-3.176875E-01
3.500E+01	3.614695E-01	-1.883750E-03	1.376538E-05	-2.307594E+00	3.200E+00	7.807623E-01	-3.247209E-02	5.532165E-03	-3.325142E-01
4.000E+01	3.528268E-01	-1.589876E-03	1.054818E-05	-2.543801E+00	3.400E+00	7.745209E-01	-2.999476E-02	4.674079E-03	-3.467394E-01
4.500E+01	3.454390E-01	-1.375749E-03	8.357667E-06	-2.785891E+00	3.600E+00	7.687447E-01	-2.781201E-02	3.994159E-03	-3.604436E-01
5.000E+01	3.389835E-01	-1.213450E-03	6.791486E-06	-3.033625E+00	3.800E+00	7.633795E-01	-2.587912E-02	3.447863E-03	-3.736944E-01
6.000E+01	3.280772E-01	-9.845589E-04	4.739537E-06	-3.544412E+00	4.000E+00	7.587036E-01	-2.415930E-02	3.003386E-03	-3.865488E-01
7.000E+01	3.190447E-01	-8.311192E-04	3.486322E-06	-4.072484E+00	4.200E+00	7.537036E-01	-2.262217E-02	2.637619E-03	-3.990550E-01
8.000E+01	3.113212E-01	-7.209510E-04	2.661684E-06	-4.614086E+00	4.400E+00	7.493196E-01	-2.124247E-02	2.333503E-03	-4.112543E-01
9.000E+01	3.045366E-01	-6.377537E-04	2.089459E-06	-5.165805E+00	4.600E+00	7.451975E-01	-1.999915E-02	2.078257E-03	-4.231820E-01
1.000E+02	2.984977E-01	-5.724677E-04	1.676299E-06	-5.724677E+00	4.800E+00	7.413120E-01	-1.887451E-02	1.862179E-03	-4.348687E-01
2.000E+02	2.589827E-01	-2.830778E-04	3.515424E-07	-1.132311E+01	5.000E+00	7.376408E-01	-1.785363E-02	1.677812E-03	-4.463409E-01
3.000E+02	2.363142E-01	-1.831466E-04	1.283474E-07	-1.648319E+01	5.500E+00	7.292785E-01	-1.567693E-02	1.320569E-03	-4.742271E-01
4.000E+02	2.207969E-01	-1.320122E-04	6.080268E-08	-2.112195E+01	6.000E+00	7.218938E-01	-1.392220E-02	1.066447E-03	-5.011992E-01
5.000E+02	2.092464E-01	-1.013144E-04	3.366969E-08	-2.532861E+01	6.500E+00	7.153036E-01	-1.248451E-02	8.796269E-04	-5.274706E-01
1.000E+03	1.768889E-01	-4.244813E-05	5.308558E-09	-4.244813E+01	7.000E+00	7.093689E-01	-1.128983E-02	7.384395E-04	-5.532014E-01
∞	0.000000E+00	0.000000E+00	0.000000E+00	∞					

ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$	ξ	θ	θ'	θ^{n+1}	$\xi^2\theta'$	$\exp(-\theta)$	$\xi^2\theta'$
7.500E+00	7.039822E-01	-1.028470E-02	6.292103E-04	-5.785144E-01	0.000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
8.000E+00	6.990592E-01	-9.429766E-03	5.429917E-04	-6.035050E-01	1.000E-01	1.665834E-03	3.330003E-02	9.983356E-01	3.330003E-02	1.000E-01	1.665834E-03	3.330003E-02	9.983356E-01	3.330003E-02	9.983356E-01	3.330003E-04
8.500E+00	6.945324E-01	-8.695481E-03	4.737431E-04	-6.282485E-01	2.000E-01	6.63367E-03	6.640101E-02	9.933687E-01	6.640101E-02	2.000E-01	6.63367E-03	6.640101E-02	9.933687E-01	6.640101E-02	9.933687E-01	2.656040E-03
9.000E+00	6.903474E-01	-8.059320E-03	4.172743E-04	-6.528049E-01	3.000E-01	1.493288E-02	9.910765E-02	9.851781E-01	9.910765E-02	3.000E-01	1.493288E-02	9.910765E-02	9.851781E-01	9.910765E-02	9.851781E-01	8.919688E-03
9.500E+00	6.864597E-01	-7.503853E-03	3.706086E-04	-6.772227E-01	4.000E-01	2.645548E-02	1.312320E-01	9.738914E-01	1.312320E-01	4.000E-01	2.645548E-02	1.312320E-01	9.738914E-01	1.312320E-01	9.738914E-01	2.099712E-02
1.000E+01	6.828324E-01	-7.015413E-03	3.315862E-04	-7.015413E-01	5.000E-01	4.115396E-02	1.625969E-01	9.596814E-01	1.625969E-01	5.000E-01	4.115396E-02	1.625969E-01	9.596814E-01	1.625969E-01	9.596814E-01	4.064923E-02
1.100E+01	6.762413E-01	-6.198380E-03	2.704805E-04	-7.500039E-01	6.000E-01	5.894407E-02	1.930388E-01	9.427593E-01	1.930388E-01	6.000E-01	5.894407E-02	1.930388E-01	9.427593E-01	1.930388E-01	9.427593E-01	6.949393E-02
1.200E+01	6.703815E-01	-5.544369E-03	2.253008E-04	-7.983891E-01	7.000E-01	7.972600E-02	2.224100E-01	9.233693E-01	2.224100E-01	7.000E-01	7.972600E-02	2.224100E-01	9.233693E-01	2.224100E-01	9.233693E-01	1.089809E-01
1.300E+01	6.651126E-01	-5.010838E-03	1.908967E-04	-8.468317E-01	8.000E-01	1.033861E-01	2.505811E-01	9.017788E-01	2.505811E-01	8.000E-01	1.033861E-01	2.505811E-01	9.017788E-01	2.505811E-01	9.017788E-01	1.603719E-01
1.400E+01	6.603295E-01	-4.568499E-03	1.640496E-04	-8.954257E-01	9.000E-01	1.297985E-01	2.774417E-01	8.782724E-01	2.774417E-01	9.000E-01	1.297985E-01	2.774417E-01	8.782724E-01	2.774417E-01	8.782724E-01	2.247278E-01
1.500E+01	6.559522E-01	-4.196607E-03	1.426644E-04	-9.442366E-01	1.000E+00	1.588277E-01	3.029014E-01	8.531434E-01	3.029014E-01	1.000E+00	1.588277E-01	3.029014E-01	8.531434E-01	3.029014E-01	8.531434E-01	3.029014E-01
1.600E+01	6.519179E-01	-3.880115E-03	1.253289E-04	-9.933095E-01	1.400E+00	2.981563E-01	3.896126E-01	7.421854E-01	3.896126E-01	1.400E+00	2.981563E-01	3.896126E-01	7.421854E-01	3.896126E-01	7.421854E-01	7.638408E-01
1.700E+01	6.481772E-01	-3.607872E-03	1.110629E-04	-1.042675E+00	1.600E+00	3.795827E-01	4.236191E-01	6.841468E-01	4.236191E-01	1.600E+00	3.795827E-01	4.236191E-01	6.841468E-01	4.236191E-01	6.841468E-01	1.084463E+00
1.800E+01	6.446902E-01	-3.371460E-03	9.916833E-05	-1.092353E+00	1.800E+00	4.671988E-01	4.515544E-01	6.267553E-01	4.515544E-01	1.800E+00	4.671988E-01	4.515544E-01	6.267553E-01	4.515544E-01	6.267553E-01	1.463039E+00
1.900E+01	6.414245E-01	-3.164419E-03	8.913673E-05	-1.142355E+00	2.000E+00	5.598230E-01	4.737714E-01	5.713102E-01	4.737714E-01	2.000E+00	5.598230E-01	4.737714E-01	5.713102E-01	4.737714E-01	5.713102E-01	1.895086E+00
2.000E+01	6.383533E-01	-2.981720E-03	8.059039E-05	-1.192688E+00	2.200E+00	6.563374E-01	4.907411E-01	5.187373E-01	4.907411E-01	2.200E+00	6.563374E-01	4.907411E-01	5.187373E-01	4.907411E-01	5.187373E-01	2.375187E+00
2.500E+01	6.252455E-01	-2.318822E-03	5.212690E-05	-1.449263E+00	2.400E+00	7.558054E-01	5.030012E-01	4.696322E-01	5.030012E-01	2.400E+00	7.558054E-01	5.030012E-01	4.696322E-01	5.030012E-01	4.696322E-01	2.897287E+00
3.000E+01	6.147606E-01	-1.903765E-03	3.654492E-05	-1.713388E+00	2.600E+00	8.572815E-01	5.11157E-01	4.243140E-01	5.11157E-01	2.600E+00	8.572815E-01	5.11157E-01	4.243140E-01	5.11157E-01	4.243140E-01	3.455142E+00
3.500E+01	6.059930E-01	-1.619462E-03	2.702838E-05	-1.983841E+00	2.800E+00	9.600126E-01	5.156433E-01	3.828881E-01	5.156433E-01	2.800E+00	9.600126E-01	5.156433E-01	3.828881E-01	5.156433E-01	3.828881E-01	4.042643E+00
4.000E+01	5.984391E-01	-1.412095E-03	2.076917E-05	-2.259352E+00	3.000E+00	1.063335E+00	5.1771153E-01	3.453022E-01	5.1771153E-01	3.000E+00	1.063335E+00	5.1771153E-01	3.453022E-01	5.1771153E-01	3.453022E-01	4.654038E+00
4.500E+01	5.917910E-01	-1.253716E-03	1.642616E-05	-2.538775E+00	3.200E+00	1.166888E+00	5.160226E-01	3.113966E-01	5.160226E-01	3.200E+00	1.166888E+00	5.160226E-01	3.113966E-01	5.160226E-01	3.113966E-01	5.284071E+00
5.000E+01	5.858471E-01	-1.128450E-03	1.328830E-05	-2.821125E+00	3.400E+00	1.269603E+00	5.128074E-01	2.809432E-01	5.128074E-01	3.400E+00	1.269603E+00	5.128074E-01	2.809432E-01	5.128074E-01	2.809432E-01	5.928053E+00
5.500E+01	5.755521E-01	-9.420720E-04	9.157498E-06	-3.391459E+00	3.600E+00	1.371696E+00	5.078613E-01	2.536763E-01	5.078613E-01	3.600E+00	1.371696E+00	5.078613E-01	2.536763E-01	5.078613E-01	2.536763E-01	6.581882E+00
6.000E+01	5.668294E-01	-8.092547E-04	6.645062E-06	-3.965348E+00	3.800E+00	1.472655E+00	5.015255E-01	2.293159E-01	5.015255E-01	3.800E+00	1.472655E+00	5.015255E-01	2.293159E-01	5.015255E-01	2.293159E-01	7.240202E+00
6.500E+01	5.582584E-01	-7.092862E-04	5.010298E-06	-4.539432E+00	4.000E+00	1.568688E+00	4.858157E-01	1.882027E-01	4.858157E-01	4.000E+00	1.568688E+00	4.858157E-01	1.882027E-01	4.858157E-01	1.882027E-01	8.569789E+00
7.000E+01	5.525717E-01	-6.310474E-04	3.891894E-06	-5.111484E+00	4.200E+00	1.670236E+00	4.769026E-01	1.709273E-01	4.769026E-01	4.200E+00	1.670236E+00	4.769026E-01	1.709273E-01	4.769026E-01	1.709273E-01	9.232834E+00
7.500E+01	5.465871E-01	-5.680053E-04	3.096327E-06	-5.680053E+00	4.400E+00	1.766517E+00	4.675506E-01	1.555223E-01	4.675506E-01	4.400E+00	1.766517E+00	4.675506E-01	1.555223E-01	4.675506E-01	1.555223E-01	9.892947E+00
8.000E+01	5.405871E-01	-5.077109E-04	2.564703E-07	-1.602119E+01	4.600E+00	1.860966E+00	4.578460E-01	1.417758E-01	4.578460E-01	4.600E+00	1.860966E+00	4.578460E-01	1.417758E-01	4.578460E-01	1.417758E-01	1.054877E+01
8.500E+01	4.854501E-01	-1.780132E-04	2.564703E-07	-1.602119E+01	4.800E+00	2.044092E+00	4.479695E-01	1.119924E+01	4.479695E-01	4.800E+00	2.044092E+00	4.479695E-01	1.119924E+01	4.479695E-01	1.119924E+01	1.19924E+01
9.000E+01	4.703420E-01	-1.289756E-04	1.320350E-07	-2.063610E+01	5.000E+00	2.261844E+00	4.320398E-01	1.041583E+01	4.320398E-01	5.000E+00	2.261844E+00	4.320398E-01	1.041583E+01	4.320398E-01	1.041583E+01	1.279696E+01
9.500E+01	4.589993E-01	-1.001212E-04	7.907753E-08	-2.503031E+01	5.200E+00	2.467210E+00	3.985636E-01	8.482117E-02	3.985636E-01	5.200E+00	2.467210E+00	3.985636E-01	8.482117E-02	3.985636E-01	8.482117E-02	1.434829E+01
1.000E+02	4.259806E-01	-4.526416E-05	1.648873E-08	-4.526416E+01	5.400E+00	2.660586E+00	3.751497E-01	6.990724E-02	3.751497E-01	5.400E+00	2.660586E+00	3.751497E-01	6.990724E-02	3.751497E-01	6.990724E-02	1.585008E+01
1.000E+03	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	7.000E+00	2.842389E+00	3.531060E-01	5.827459E-02	3.531060E-01	7.000E+00	2.842389E+00	3.531060E-01	5.827459E-02	3.531060E-01	5.827459E-02	1.730219E+01

ξ	θ	θ'	$\exp(-\theta)$	$\xi^2\theta'$	ξ	θ	θ'	$\exp(-\theta)$	$\xi^2\theta'$
7.500E+00	3.013942E+00	3.325560E-01	4.909778E-02	1.870627E+01	1.600E+03	1.405756E+01	1.275496E-03	7.850172E-07	3.265269E+03
8.000E+00	3.175397E+00	3.135140E-01	4.177752E-02	2.006490E+01	1.800E+03	1.429770E+01	1.131557E-03	6.174289E-07	3.666246E+03
8.500E+00	3.327699E+00	2.959319E-01	3.587556E-02	2.138108E+01	2.000E+03	1.451209E+01	1.016395E-03	4.982857E-07	4.065580E+03
9.000E+00	3.471559E+00	2.797279E-01	3.106857E-02	2.265796E+01	2.200E+03	1.470565E+01	9.222316E-04	4.105967E-07	4.463601E+03
9.500E+00	3.607640E+00	2.648050E-01	2.711575E-02	2.389865E+01					
1.000E+01	3.736560E+00	2.510611E-01	2.383596E-02	2.510611E+01					
1.100E+01	3.975120E+00	2.267131E-01	1.877706E-02	2.743228E+01					
1.200E+01	4.191178E+00	2.059457E-01	1.512845E-02	2.965617E+01					
1.300E+01	4.387994E+00	1.881317E-01	1.242563E-02	3.179426E+01					
1.400E+01	4.568253E+00	1.727558E-01	1.037607E-02	3.386015E+01					
1.500E+01	4.734178E+00	1.594003E-01	8.789668E-03	3.586507E+01					
1.600E+01	4.887614E+00	1.477278E-01	7.539388E-03	3.781831E+01					
1.700E+01	5.030103E+00	1.374657E-01	6.538135E-03	3.972757E+01					
1.800E+01	5.162942E+00	1.283929E-01	5.724835E-03	4.159931E+01					
1.900E+01	5.287225E+00	1.203294E-01	5.065769E-03	4.343882E+01					
2.000E+01	5.403887E+00	1.131274E-01	4.499058E-03	4.525097E+01					
2.500E+01	5.897310E+00	8.642557E-02	2.746824E-03	5.401598E+01					
3.000E+01	6.284061E+00	6.943642E-02	1.865803E-03	6.249278E+01					
3.500E+01	6.600543E+00	5.783467E-02	1.359629E-03	7.084747E+01					
4.000E+01	6.867771E+00	4.948146E-02	1.040794E-03	7.917033E+01					
4.500E+01	7.098813E+00	4.321634E-02	8.260852E-04	8.751309E+01					
5.000E+01	7.302273E+00	3.836249E-02	6.740049E-04	9.590622E+01					
6.000E+01	7.648490E+00	3.136331E-02	4.767637E-04	1.129079E+02					
7.000E+01	7.936828E+00	2.658033E-02	3.573383E-04	1.302436E+02					
8.000E+01	8.184440E+00	2.311346E-02	2.789606E-04	1.479262E+02					
9.000E+01	8.401877E+00	2.048700E-02	2.244456E-04	1.659447E+02					
1.000E+02	8.598060E+00	1.842785E-02	1.848326E-04	1.842785E+02					
2.000E+02	9.887835E+00	9.496032E-03	5.078880E-05	3.798413E+02					
3.000E+02	1.066923E+01	6.518055E-03	2.324940E-05	5.866250E+02					
4.000E+02	1.123720E+01	4.980420E-03	1.317482E-05	7.968672E+02					
5.000E+02	1.168443E+01	4.030609E-03	8.423977E-06	1.007652E+03					
6.000E+02	1.205324E+01	3.382905E-03	5.825635E-06	1.217846E+03					
7.000E+02	1.236685E+01	2.912291E-03	4.257402E-06	1.427023E+03					
8.000E+02	1.263944E+01	2.554780E-03	3.241604E-06	1.635059E+03					
9.000E+02	1.288035E+01	2.274029E-03	2.547619E-06	1.841963E+03					
1.000E+03	1.309606E+01	2.047800E-03	2.053310E-06	2.047800E+03					
1.200E+03	1.346941E+01	1.705987E-03	1.413549E-06	2.456622E+03					
1.400E+03	1.378479E+01	1.460323E-03	1.031193E-06	2.862234E+03					

TABLE 2. BOUNDARY VALUES OF LANE-EMDEN FUNCTIONS.

N=1, Slab				
n	ξ_1	θ_1^1	$-\xi_1^2/\theta_1^1$	
-0.9	1.26907230	-4.472136	2.837732x10 ⁻¹	
-0.8	1.28498902	-3.162278	4.063492x10 ⁻¹	
-0.5	1.33333333	-2.000000	6.666667x10 ⁻¹	
-0.2	1.38194567	-1.581139	8.740192x10 ⁻¹	
0	1.41421356	-1.414214	1.000000	
0.5	1.49368640	-1.154701	1.293555	
1	1.57079633	-1.000000	1.570796	
1.5	1.64534085	-8.944272x10 ⁻¹	1.839547	
2	1.71731534	-8.164966x10 ⁻¹	2.103273	
3	1.85407468	-7.071068x10 ⁻¹	2.622058	
4	1.98232217	-6.324555x10 ⁻¹	3.134327	
5	2.10327316	-5.773503x10 ⁻¹	3.642976	
6	2.21794979	-5.345225x10 ⁻¹	4.149404	
10	2.62843161	-4.264014x10 ⁻¹	6.164219	
20	3.44944436	-3.086067x10 ⁻¹	1.117748x10 ¹	

N=2, Cylinder				
n	ξ_1	θ_1^1	$-\xi_1^2/\theta_1^1$	
-0.9	1.71782384	-4.307079	7.398803	1.994187x10 ⁻¹
-0.8	1.74556088	-2.935681	5.124411	2.973008x10 ⁻¹
-0.5	1.83413266	-1.669972	3.062950	5.491508x10 ⁻¹
-0.2	1.93087184	-1.193383	2.304271	8.089905x10 ⁻¹
0	2.00000000	-1.000000	2.000000	1.000000
0.5	2.18866219	-6.975389x10 ⁻¹	1.527374	1.569563
1	2.40482356	-5.191475x10 ⁻¹	1.248459	2.316129
1.5	2.64777677	-4.007569x10 ⁻¹	1.061115	3.303470
2	2.92132072	-3.167585x10 ⁻¹	9.253533x10 ⁻¹	4.611274
3	3.57390098	-2.070908x10 ⁻¹	7.401221x10 ⁻¹	8.628825

N=3, Sphere				
n	ξ_1	θ_1^1	$-\xi_1^2/\theta_1^1$	
-0.9	2.05040073	-4.232444	1.779380x10 ¹	1.614828x10 ⁻¹
-0.8	2.08744257	-2.83273	1.234575x10 ¹	2.455867x10 ⁻¹
-0.5	2.20838842	-1.521468	7.421510	4.838724x10 ⁻¹
-0.2	2.34663965	-1.020325	5.618645	7.666312x10 ⁻¹
0	2.44948974	-8.164966x10 ⁻¹	4.898979	1.000000
0.5	2.75269805	-4.99971x10 ⁻¹	3.788651	1.835143
1	3.14159265	-3.183099x10 ⁻¹	3.141593	3.289868
1.5	3.65375374	-2.03013x10 ⁻¹	2.714054	5.990705
2	4.35287460	-1.272487x10 ⁻¹	2.411046	1.40254x10 ¹
2.5	5.35527546	-7.626491x10 ⁻²	2.187200	2.340646x10 ¹
3	6.89684862	-4.242976x10 ⁻²	2.018236	5.418248x10 ¹
3.5	9.53580534	-2.079096x10 ⁻²	1.890557	1.528837x10 ²
4	1.49715463x10 ¹	-8.018079x10 ⁻³	1.797230	6.224079x10 ²
4.5	3.18364632x10 ¹	-1.714549x10 ⁻³	1.737799	6.189473x10 ³
4.75	6.63870955x10 ¹	-3.912321x10 ⁻⁴	1.724256470	5.656241x10 ⁴
4.80	8.36128388x10 ¹	-2.453446x10 ⁻⁴	1.723445575	1.138709x10 ⁵
4.81	8.84074527x10 ¹	-2.204976x10 ⁻⁴	1.723381986	1.336484x10 ⁵
4.82	9.35160367x10 ¹	-1.970617x10 ⁻⁴	1.723354059	1.581840x10 ⁵
4.821	9.40584862x10 ¹	-1.947952x10 ⁻⁴	1.723353281	1.609527x10 ⁵
4.822	9.46070673x10 ¹	-1.925427x10 ⁻⁴	1.723352876	1.637854x10 ⁵
4.823	9.51618844x10 ¹	-1.903041x10 ⁻⁴	1.723352843	1.666839x10 ⁵
4.824	9.57230437x10 ¹	-1.880794x10 ⁻⁴	1.723353186	1.696500x10 ⁵
4.825	9.62906542x10 ¹	-1.858687x10 ⁻⁴	1.723353906	1.726858x10 ⁵

n	ξ_1	θ_1'	$-\xi_1^2\theta_1'$	$-\xi_1/3\theta_1'$
4.826	9.68648274×10^1	-1.836718×10^{-4}	1.723355004	1.757933×10^5
4.827	9.74456773×10^1	-1.814889×10^{-4}	1.723356482	1.789746×10^5
4.828	9.80333204×10^1	-1.793198×10^{-4}	1.723358340	1.822319×10^5
4.829	9.86278761×10^1	-1.771645×10^{-4}	1.723360582	1.855674×10^5
4.83	9.92294667×10^1	-1.750232×10^{-4}	1.723363209	1.889835×10^5
4.84	1.05661355×10^2	-1.543677×10^{-4}	1.723410981	2.281594×10^5
4.85	1.12955631×10^2	-1.350813×10^{-4}	1.723499081	2.787349×10^5
4.90	1.71433450×10^2	-5.868160×10^{-5}	1.724618596	9.738058×10^5
4.95	3.47400371×10^2	-1.431107×10^{-5}	1.727160845	8.091645×10^6
4.99	1.75818915×10^3	-5.598955×10^{-7}	1.730765298	1.046736×10^9
5	∞	0.000000	1.732050808	∞

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